

Bitwise differential trail through **KATAN32** found by SAT

CiVerLy

June 15, 2026

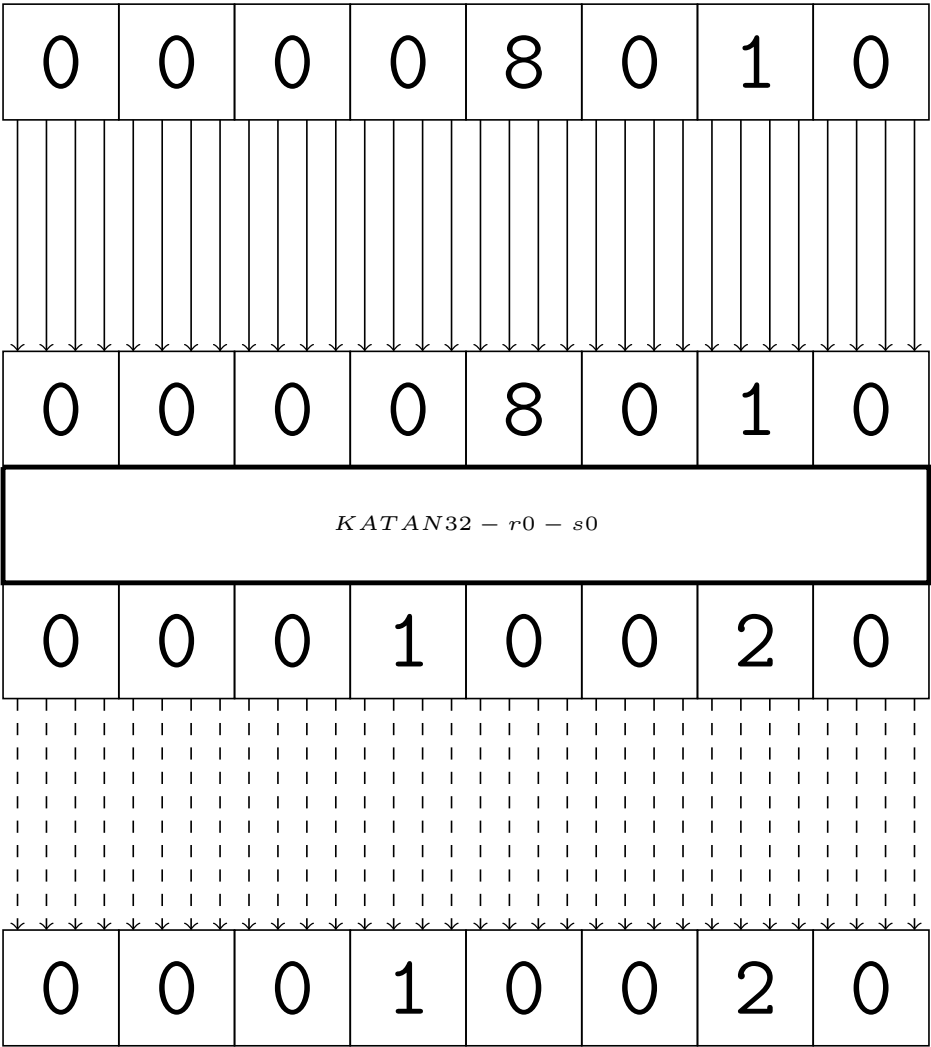
1 KATAN32

Maximal differential probability: 2^{-30}



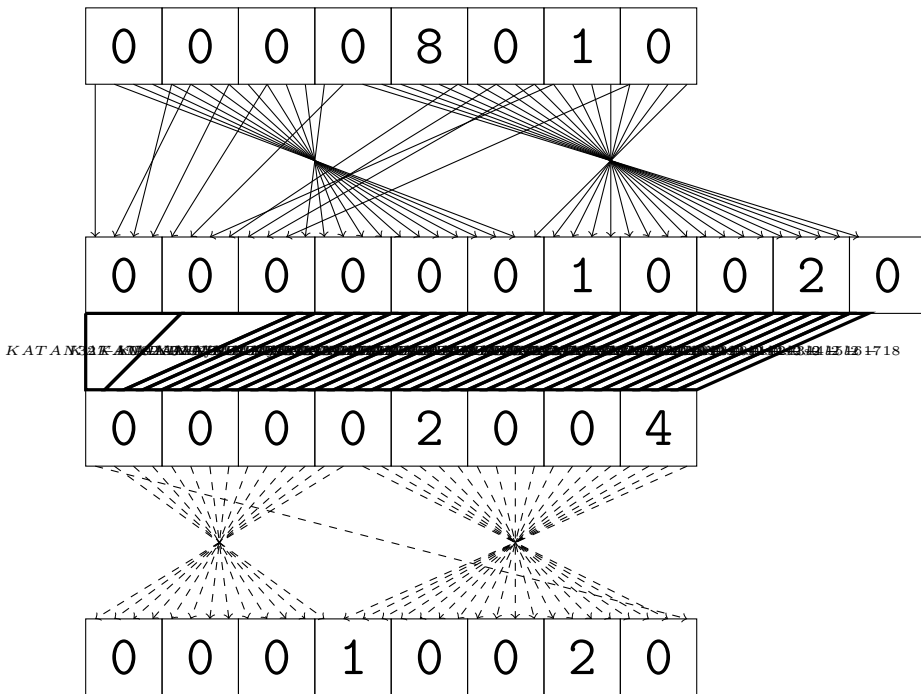
2 KATAN32-r0

Maximal differential probability: $2^{-0.0}$



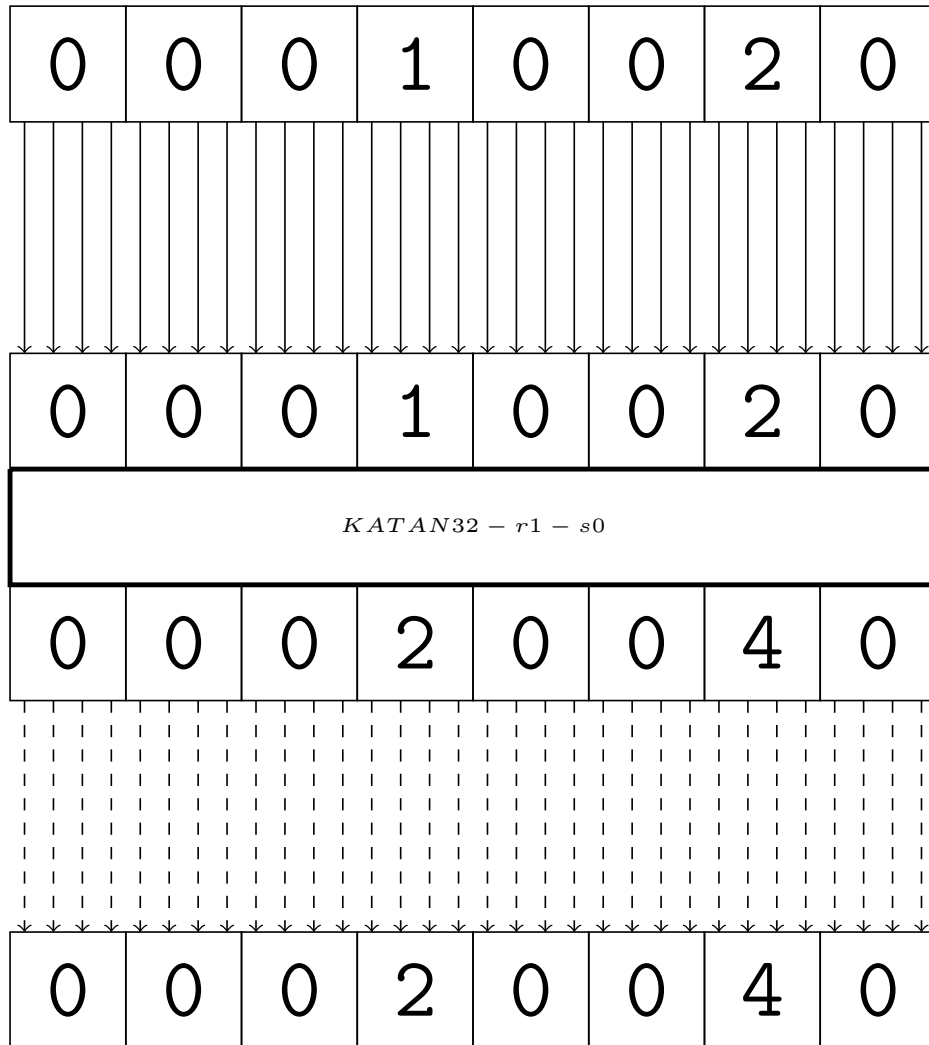
3 KATAN32-r0-s0

Maximal differential probability: $2^{-0.0}$



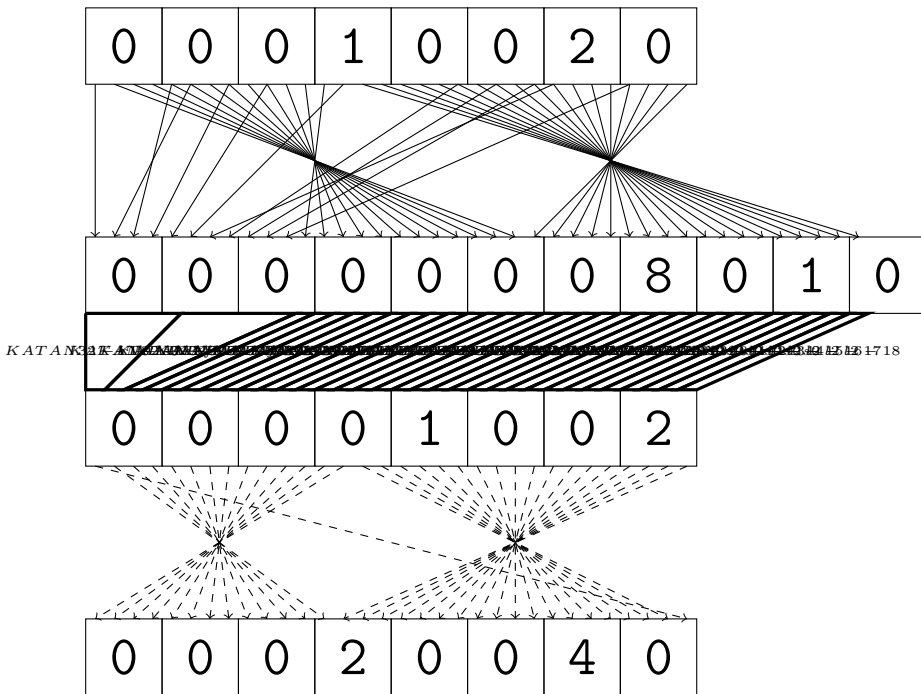
4 KATAN32-r1

Maximal differential probability: $2^{-0.0}$



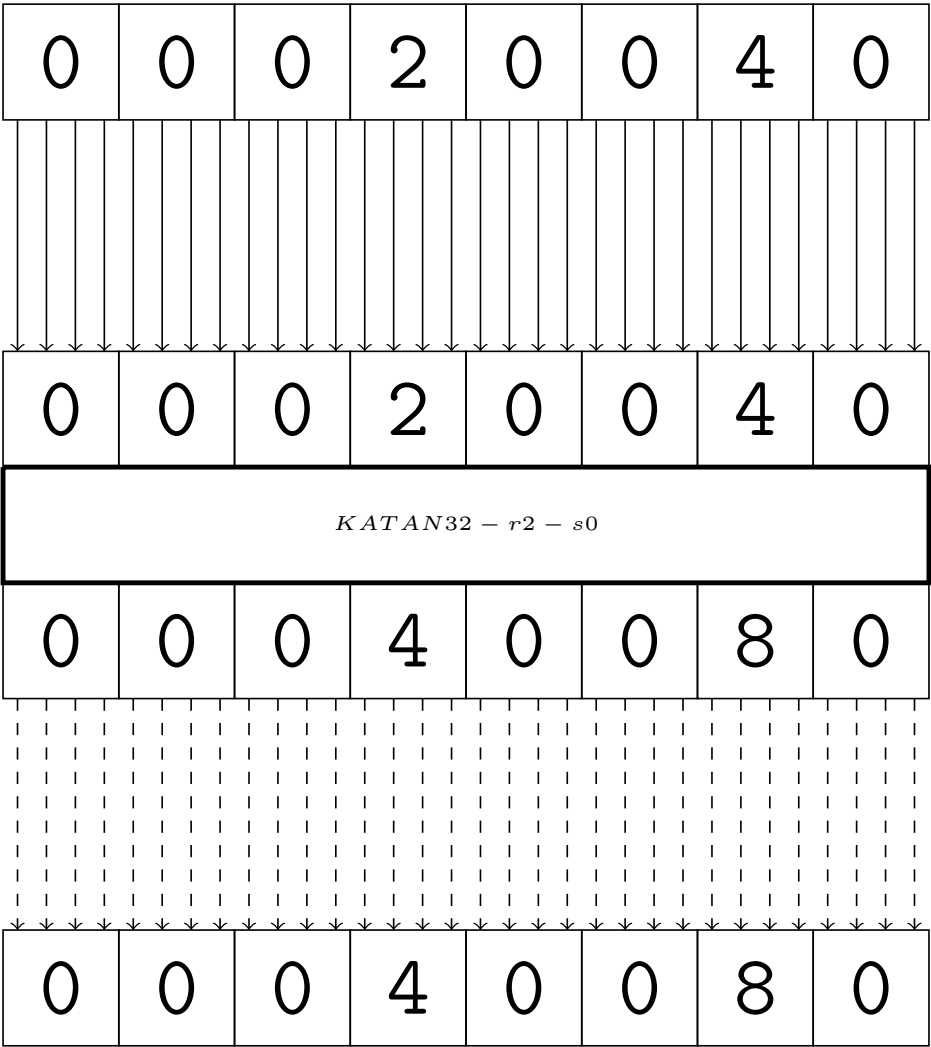
5 KATAN32-r1-s0

Maximal differential probability: $2^{-0.0}$



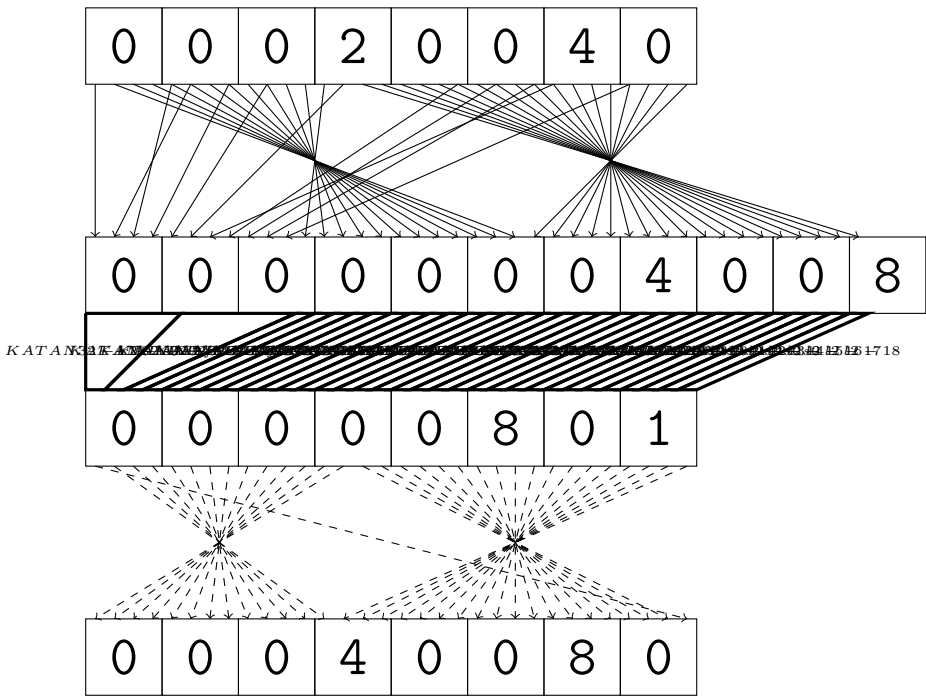
6 KATAN32-r2

Maximal differential probability: $2^{-0.0}$



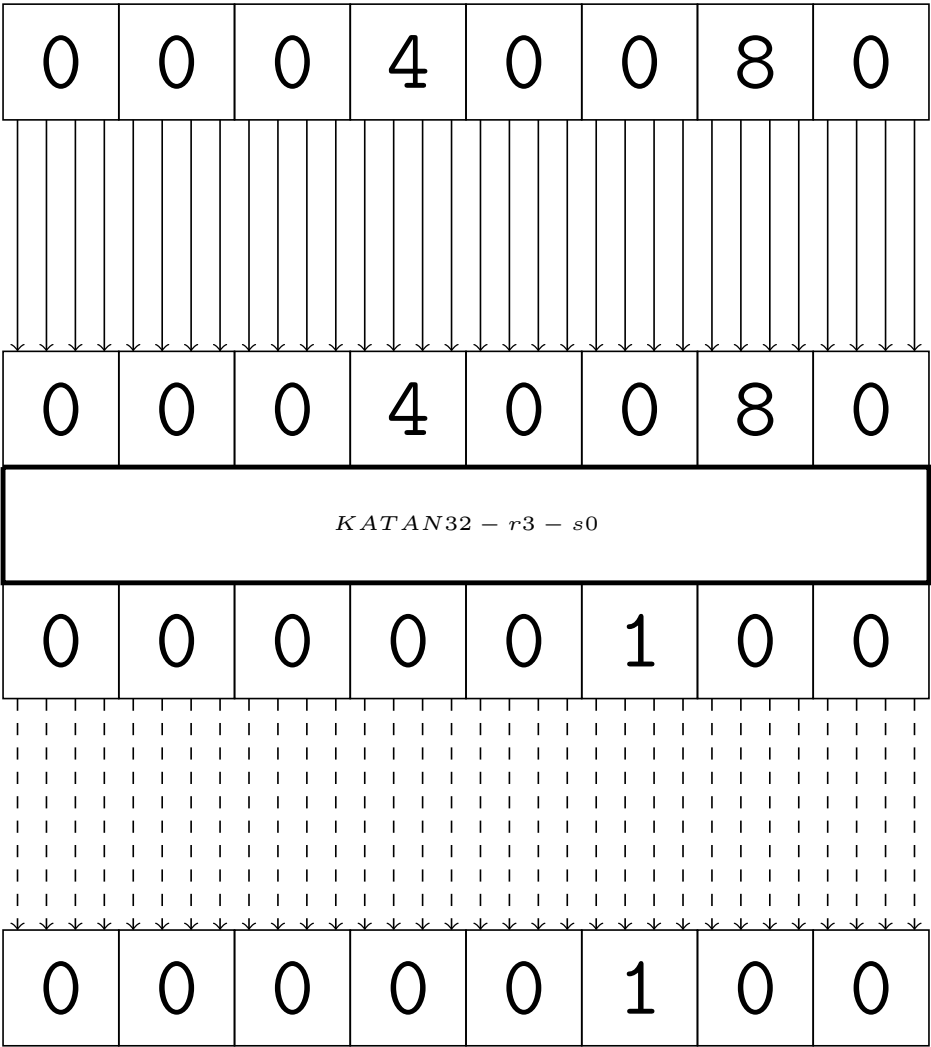
7 KATAN32-r2-s0

Maximal differential probability: $2^{-0.0}$



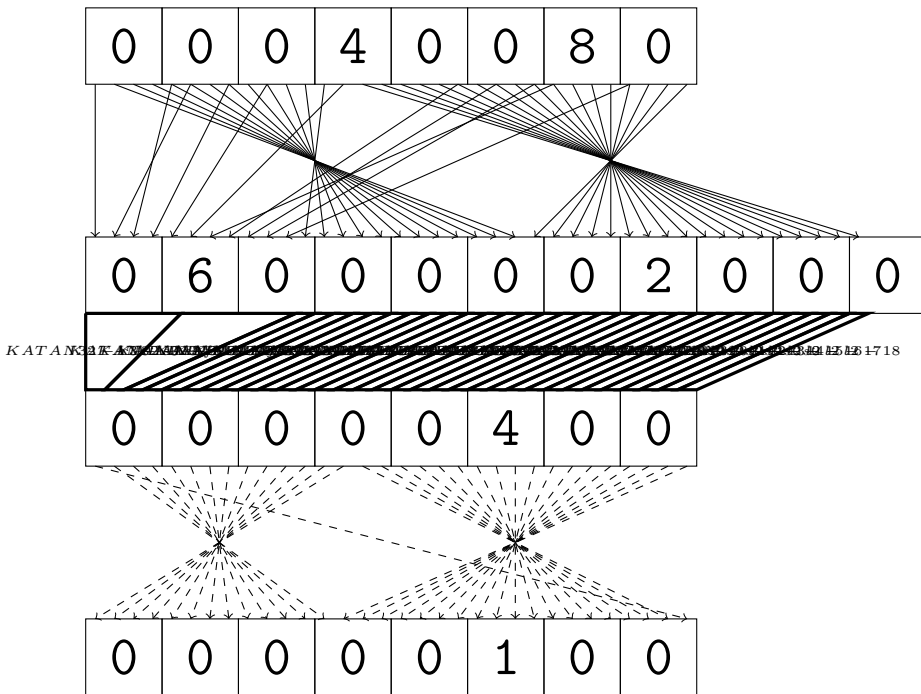
8 KATAN32-r3

Maximal differential probability: $2^{-0.0}$



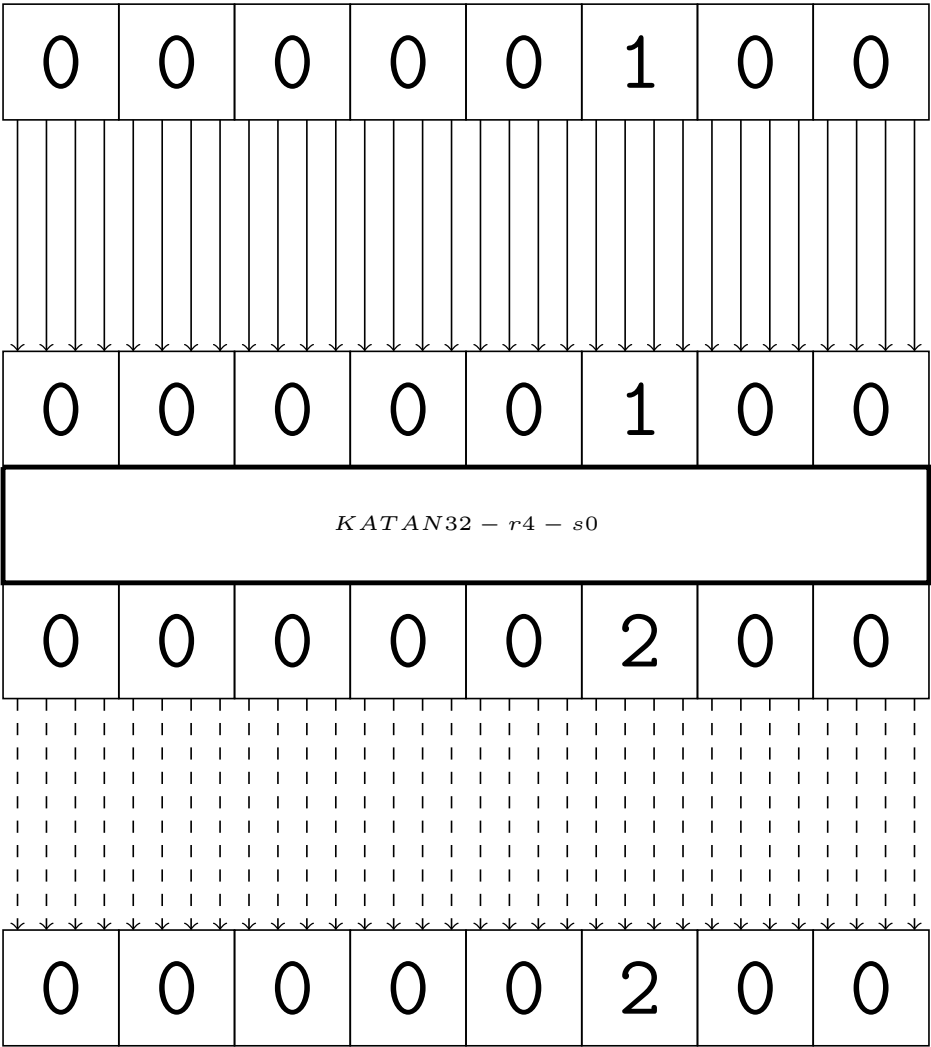
9 KATAN32-r3-s0

Maximal differential probability: $2^{-0.0}$



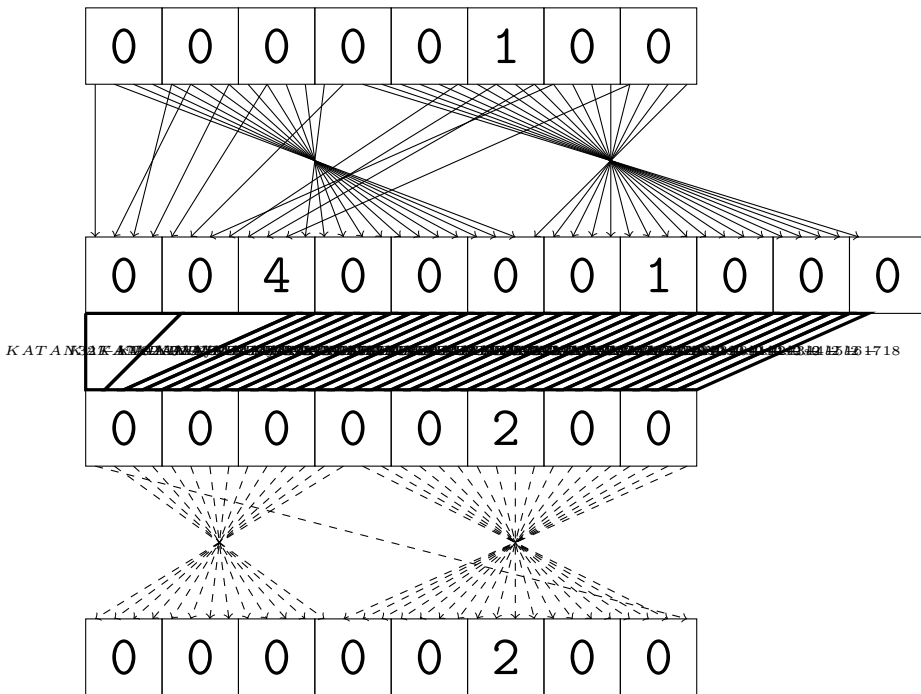
10 KATAN32-r4

Maximal differential probability: $2^{-1.0}$



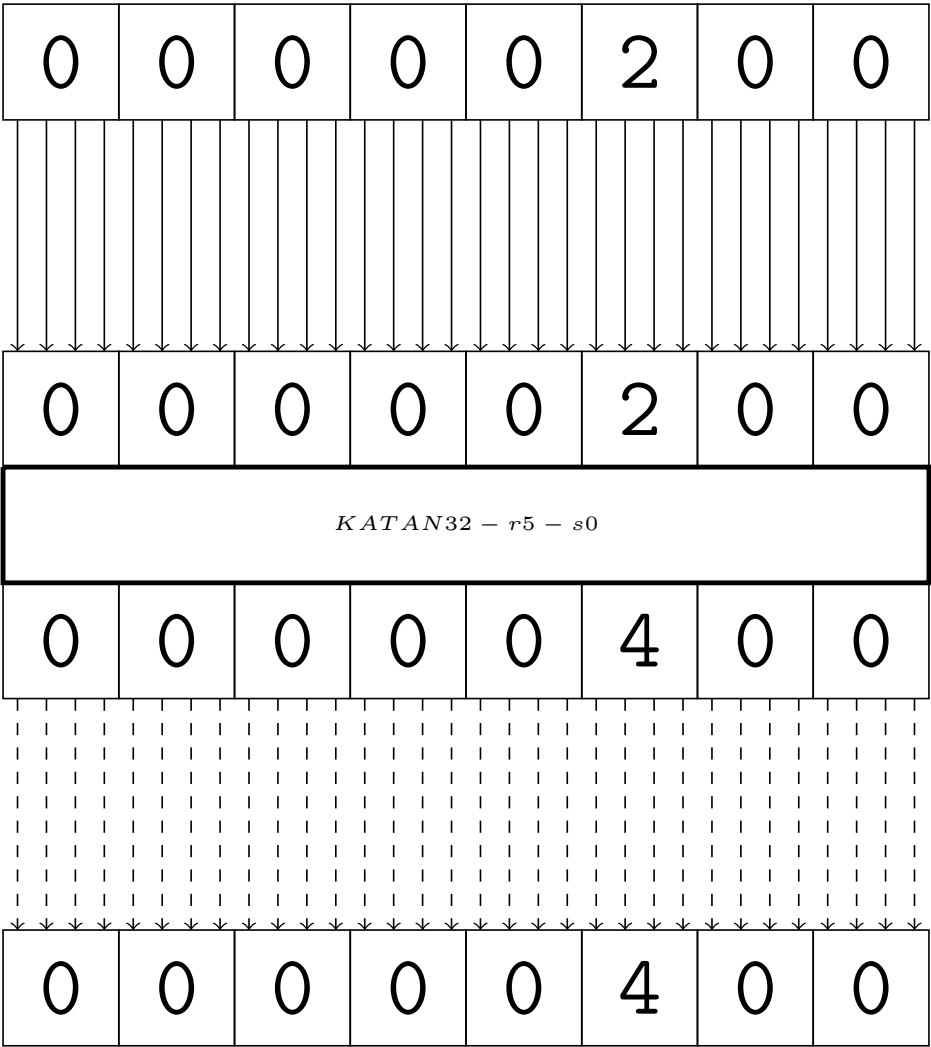
11 KATAN32-r4-s0

Maximal differential probability: $2^{-1.0}$



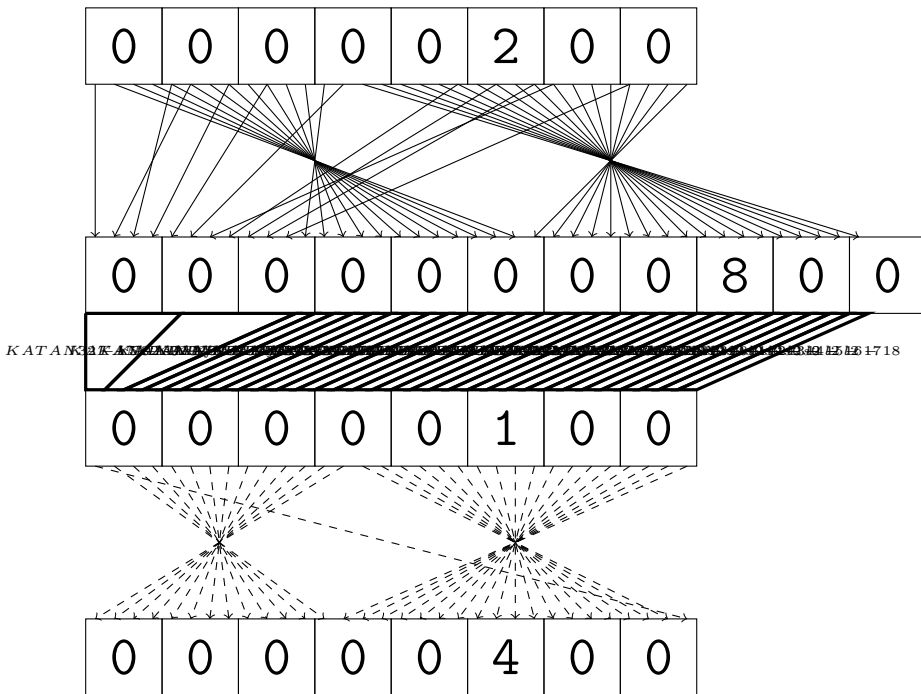
12 KATAN32-r5

Maximal differential probability: $2^{-0.0}$



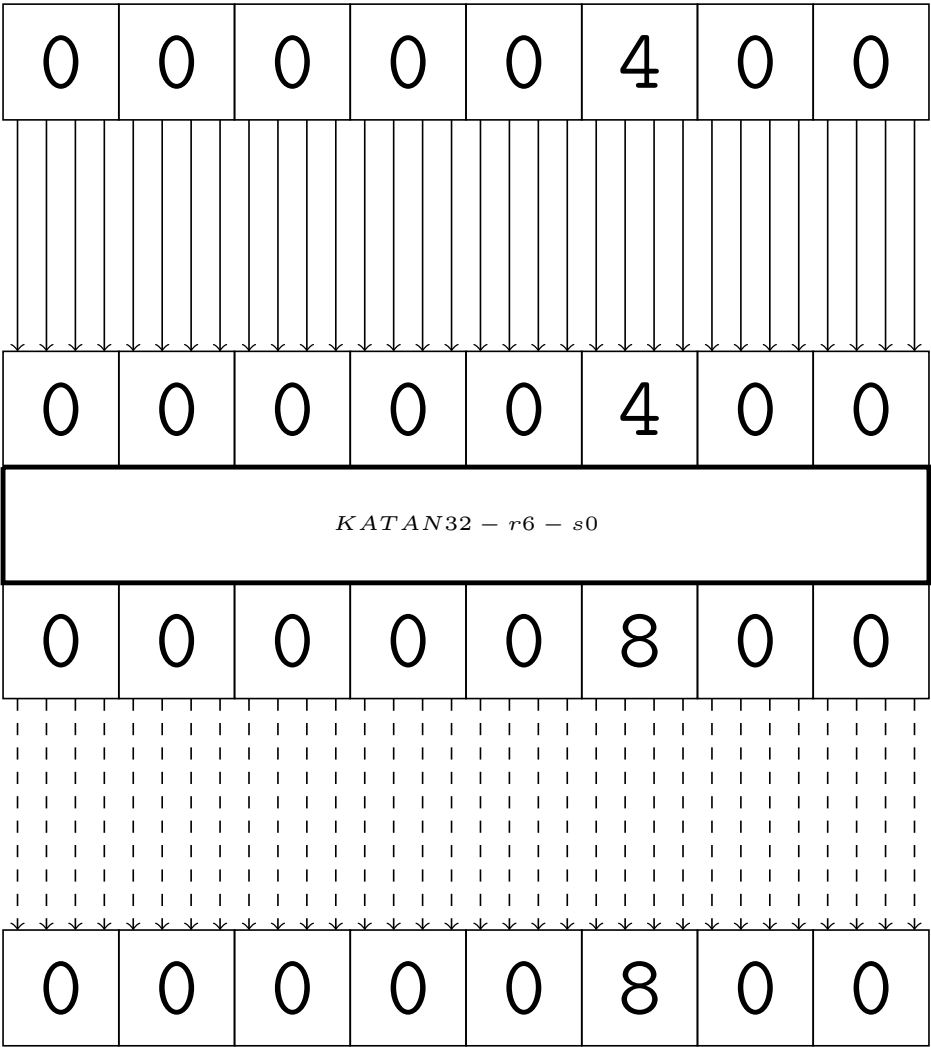
13 KATAN32-r5-s0

Maximal differential probability: $2^{-0.0}$



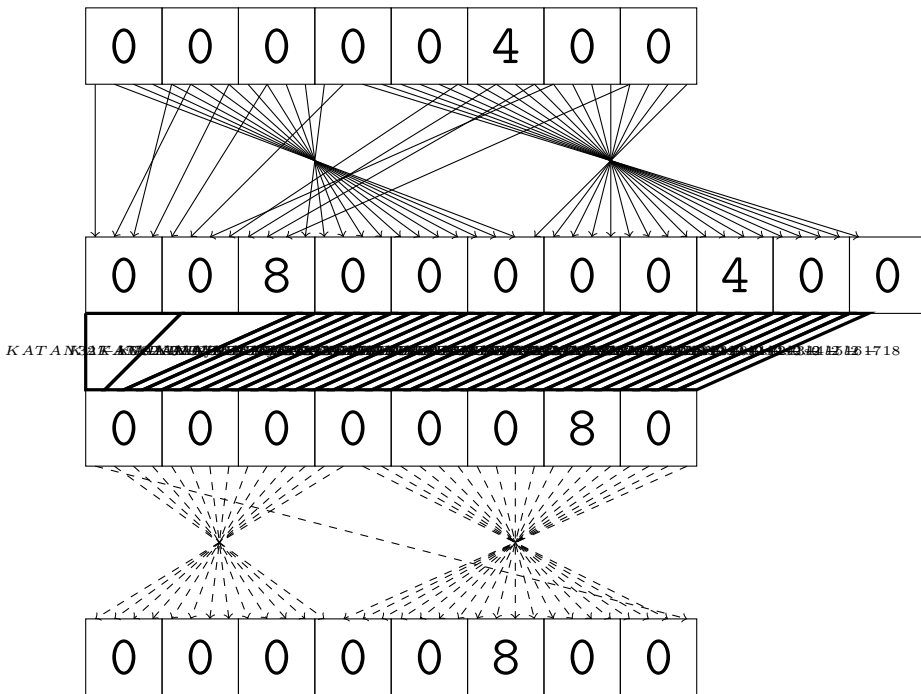
14 KATAN32-r6

Maximal differential probability: $2^{-1.0}$



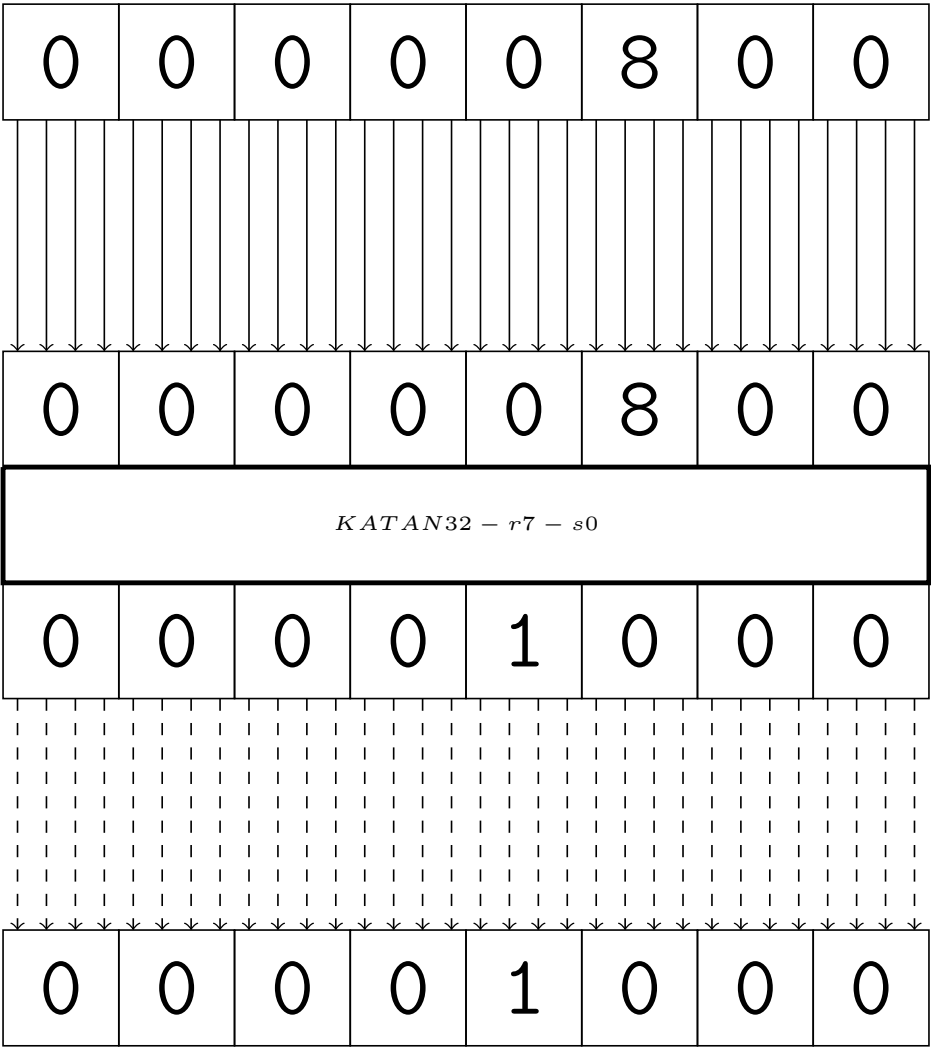
15 KATAN32-r6-s0

Maximal differential probability: $2^{-1.0}$



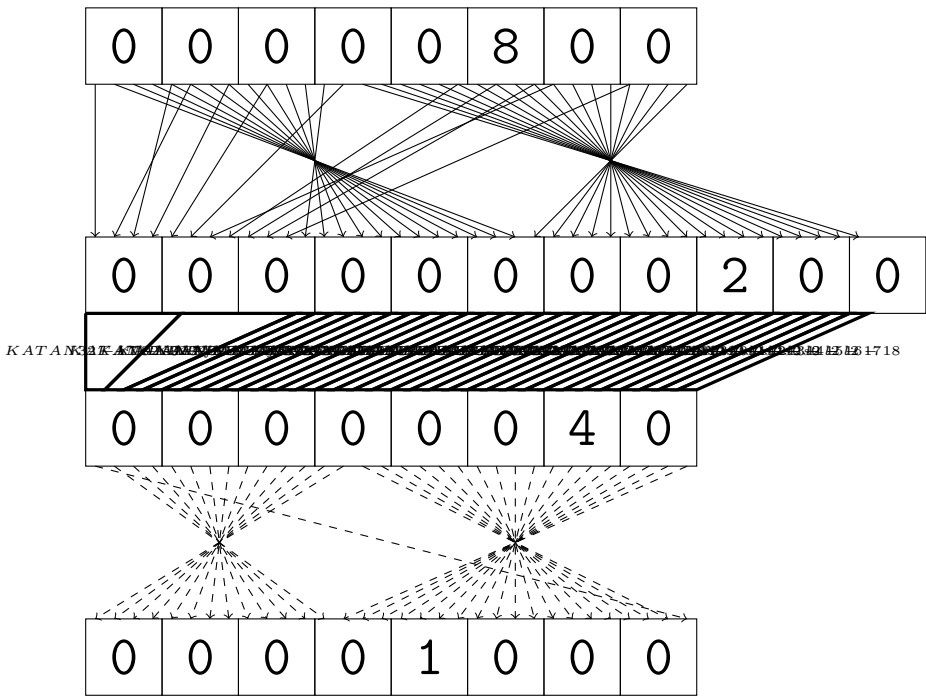
16 KATAN32-r7

Maximal differential probability: $2^{-0.0}$



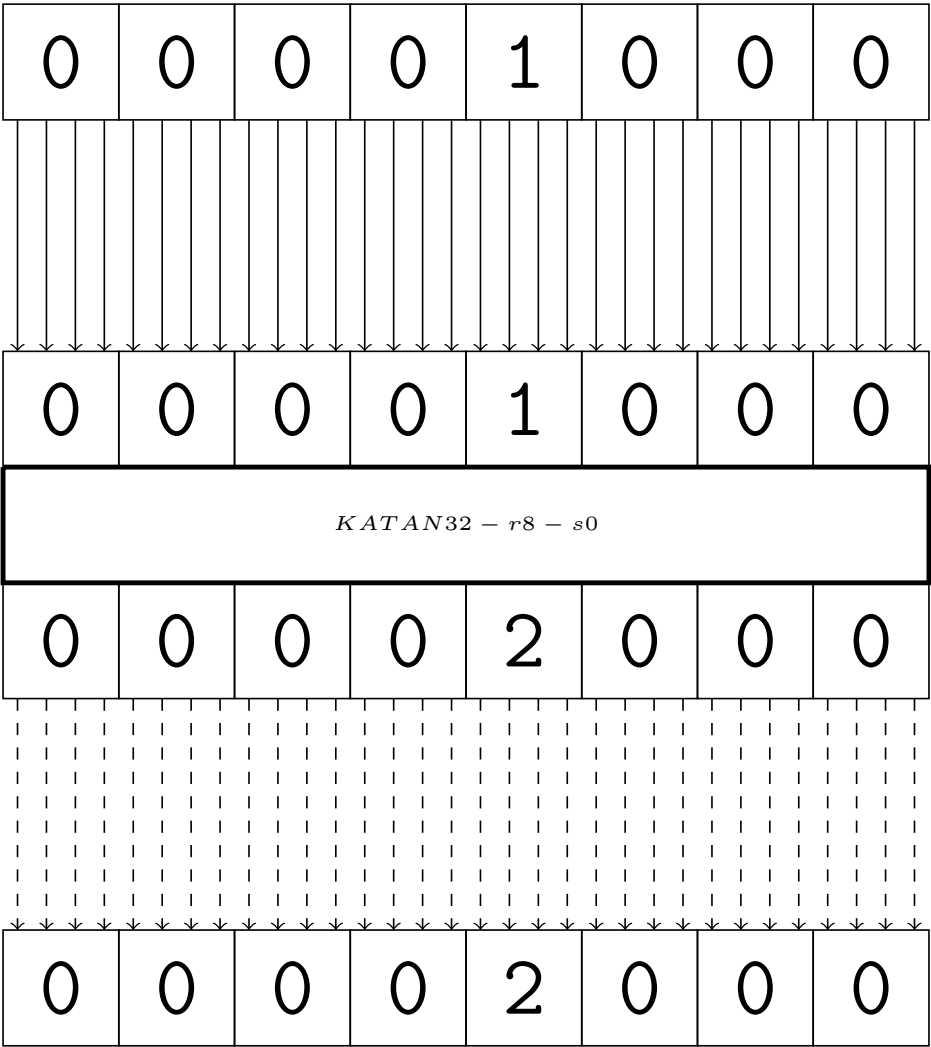
17 KATAN32-r7-s0

Maximal differential probability: $2^{-0.0}$



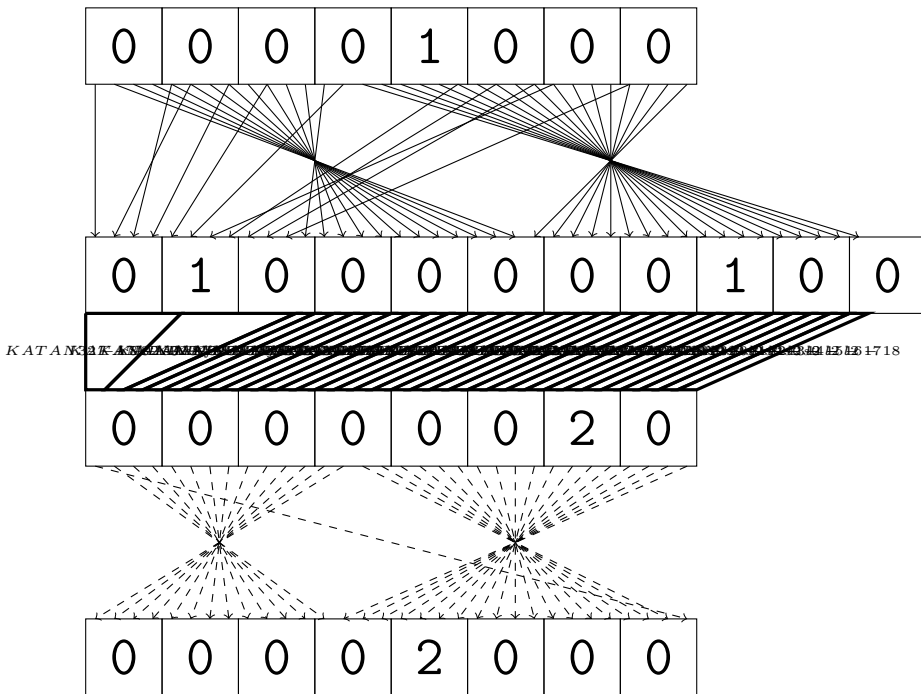
18 KATAN32-r8

Maximal differential probability: $2^{-1.0}$



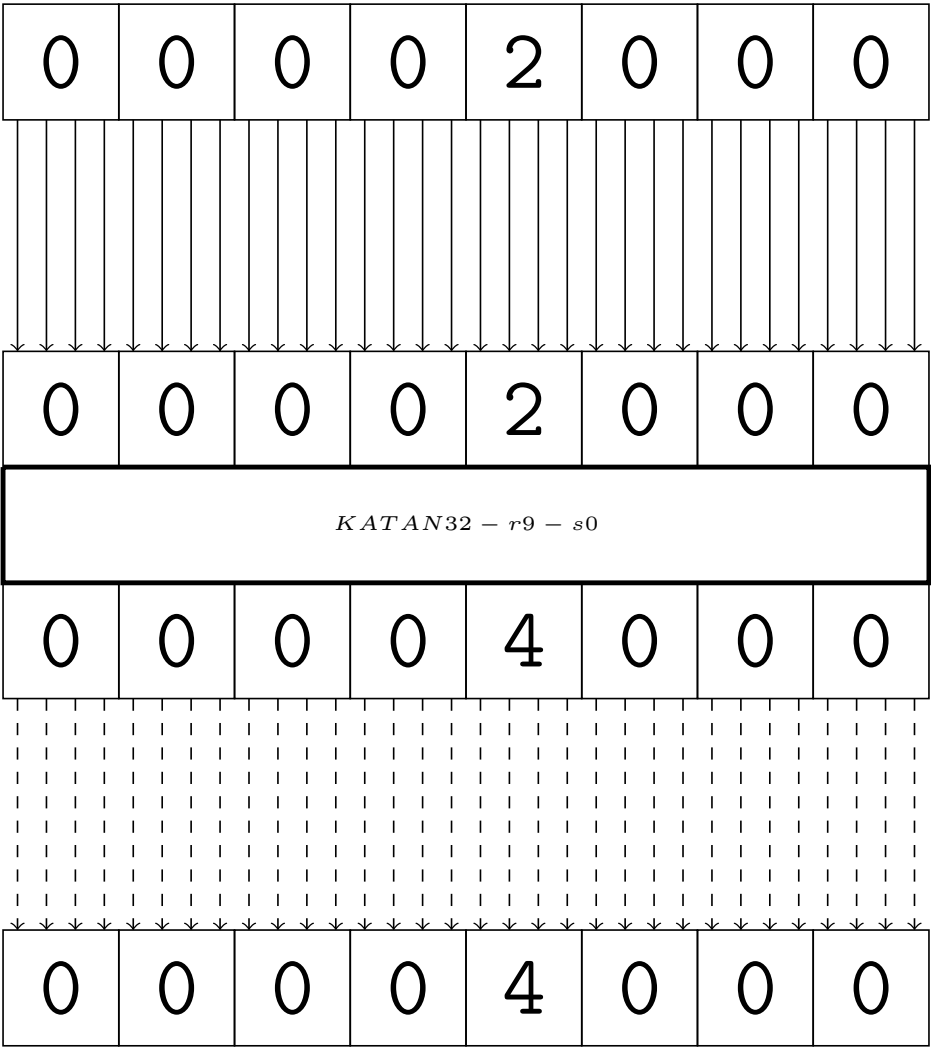
19 KATAN32-r8-s0

Maximal differential probability: $2^{-1.0}$



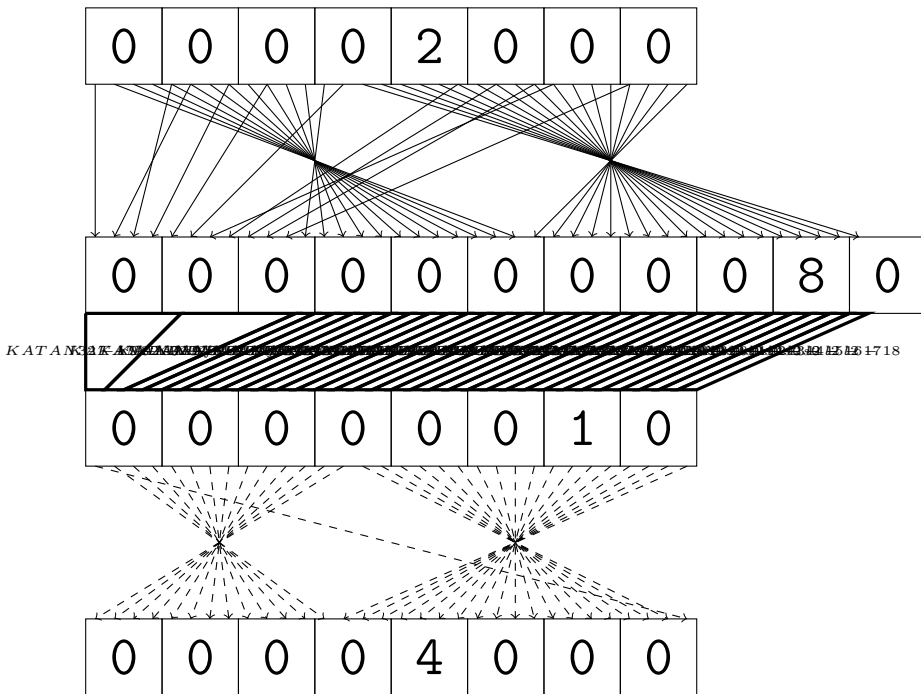
20 KATAN32-r9

Maximal differential probability: $2^{-0.0}$



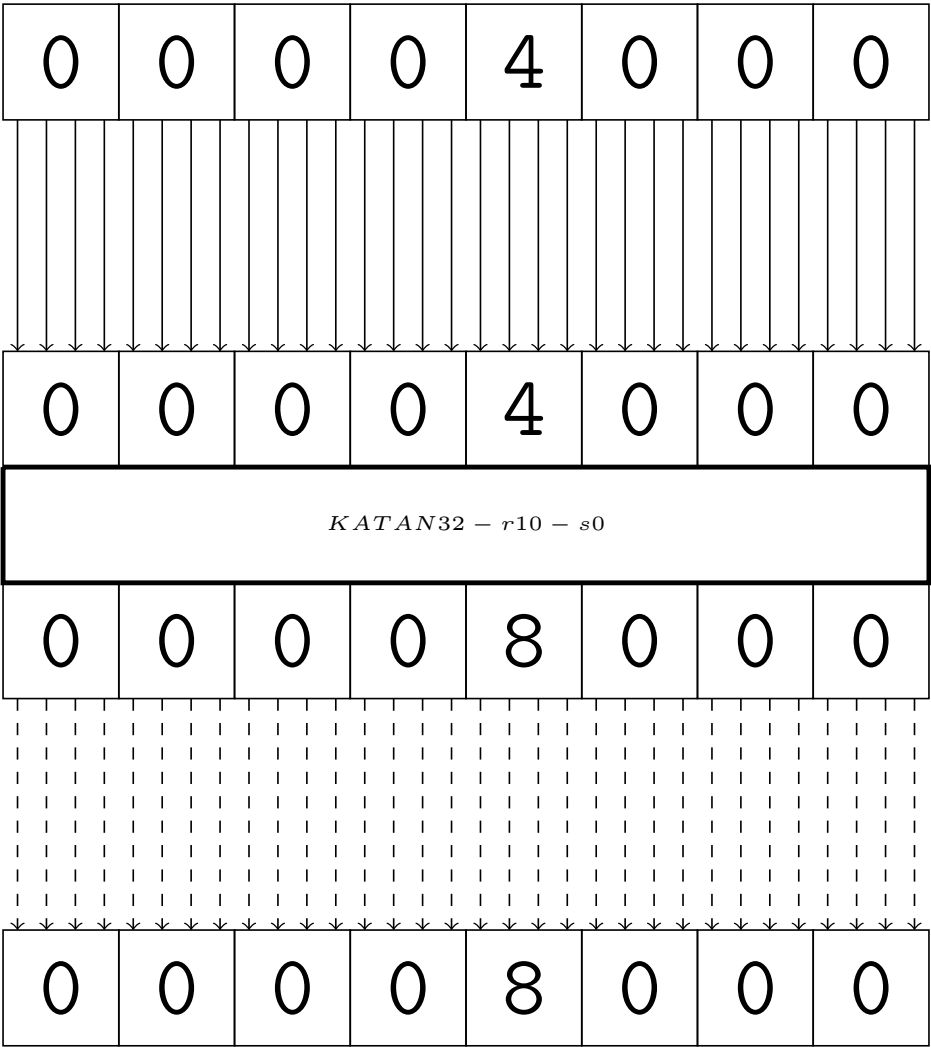
21 KATAN32-r9-s0

Maximal differential probability: $2^{-0.0}$



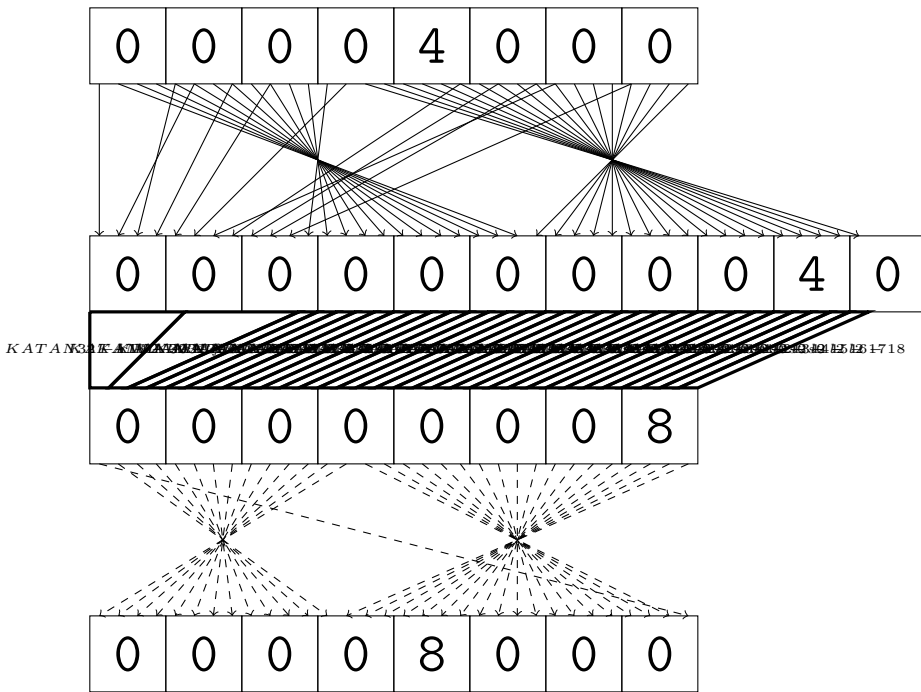
22 KATAN32-r10

Maximal differential probability: $2^{-0.0}$



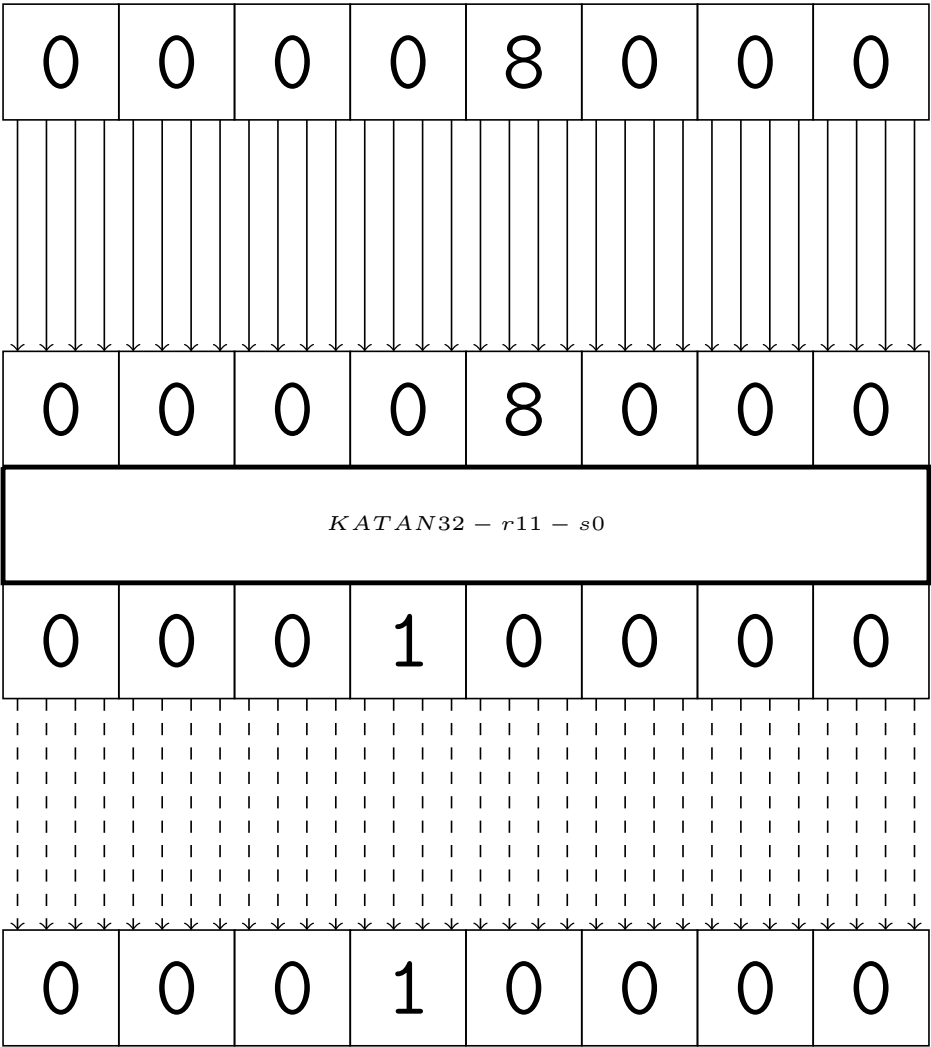
23 KATAN32-r10-s0

Maximal differential probability: $2^{-0.0}$



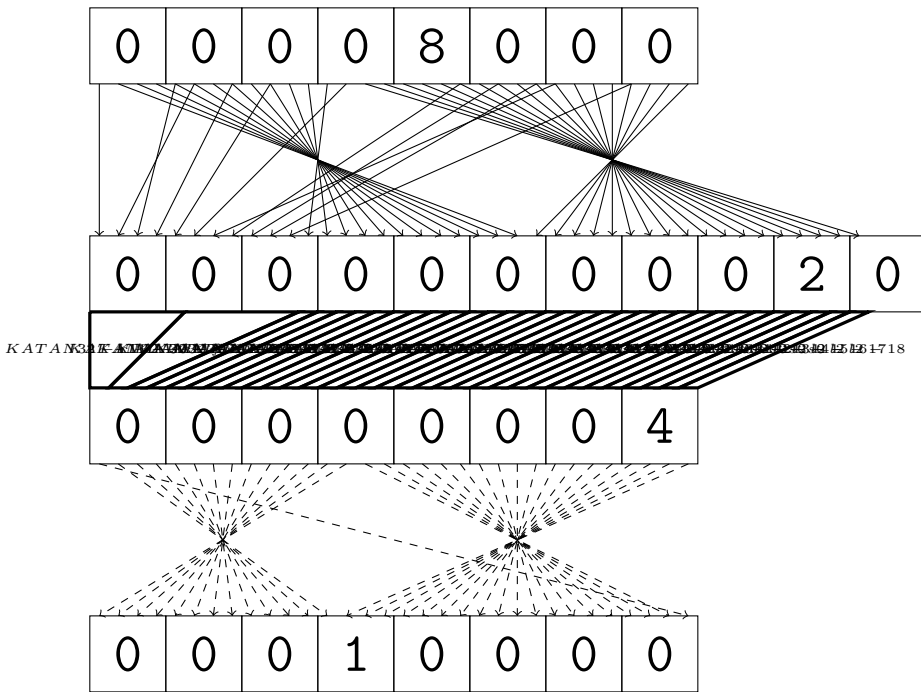
24 KATAN32-r11

Maximal differential probability: $2^{-0.0}$



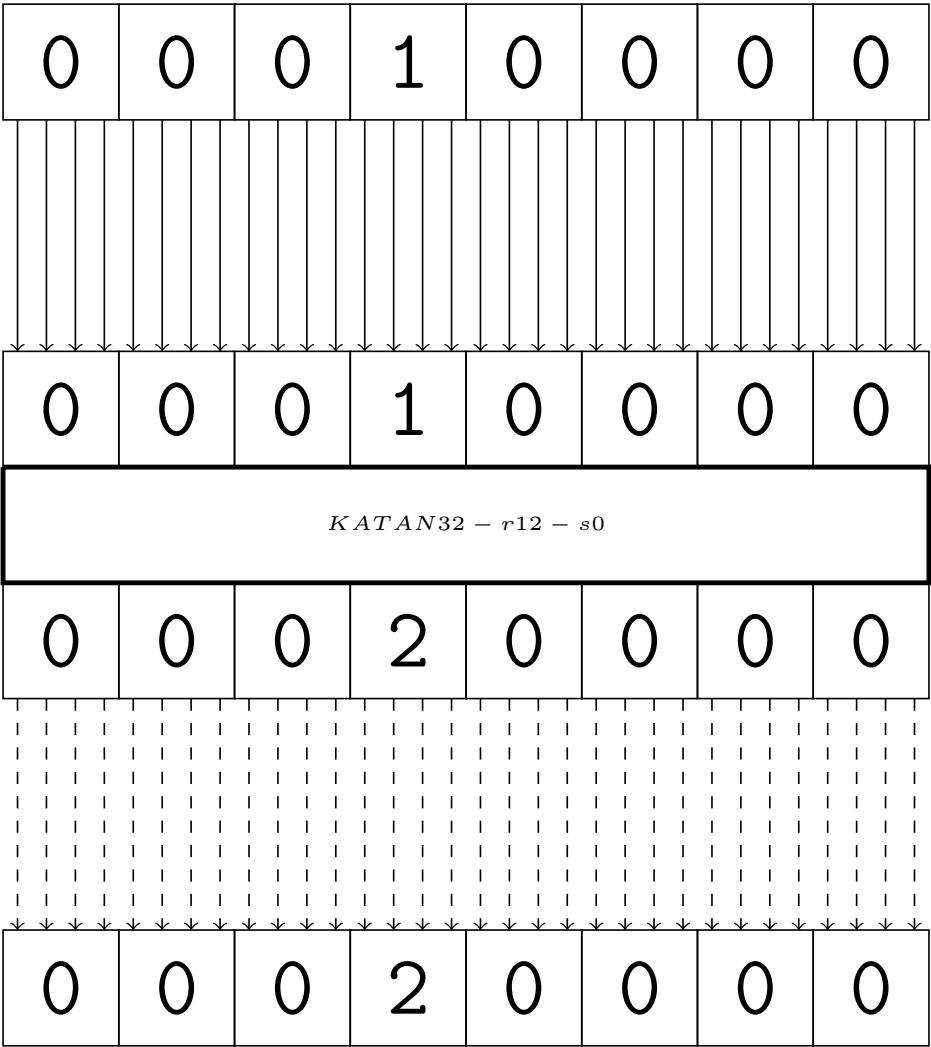
25 KATAN32-r11-s0

Maximal differential probability: $2^{-0.0}$



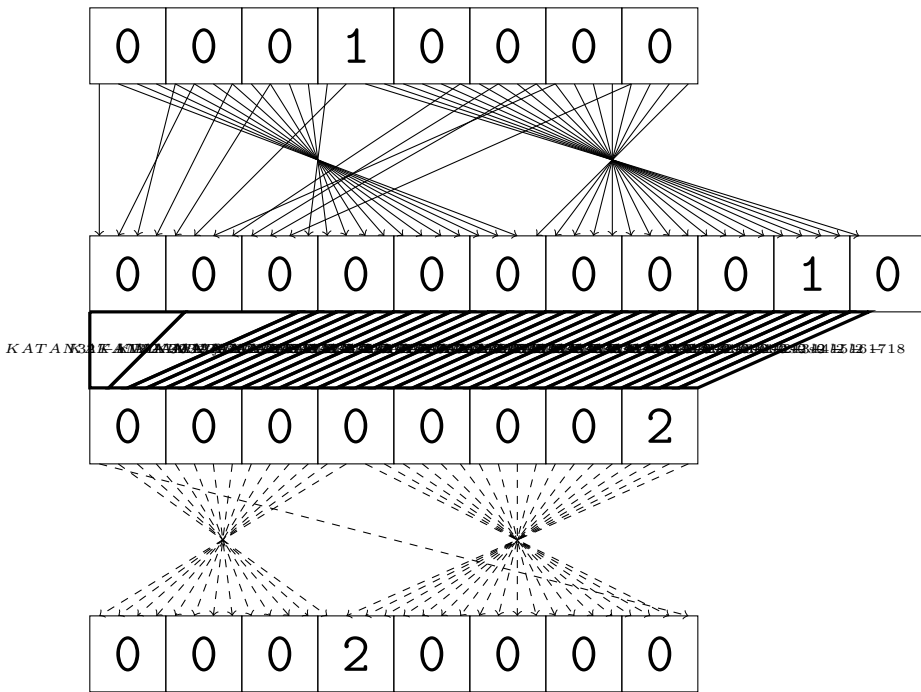
26 KATAN32-r12

Maximal differential probability: $2^{-0.0}$



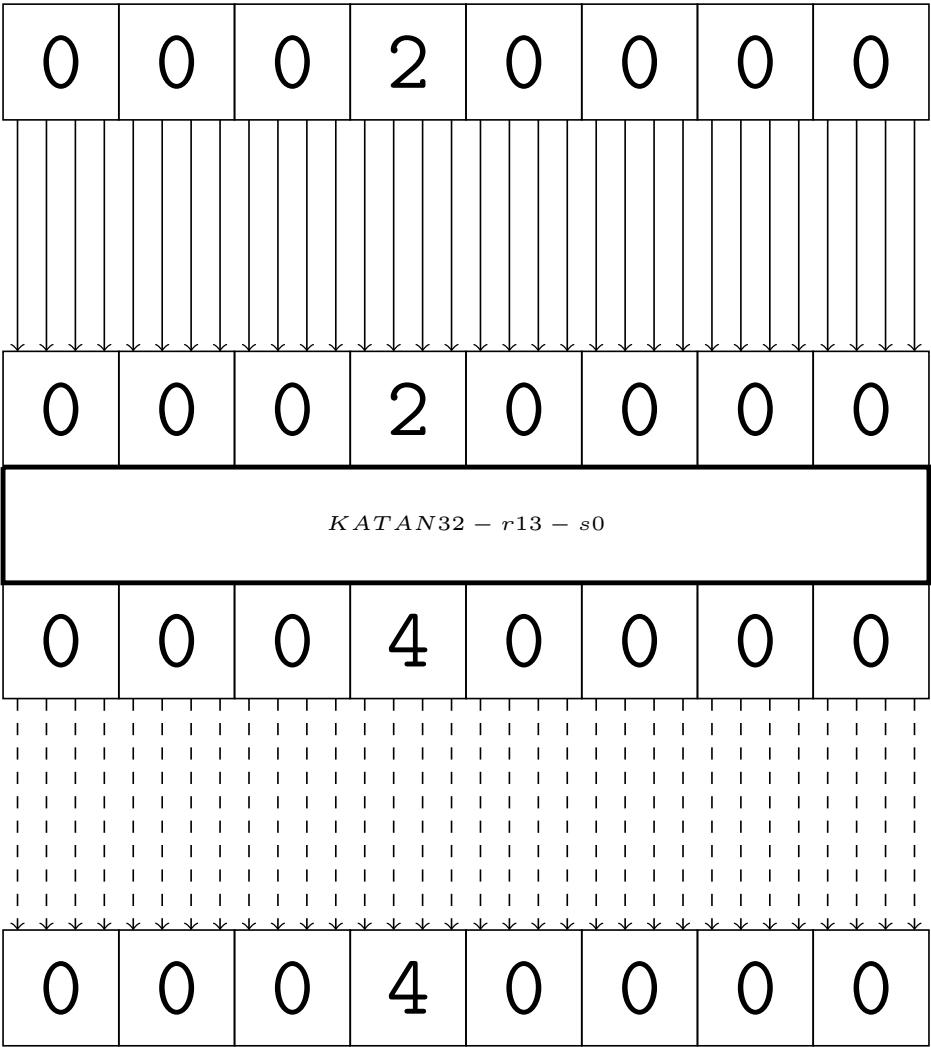
27 KATAN32-r12-s0

Maximal differential probability: $2^{-0.0}$



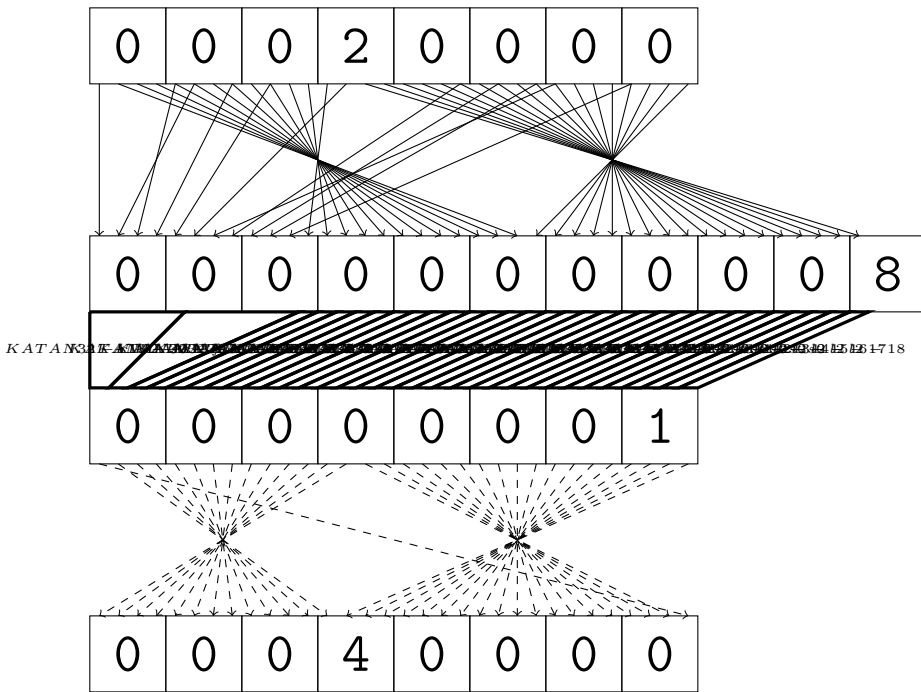
28 KATAN32-r13

Maximal differential probability: $2^{-0.0}$



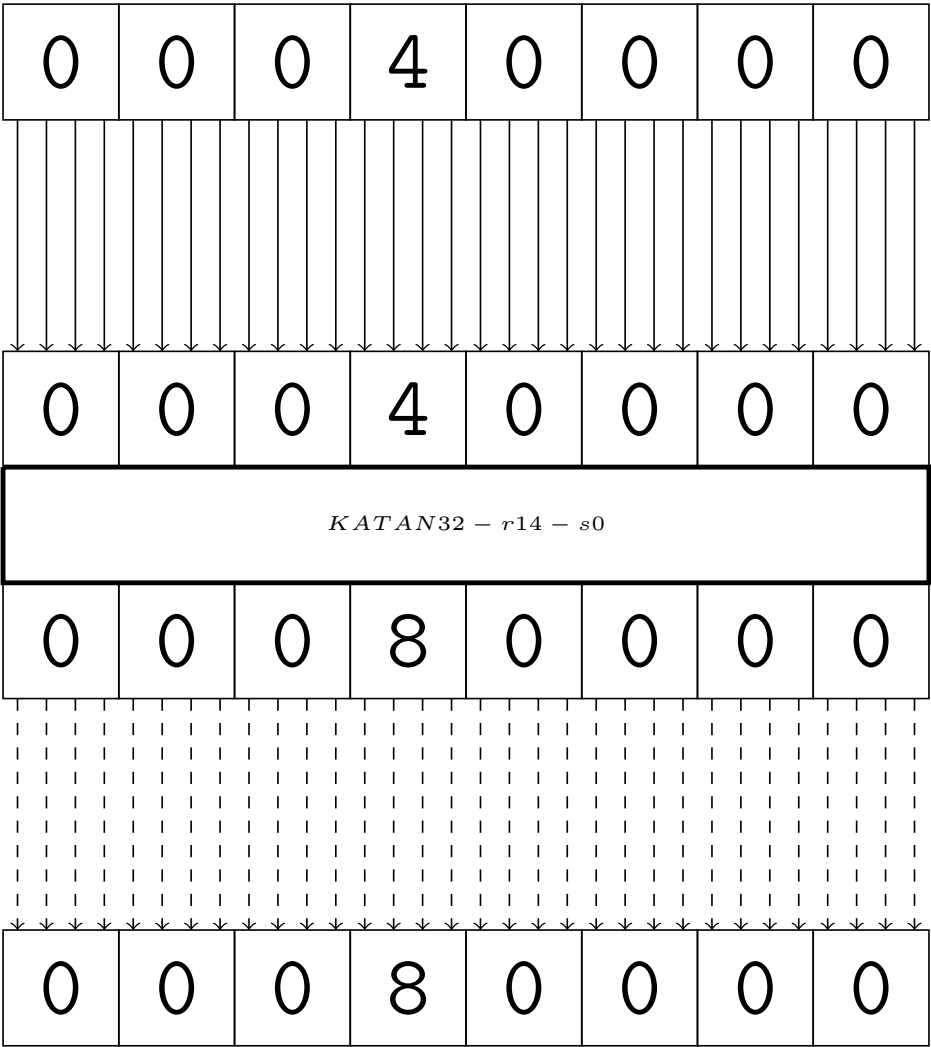
29 KATAN32-r13-s0

Maximal differential probability: $2^{-0.0}$



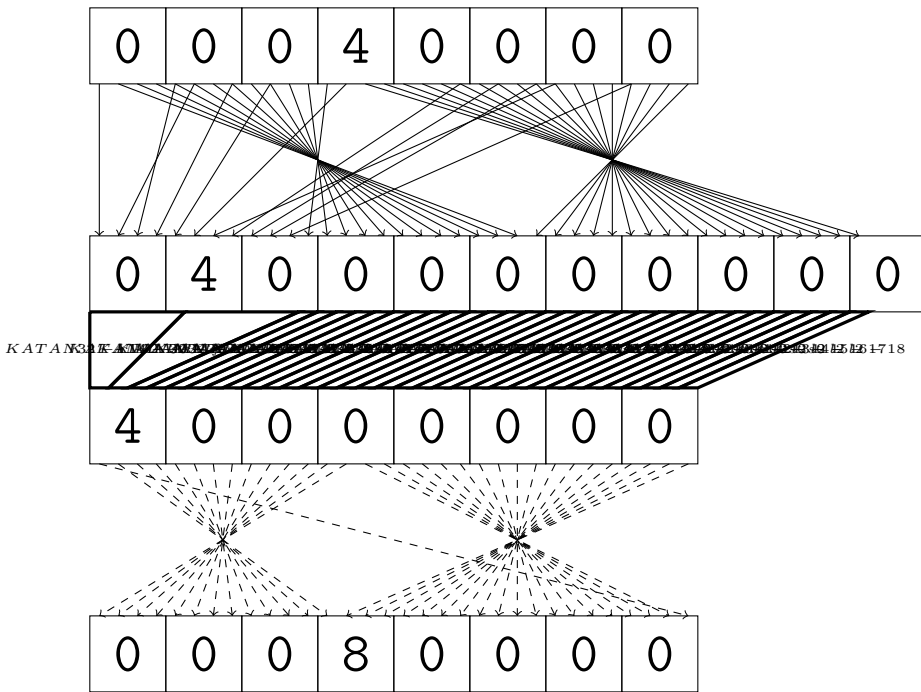
30 KATAN32-r14

Maximal differential probability: $2^{-0.0}$



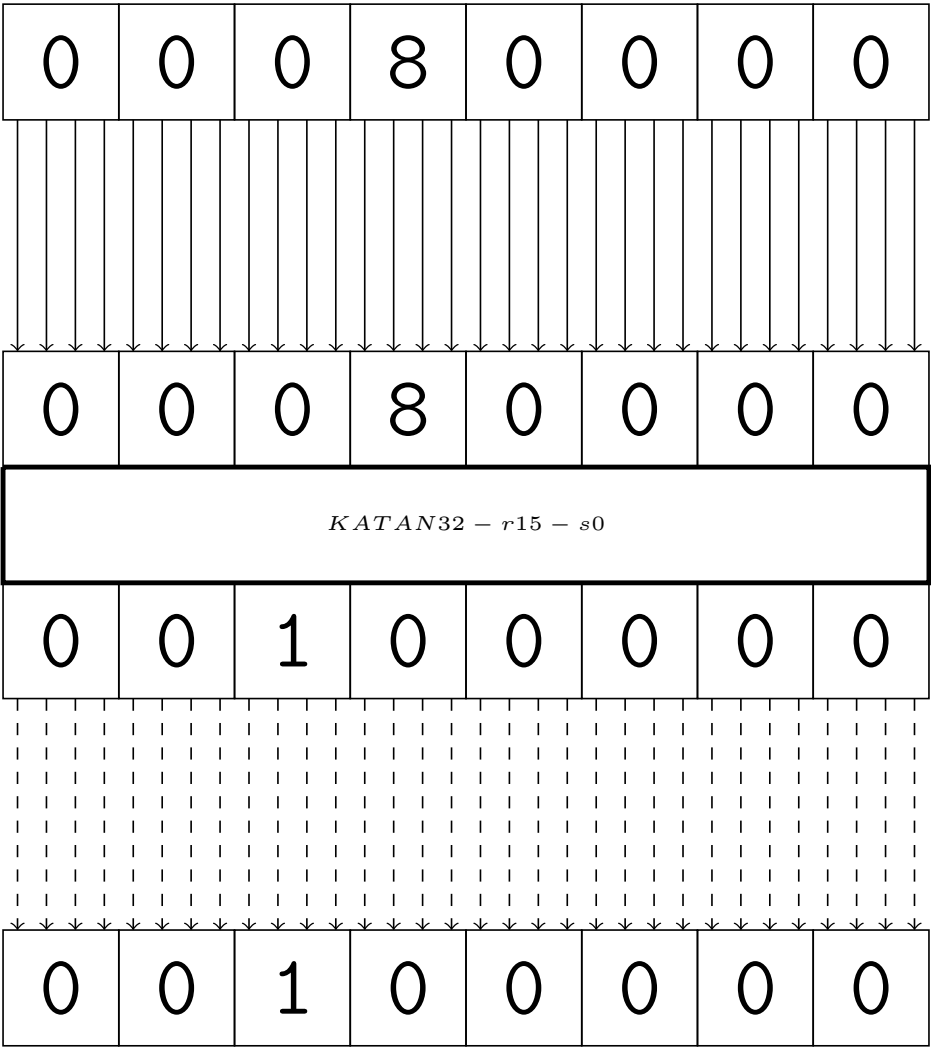
31 KATAN32-r14-s0

Maximal differential probability: $2^{-0.0}$



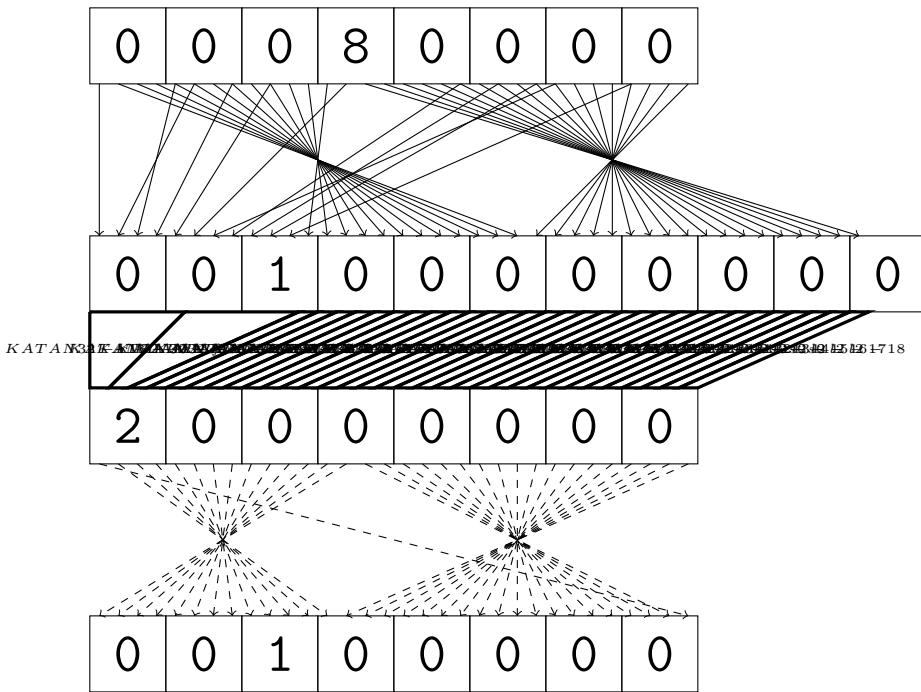
32 KATAN32-r15

Maximal differential probability: $2^{-0.0}$



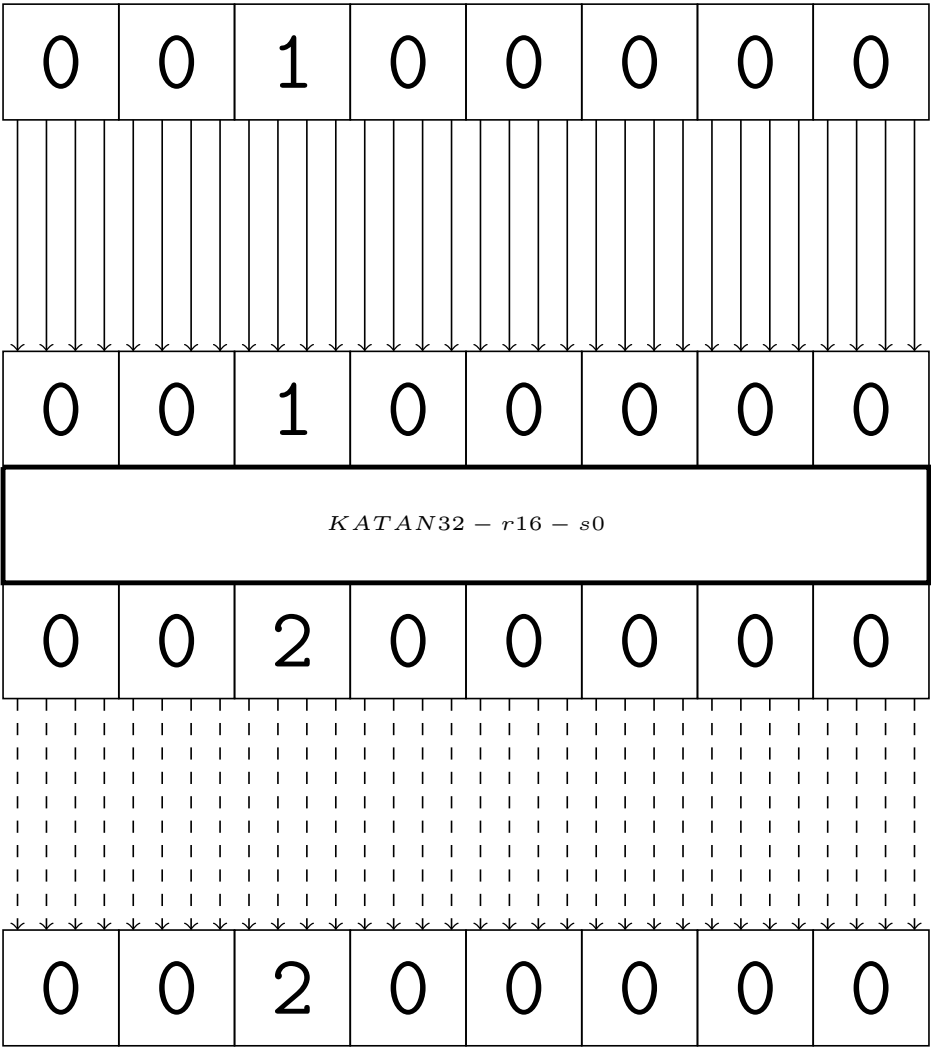
33 KATAN32-r15-s0

Maximal differential probability: $2^{-0.0}$



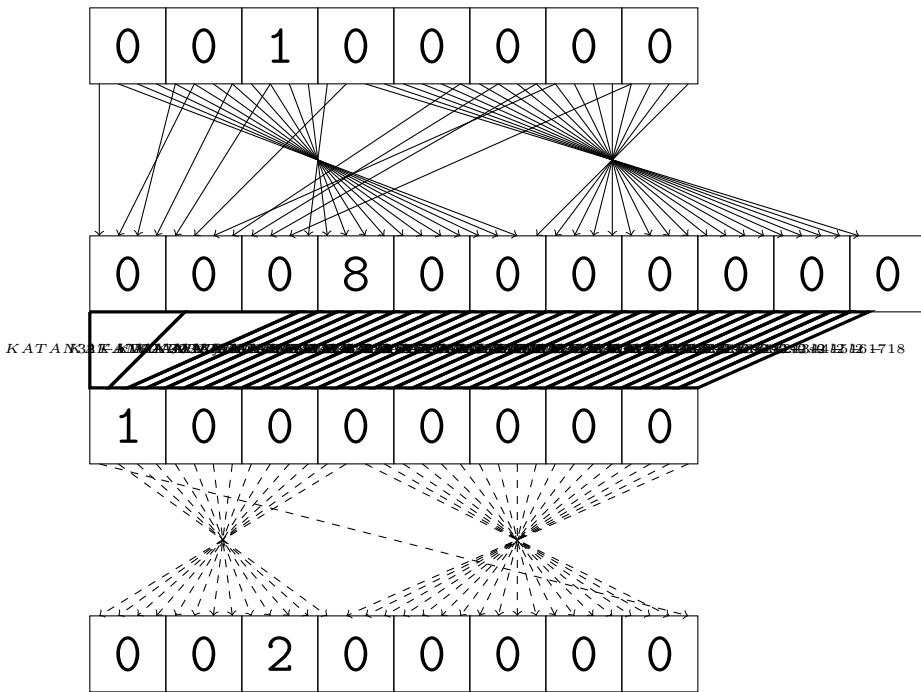
34 KATAN32-r16

Maximal differential probability: $2^{-0.0}$



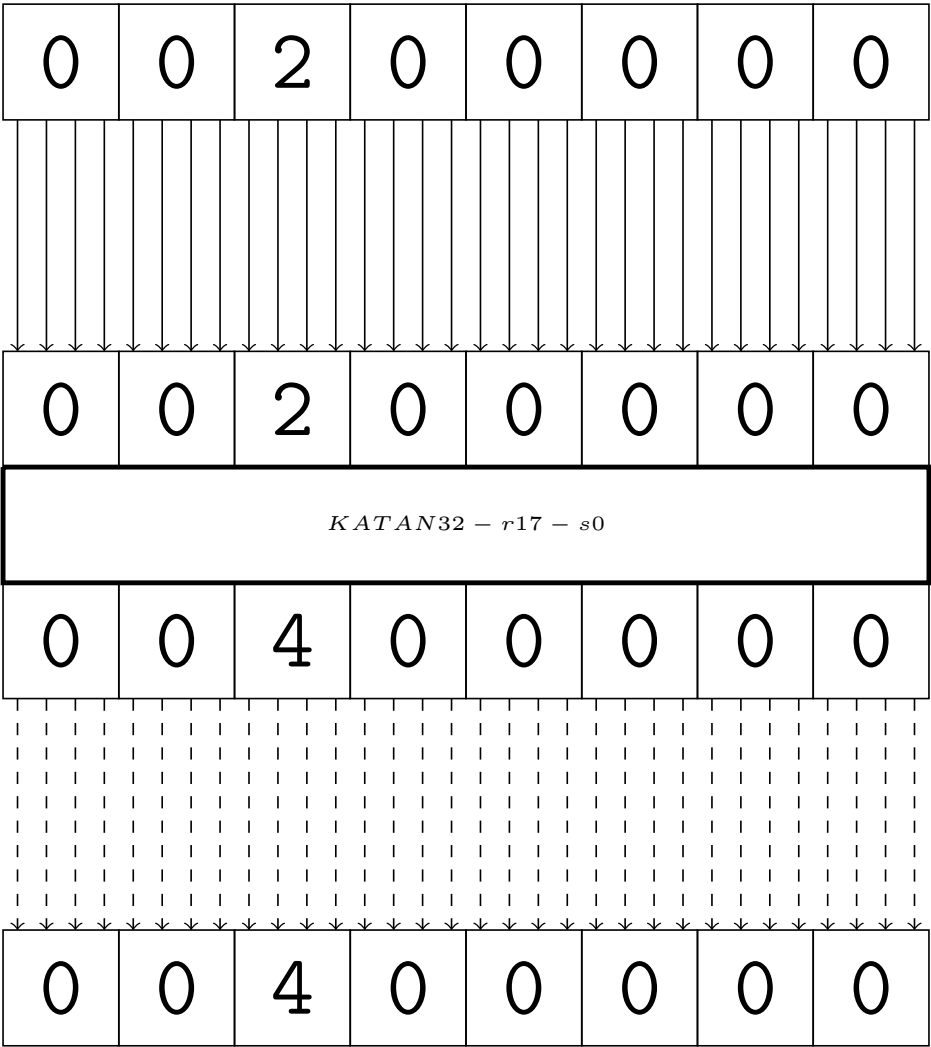
35 KATAN32-r16-s0

Maximal differential probability: $2^{-0.0}$



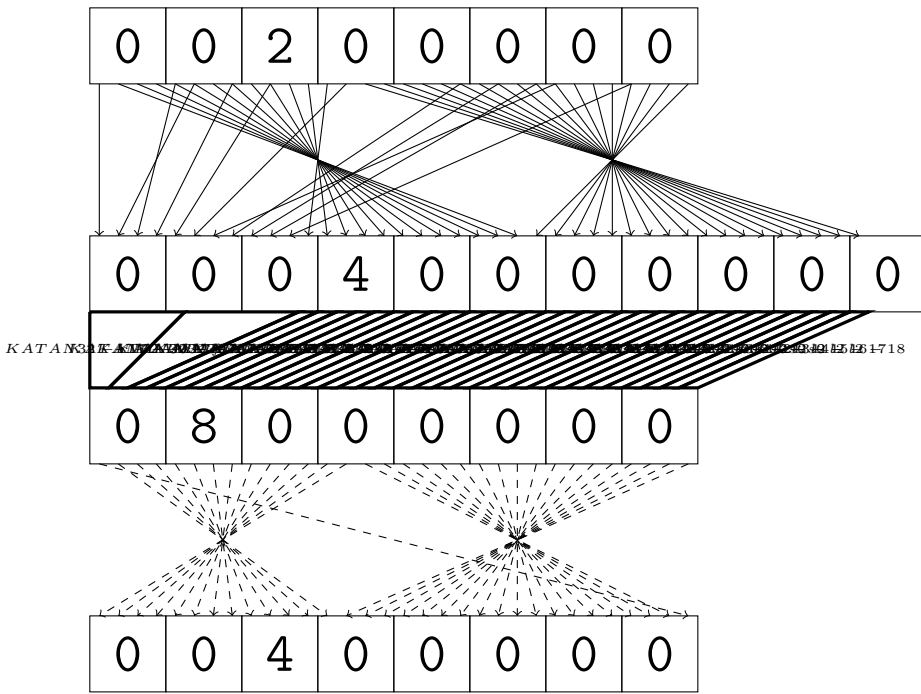
36 KATAN32-r17

Maximal differential probability: $2^{-0.0}$



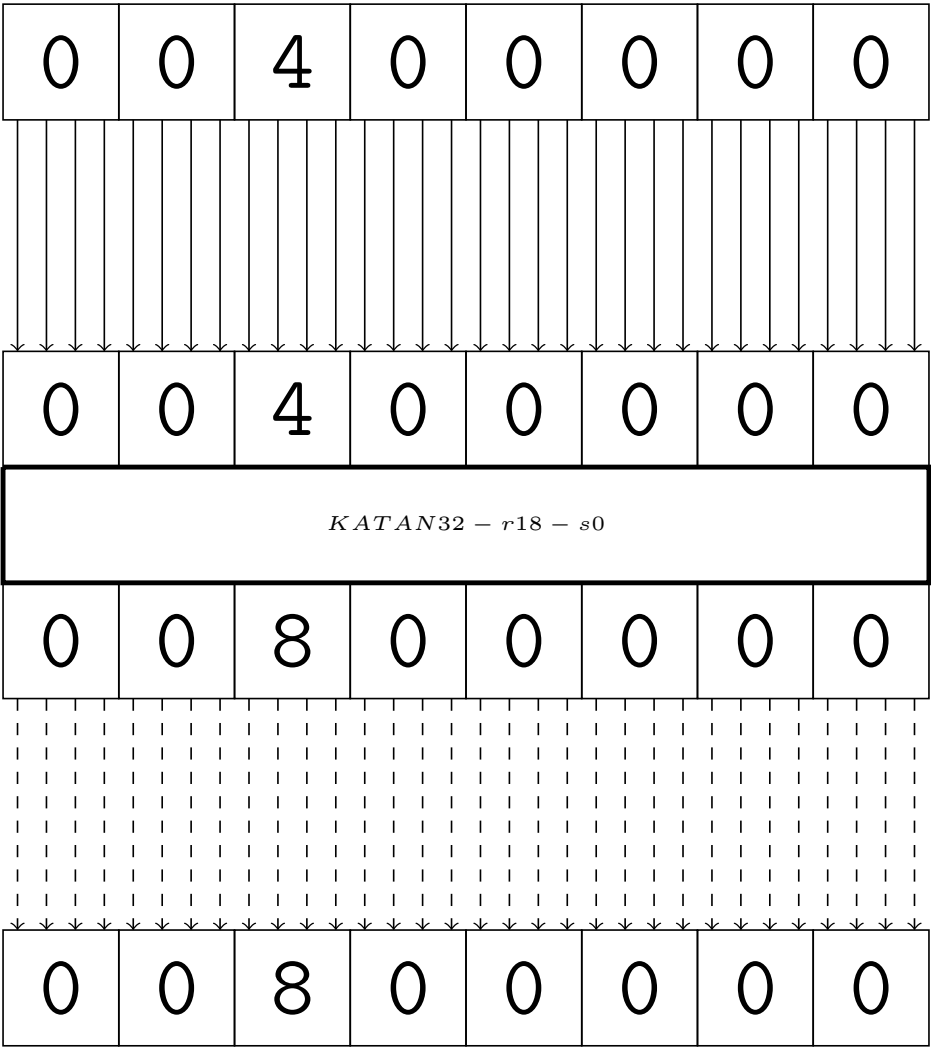
37 KATAN32-r17-s0

Maximal differential probability: $2^{-0.0}$



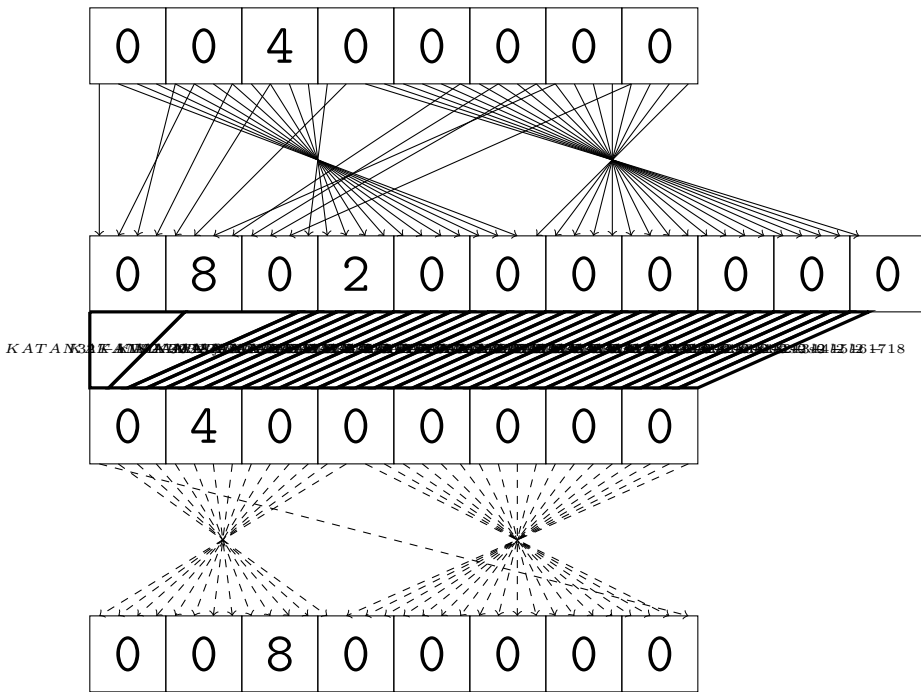
38 KATAN32-r18

Maximal differential probability: $2^{-0.0}$



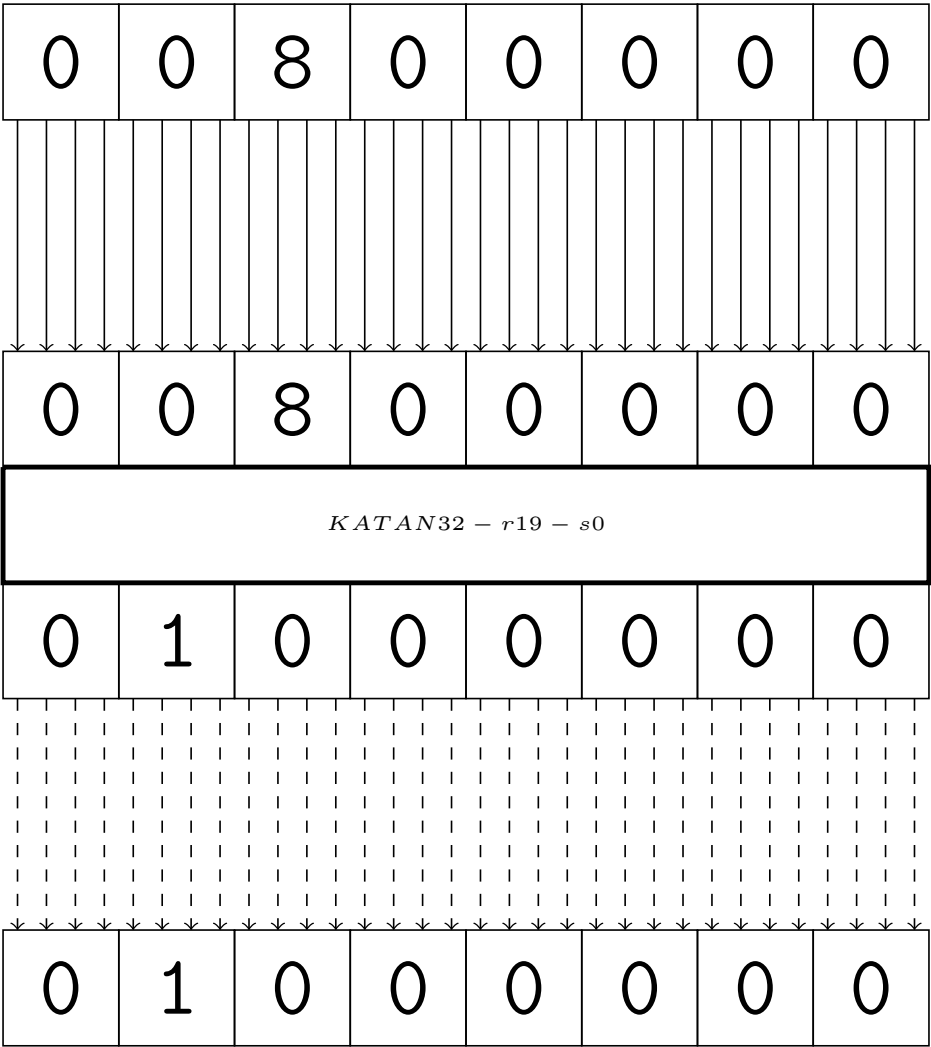
39 KATAN32-r18-s0

Maximal differential probability: $2^{-0.0}$



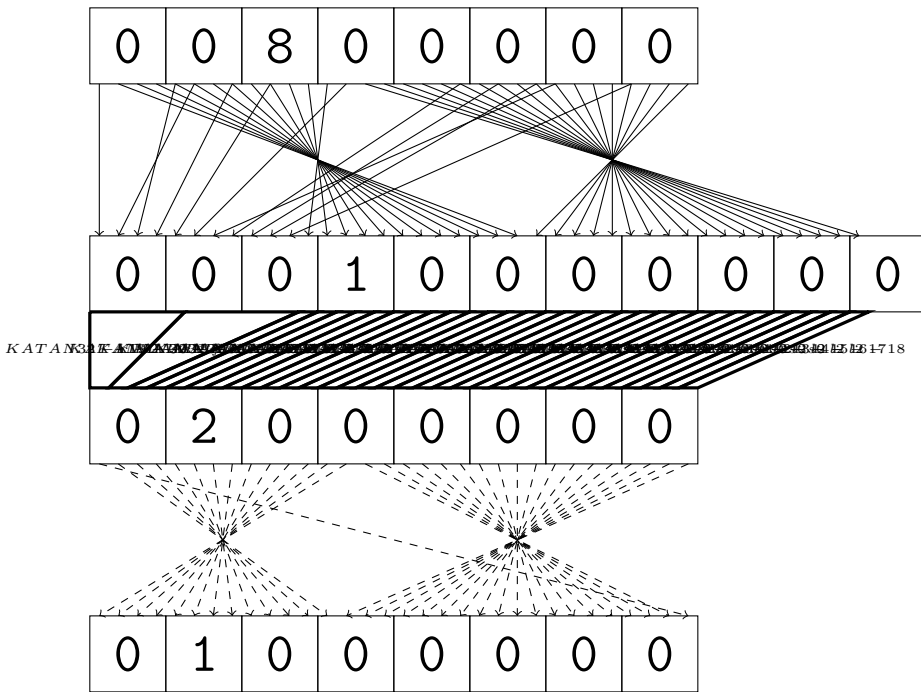
40 KATAN32-r19

Maximal differential probability: $2^{-0.0}$



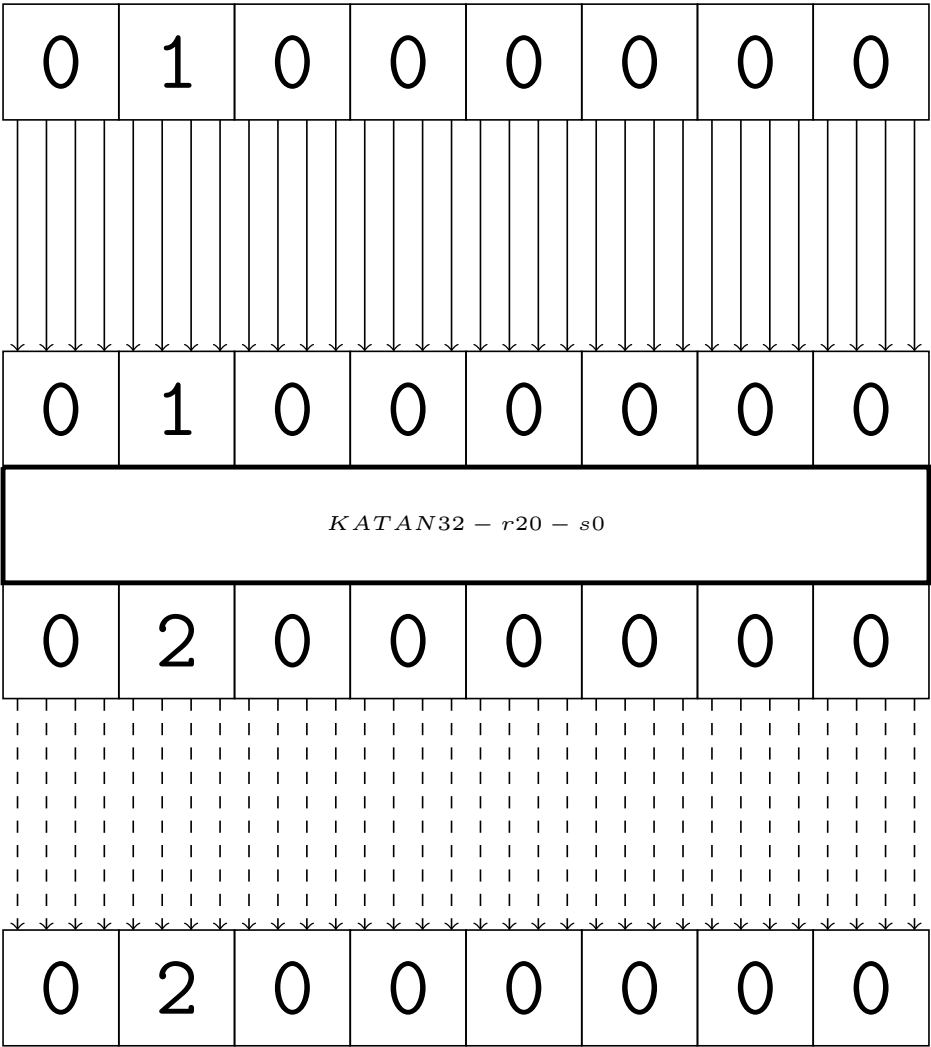
41 KATAN32-r19-s0

Maximal differential probability: $2^{-0.0}$



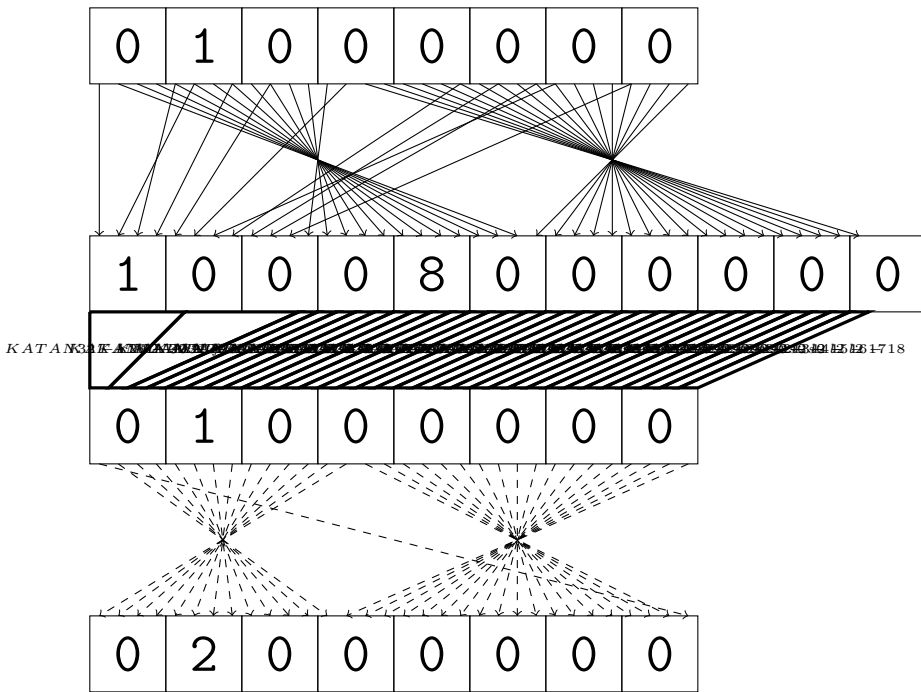
42 KATAN32-r20

Maximal differential probability: $2^{-1.0}$



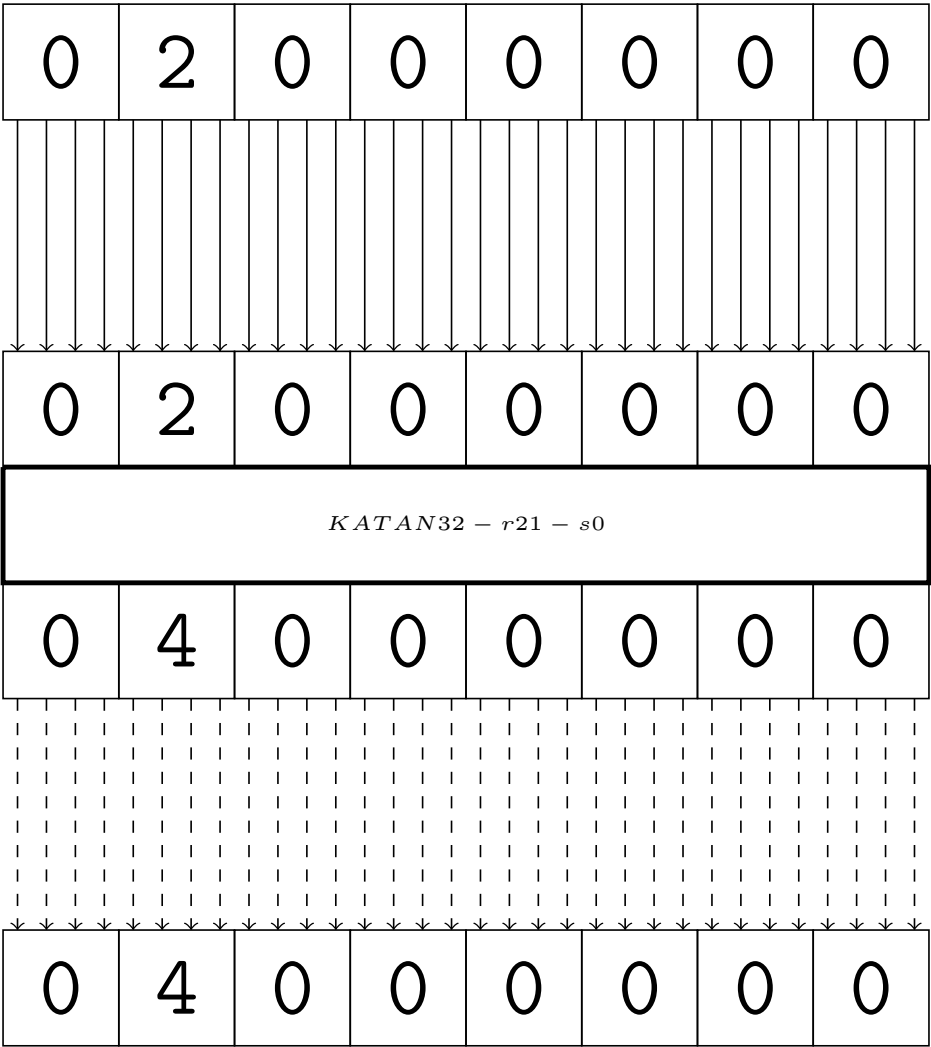
43 KATAN32-r20-s0

Maximal differential probability: $2^{-1.0}$



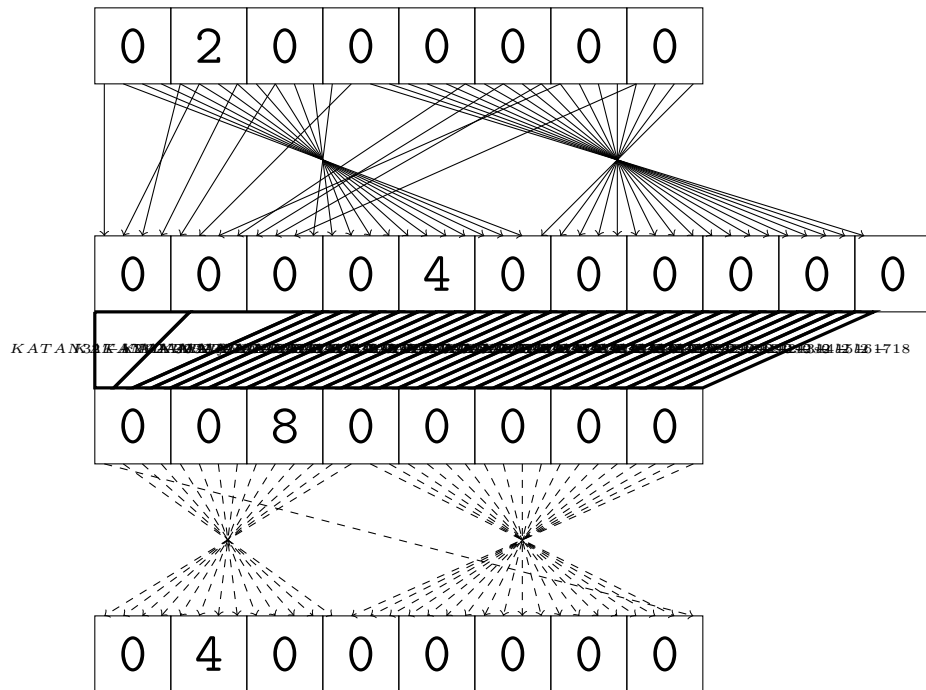
44 **KATAN32-r21**

Maximal differential probability: $2^{-0.0}$



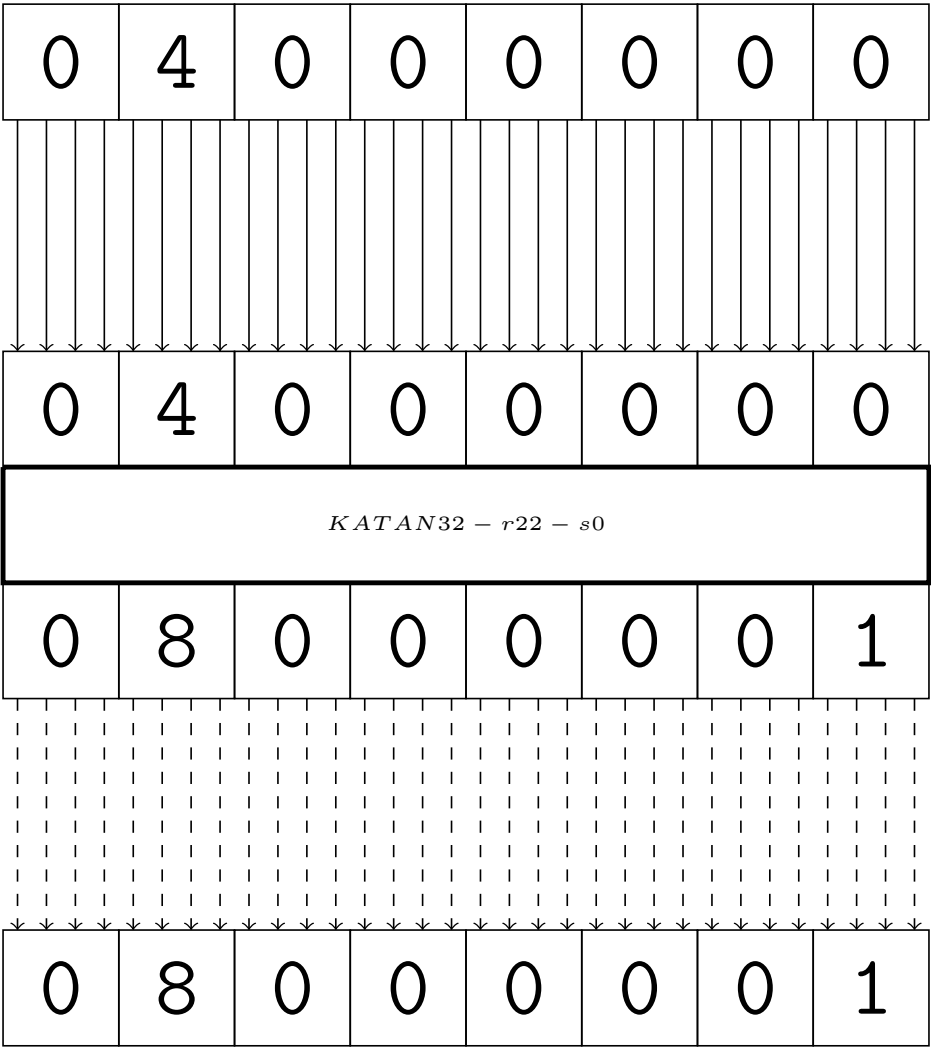
45 KATAN32-r21-s0

Maximal differential probability: $2^{-0.0}$



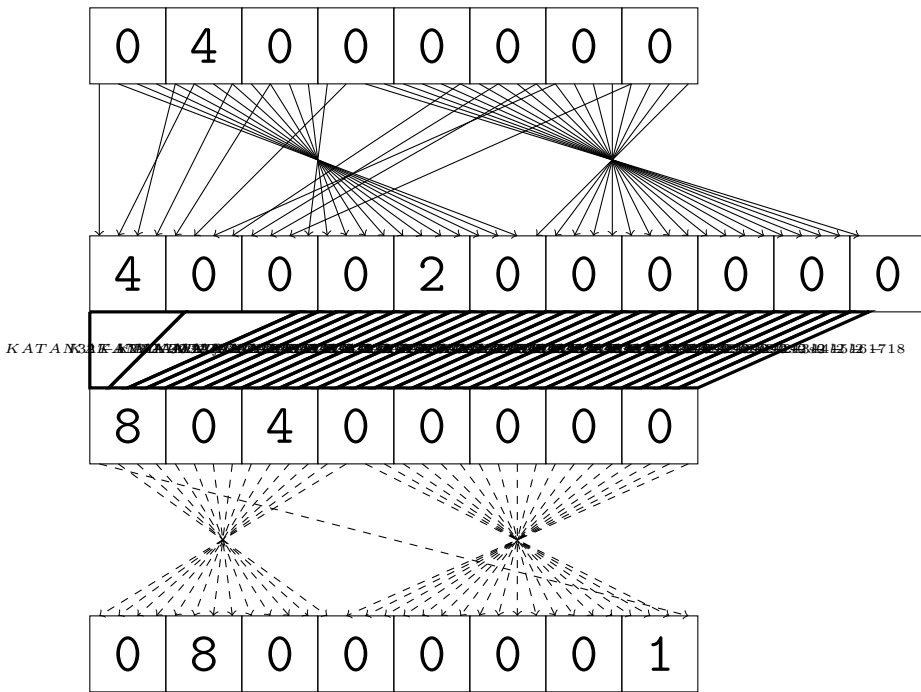
46 KATAN32-r22

Maximal differential probability: $2^{-0.0}$



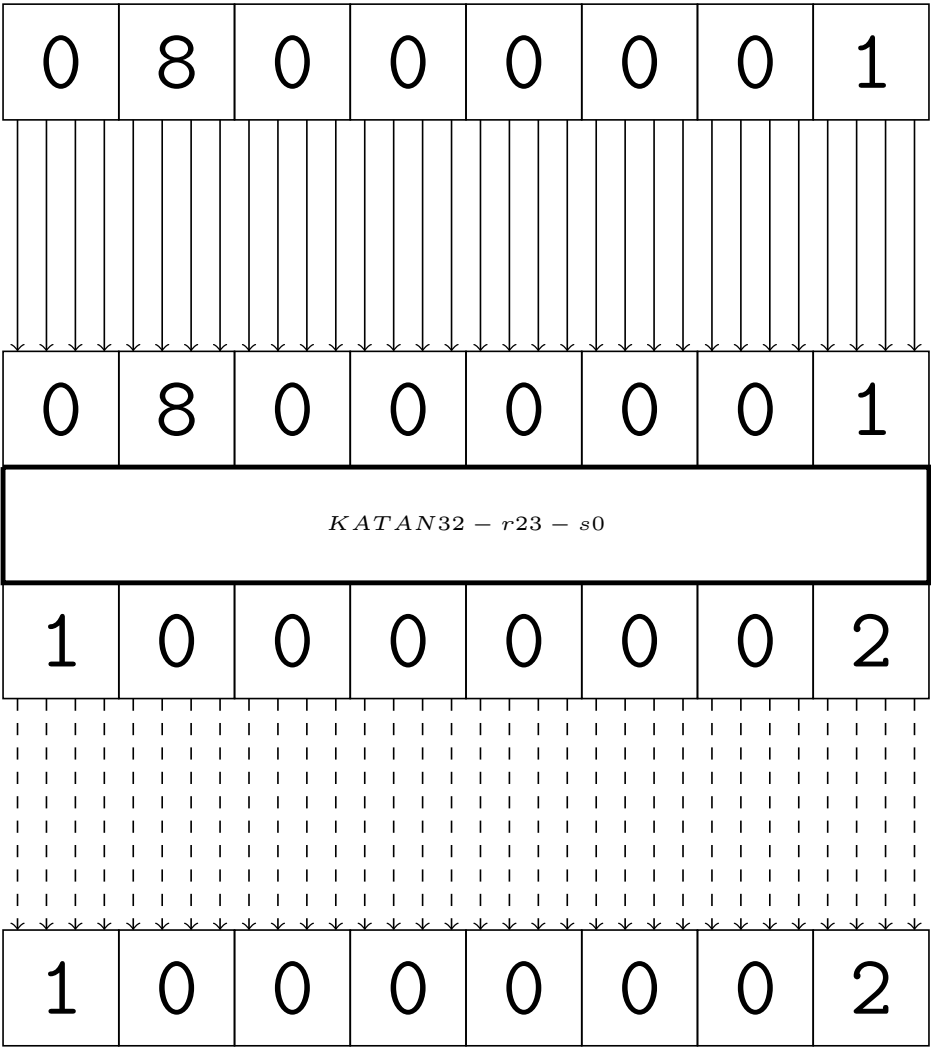
47 KATAN32-r22-s0

Maximal differential probability: $2^{-0.0}$



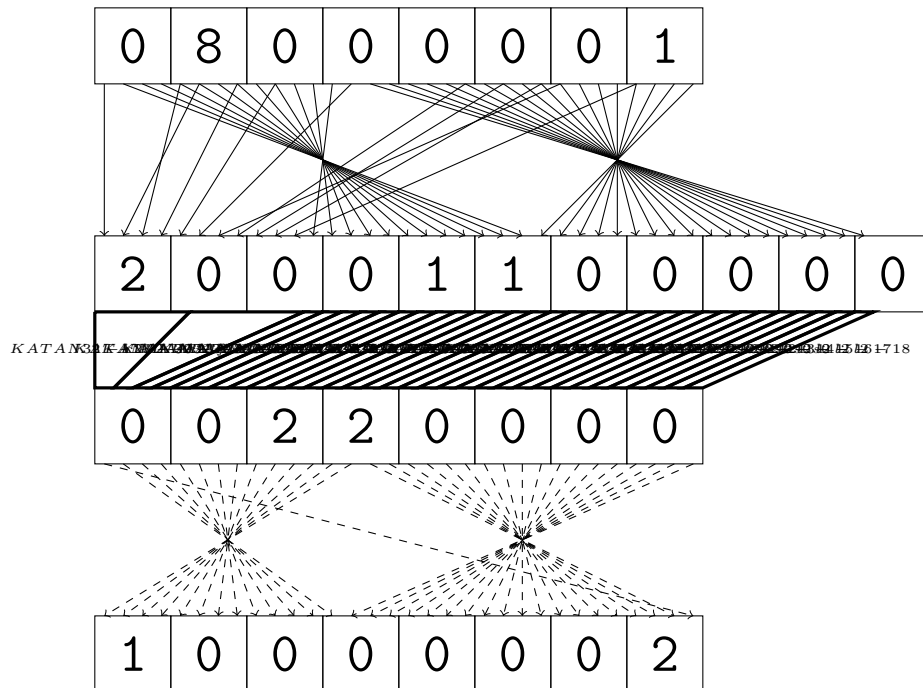
48 **KATAN32-r23**

Maximal differential probability: $2^{-1.0}$



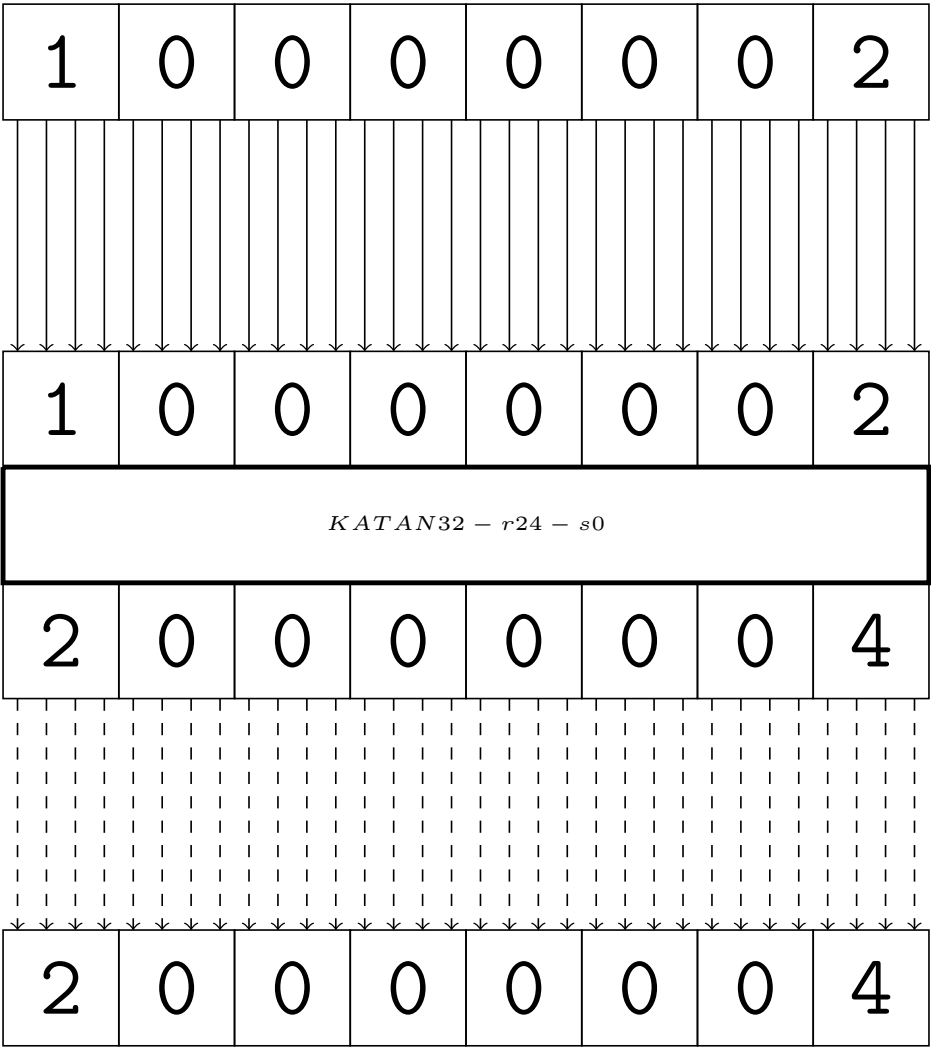
49 KATAN32-r23-s0

Maximal differential probability: $2^{-1.0}$



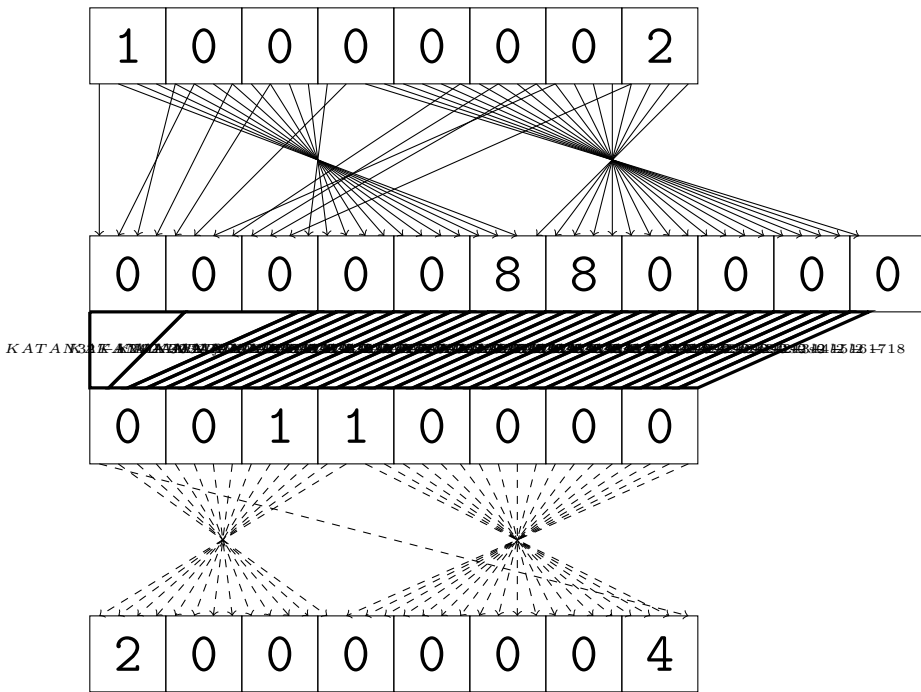
50 **KATAN32-r24**

Maximal differential probability: $2^{-0.0}$



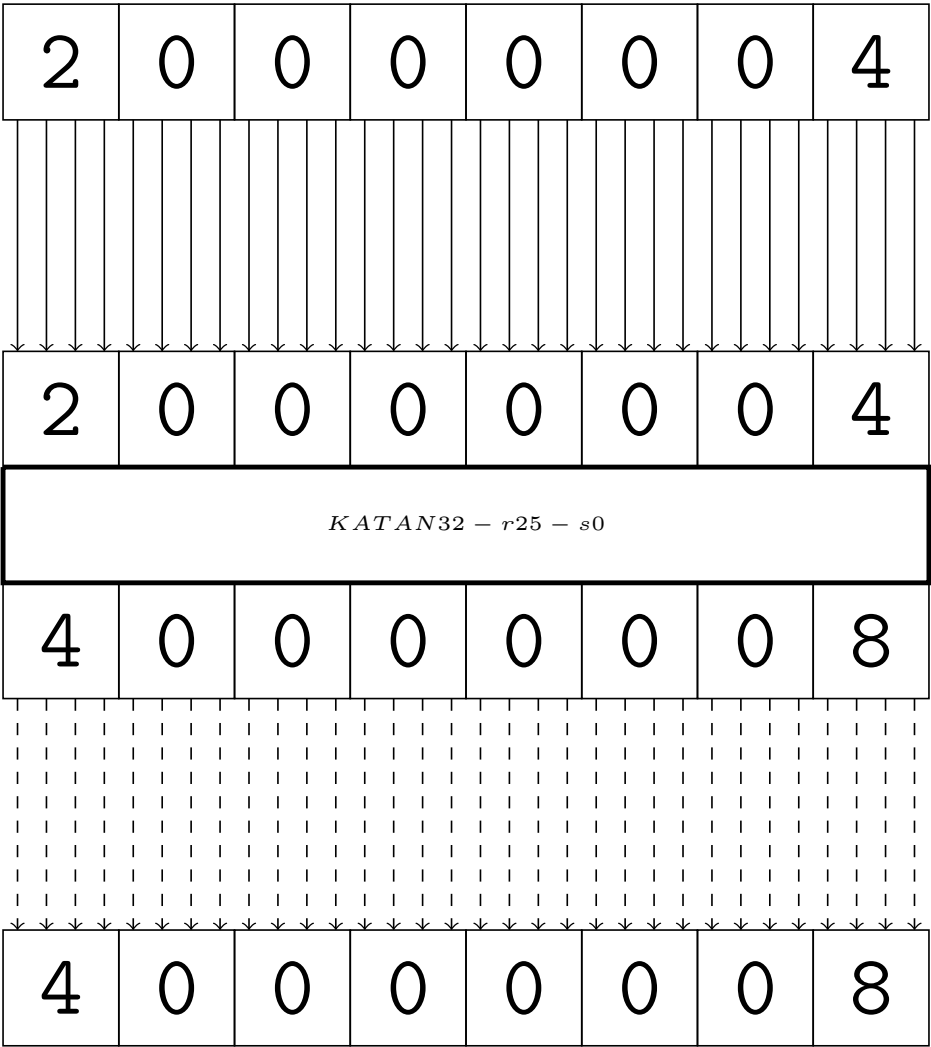
51 KATAN32-r24-s0

Maximal differential probability: $2^{-0.0}$



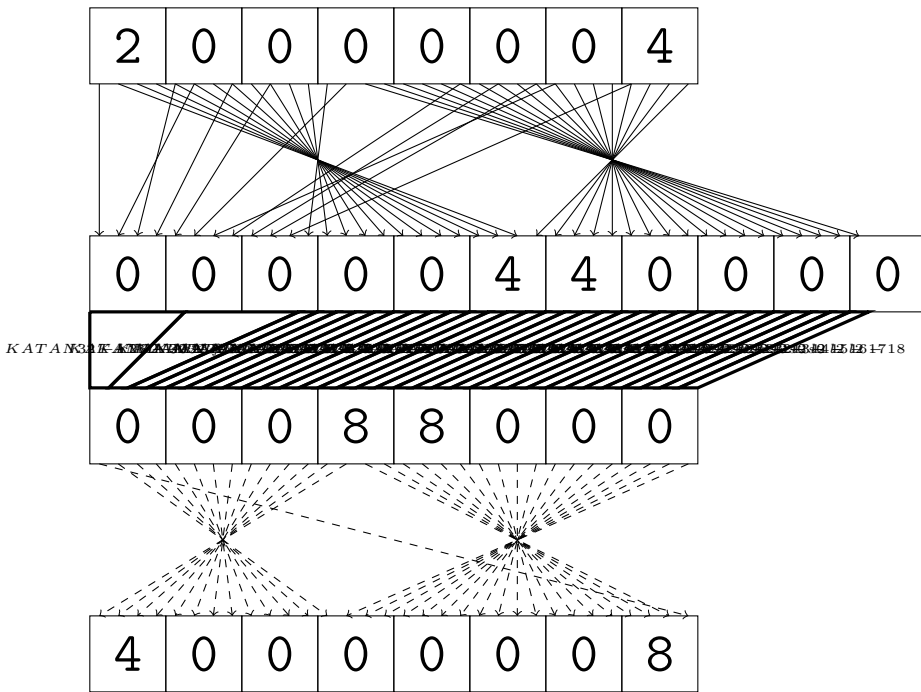
52 KATAN32-r25

Maximal differential probability: $2^{-0.0}$



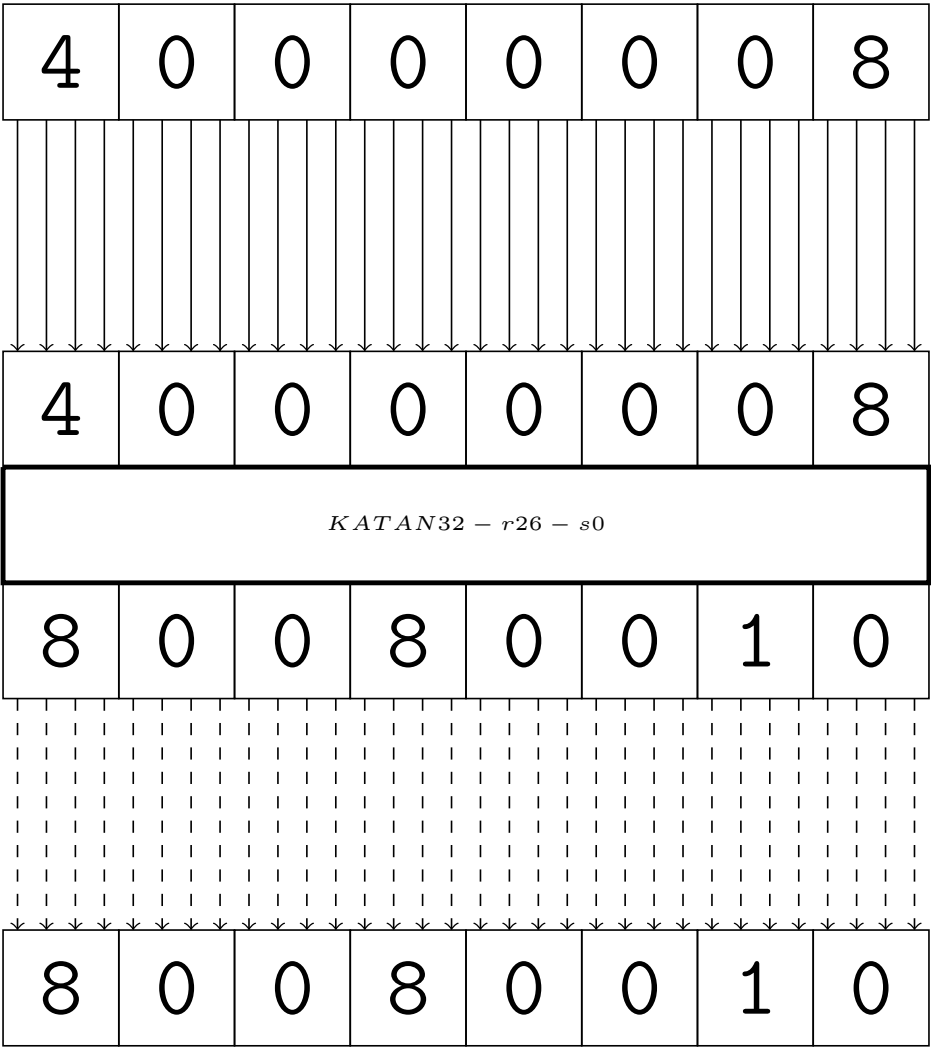
53 KATAN32-r25-s0

Maximal differential probability: $2^{-0.0}$



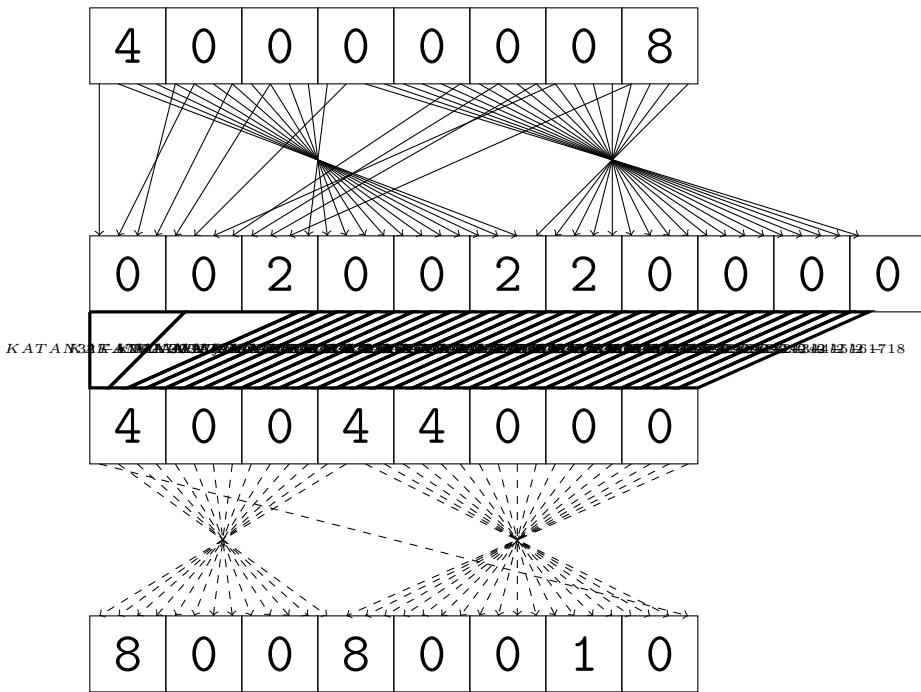
54 KATAN32-r26

Maximal differential probability: $2^{-1.0}$



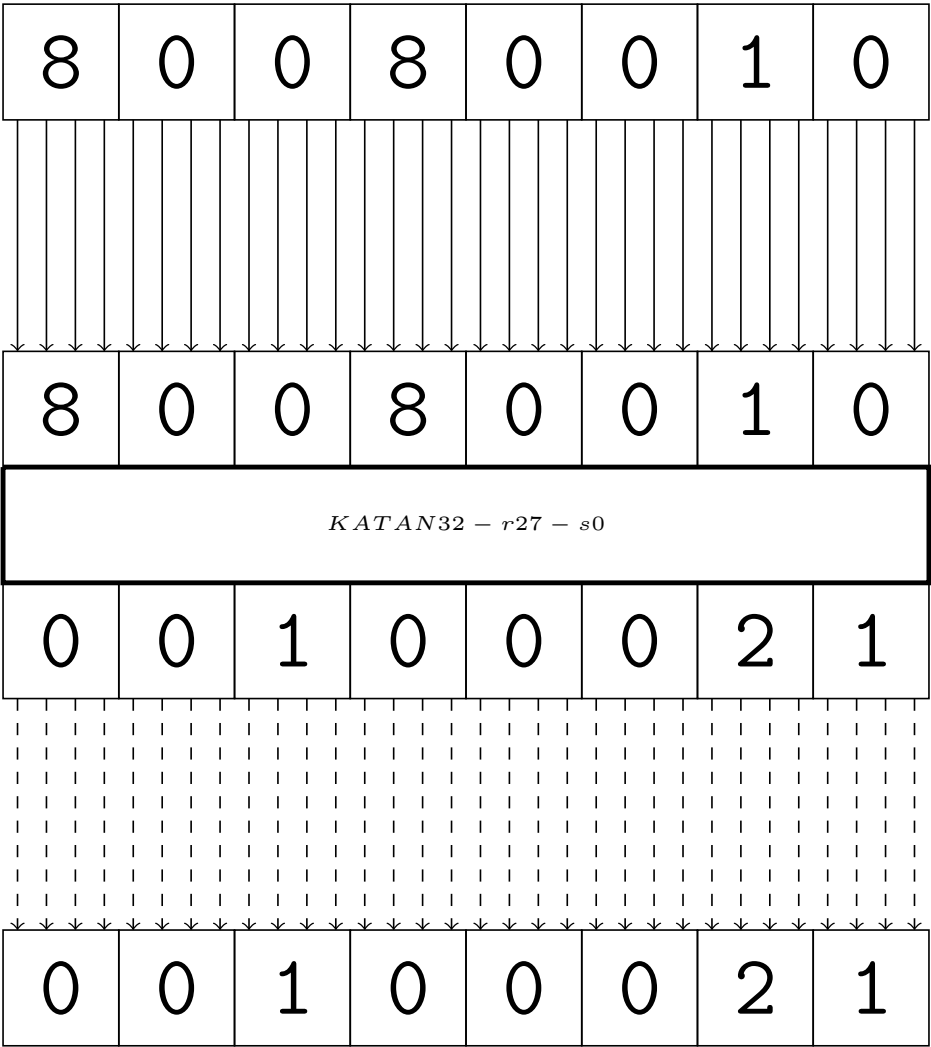
55 KATAN32-r26-s0

Maximal differential probability: $2^{-1.0}$



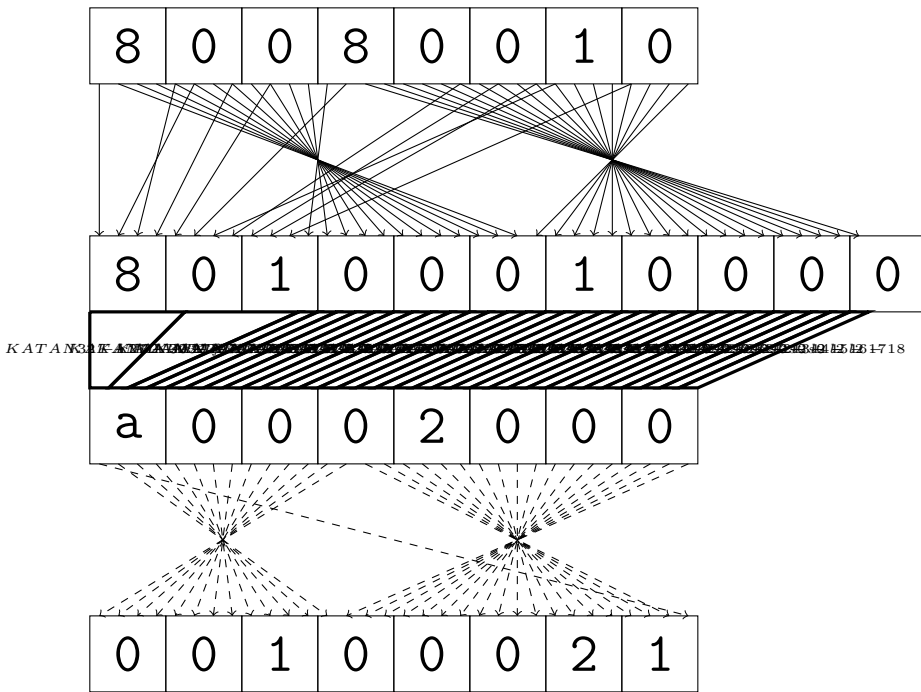
56 KATAN32-r27

Maximal differential probability: $2^{-0.0}$



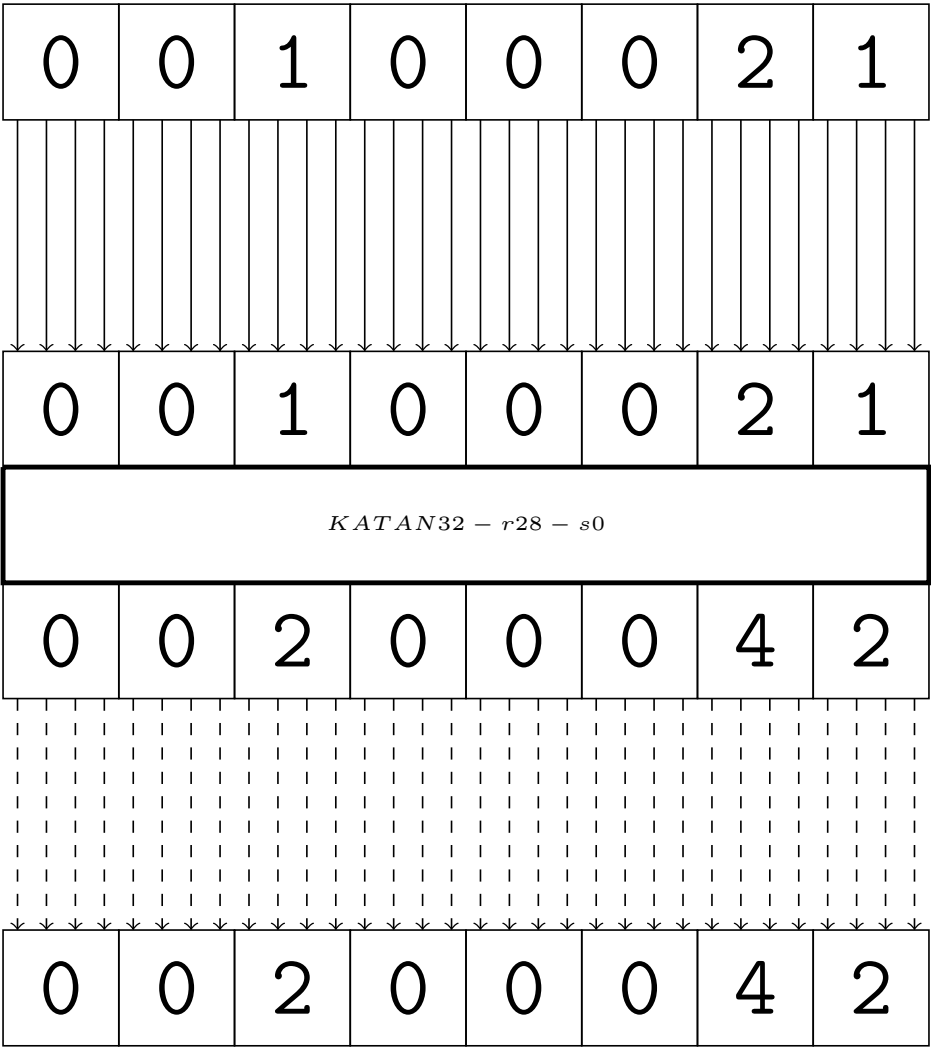
57 KATAN32-r27-s0

Maximal differential probability: $2^{-0.0}$



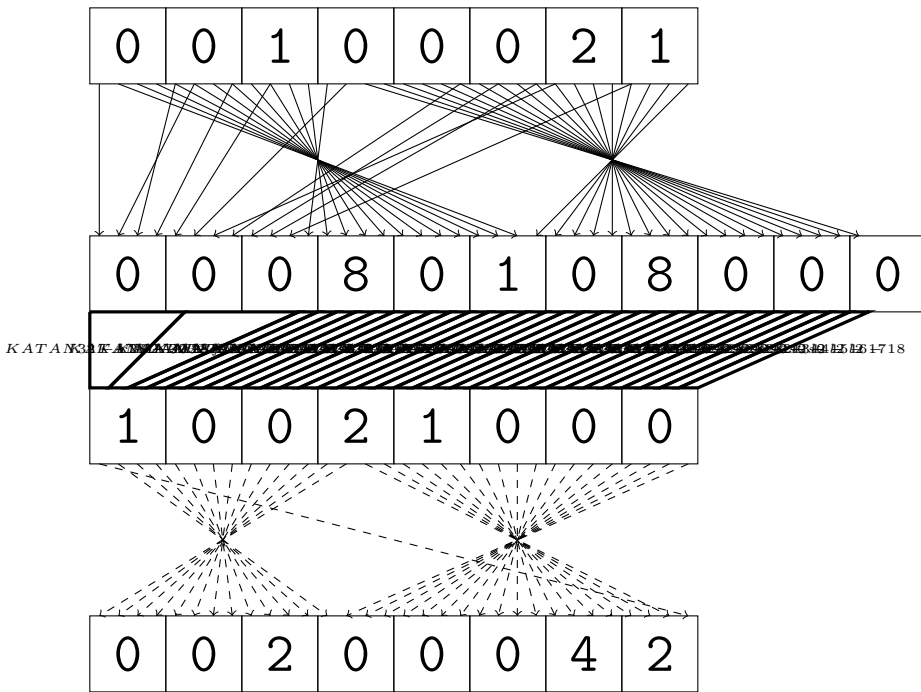
58 KATAN32-r28

Maximal differential probability: $2^{-0.0}$



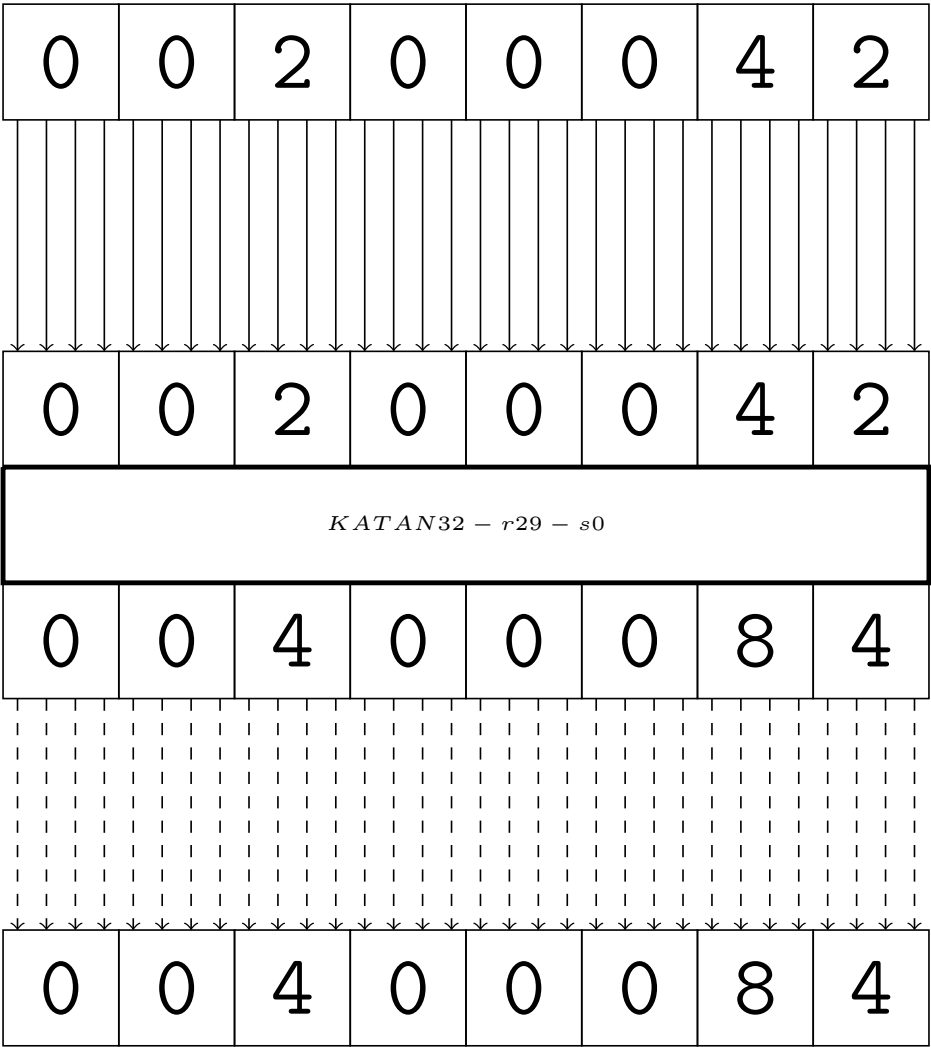
59 KATAN32-r28-s0

Maximal differential probability: $2^{-0.0}$



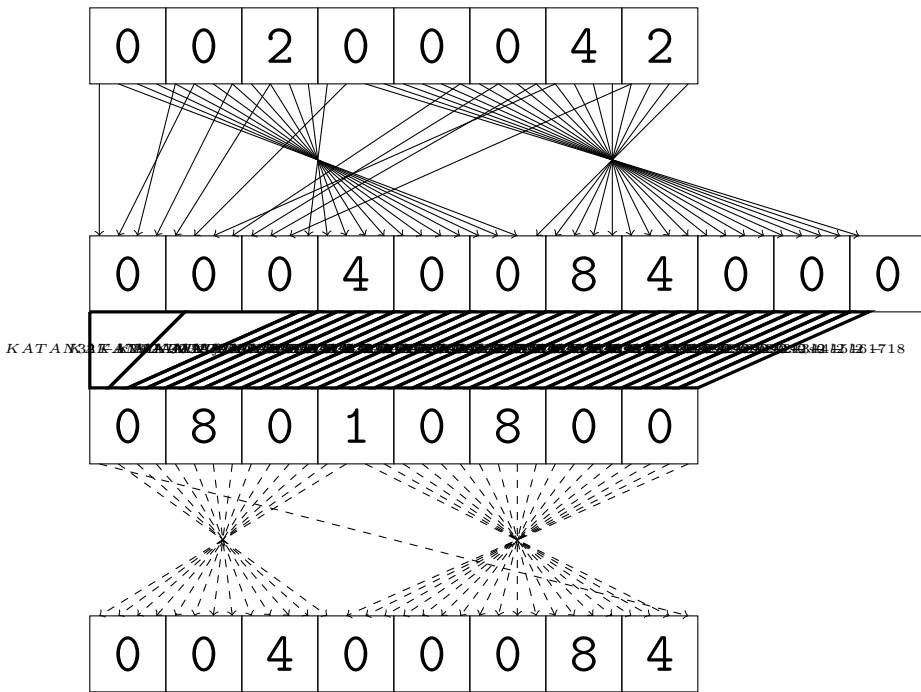
60 KATAN32-r29

Maximal differential probability: $2^{-0.0}$



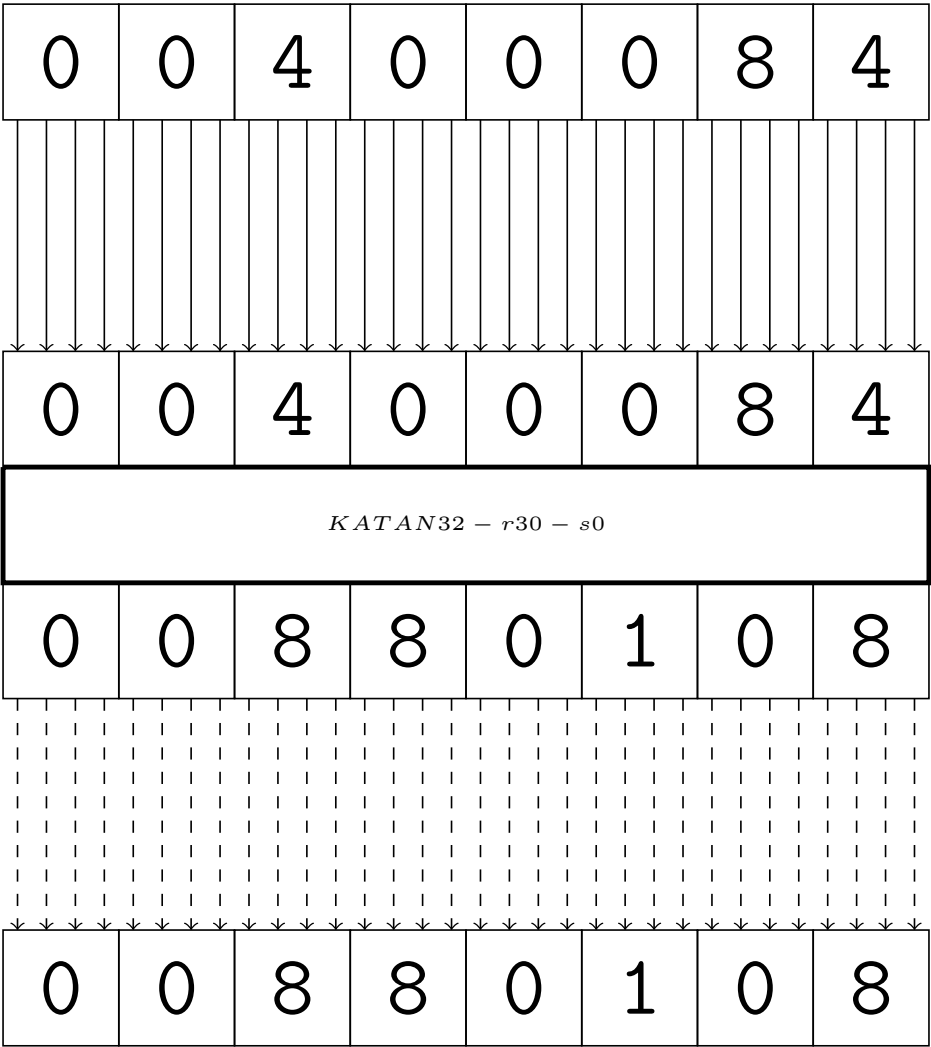
61 KATAN32-r29-s0

Maximal differential probability: $2^{-0.0}$



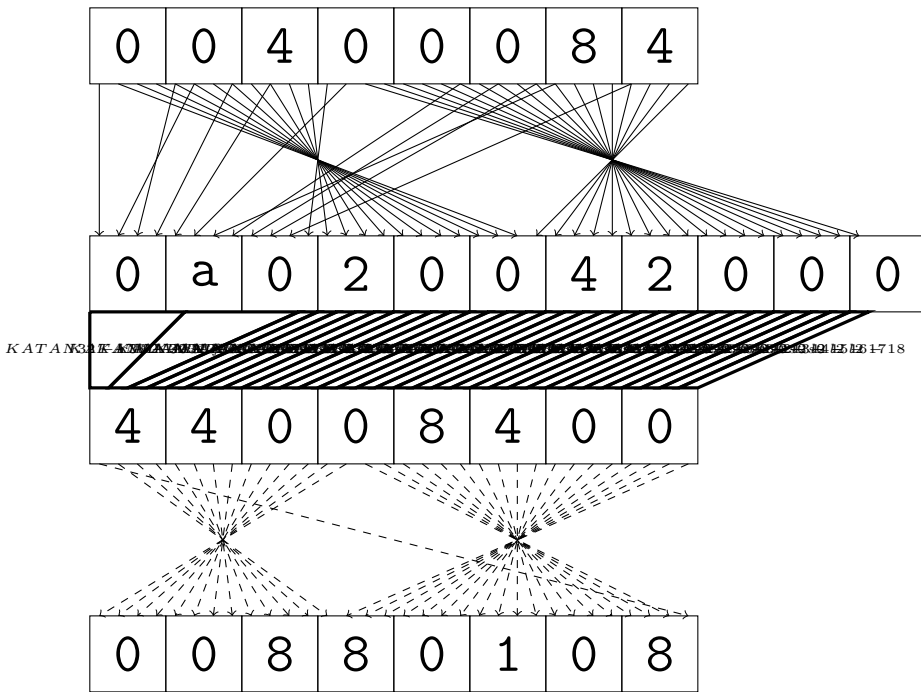
62 KATAN32-r30

Maximal differential probability: $2^{-0.0}$



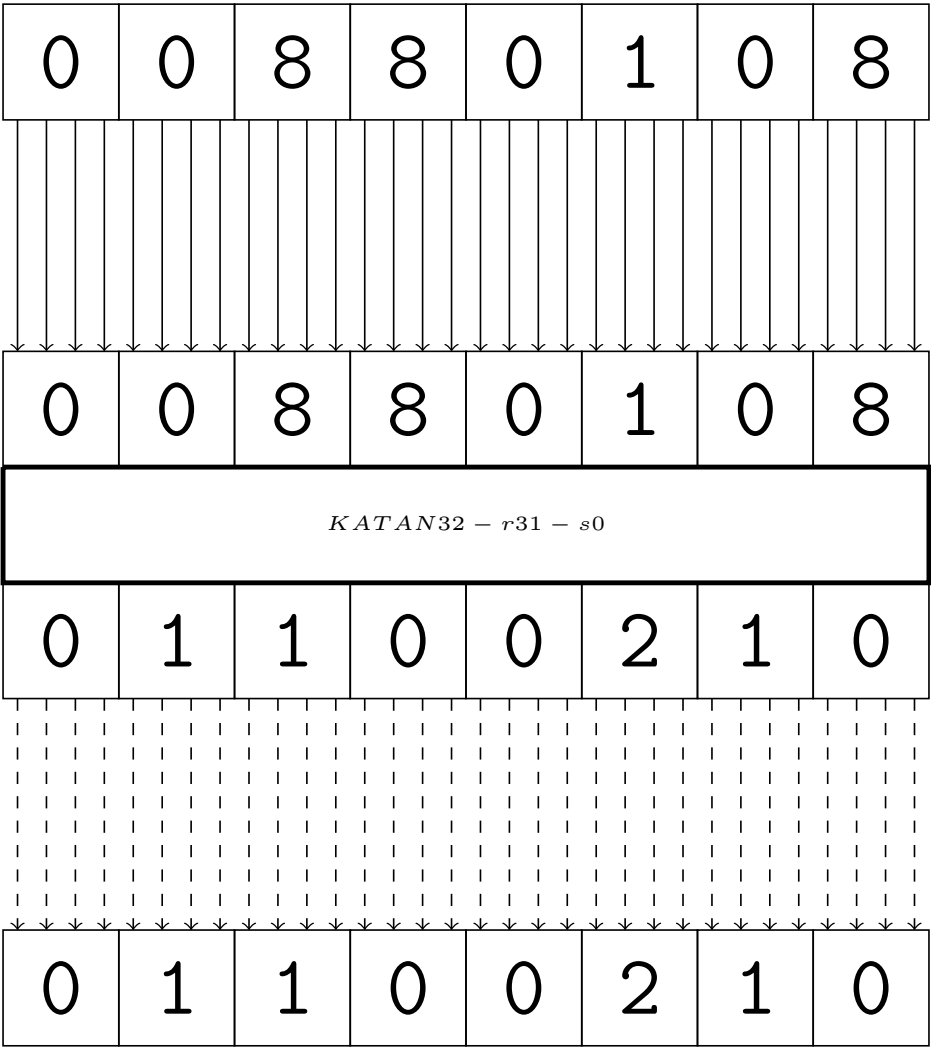
63 KATAN32-r30-s0

Maximal differential probability: $2^{-0.0}$



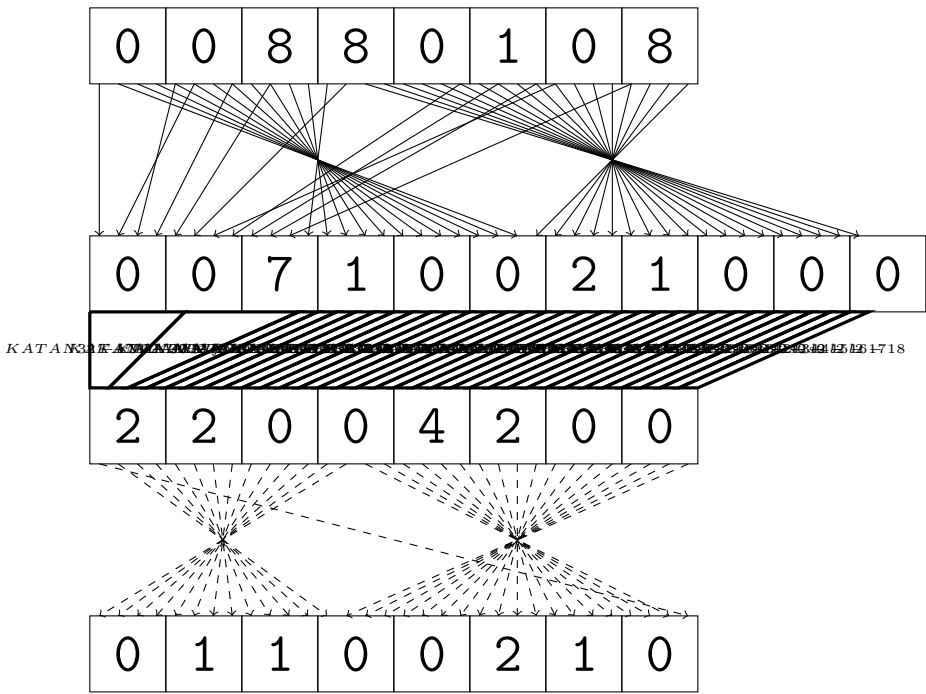
64 KATAN32-r31

Maximal differential probability: $2^{-1.0}$



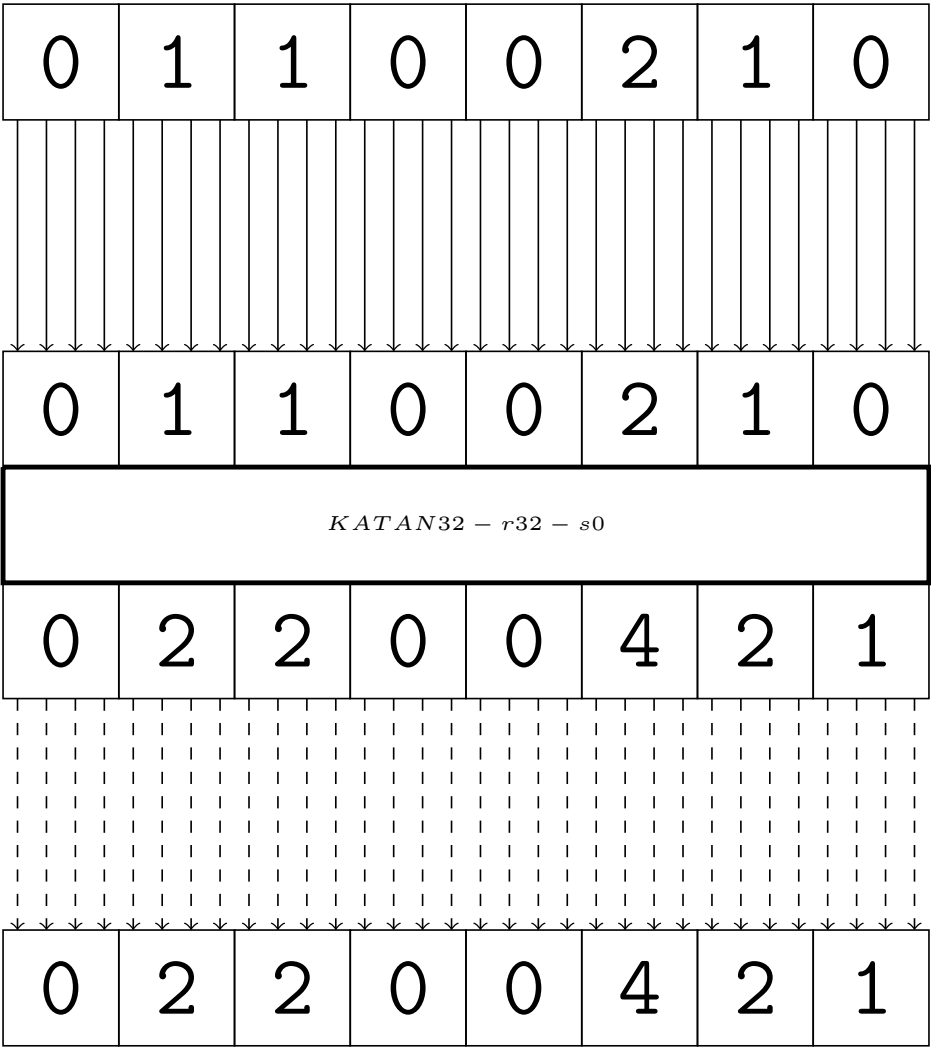
65 KATAN32-r31-s0

Maximal differential probability: $2^{-1.0}$



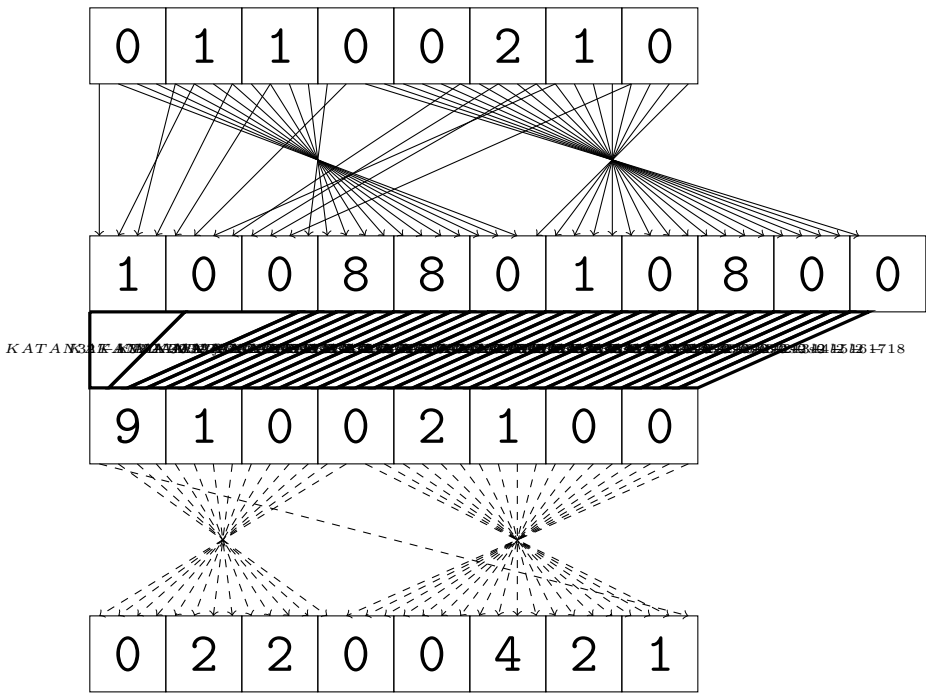
66 KATAN32-r32

Maximal differential probability: $2^{-1.0}$



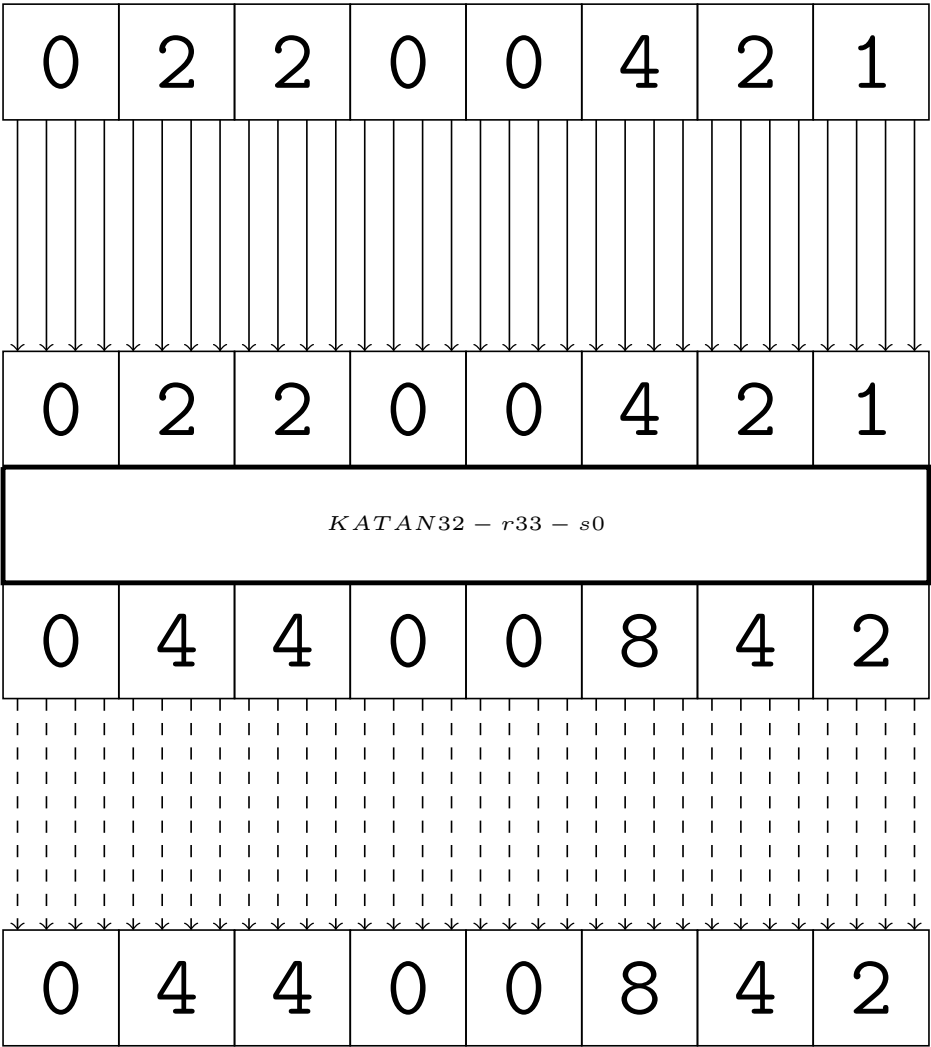
67 KATAN32-r32-s0

Maximal differential probability: $2^{-1.0}$



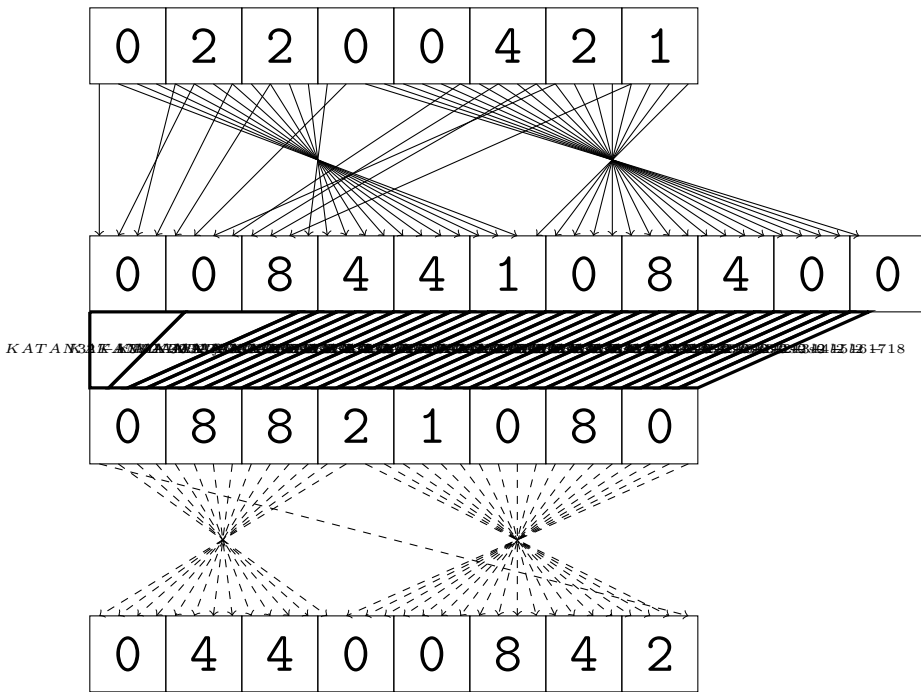
68 KATAN32-r33

Maximal differential probability: $2^{-1.0}$



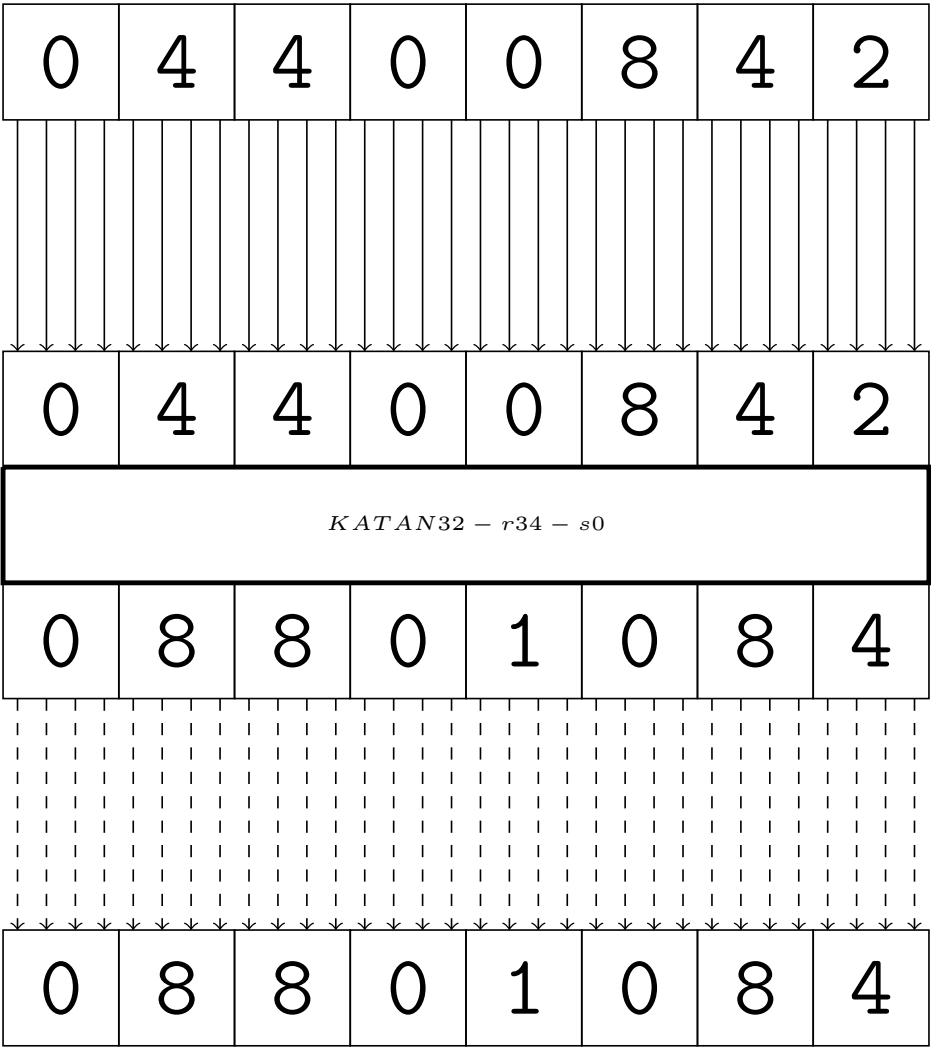
69 KATAN32-r33-s0

Maximal differential probability: $2^{-1.0}$



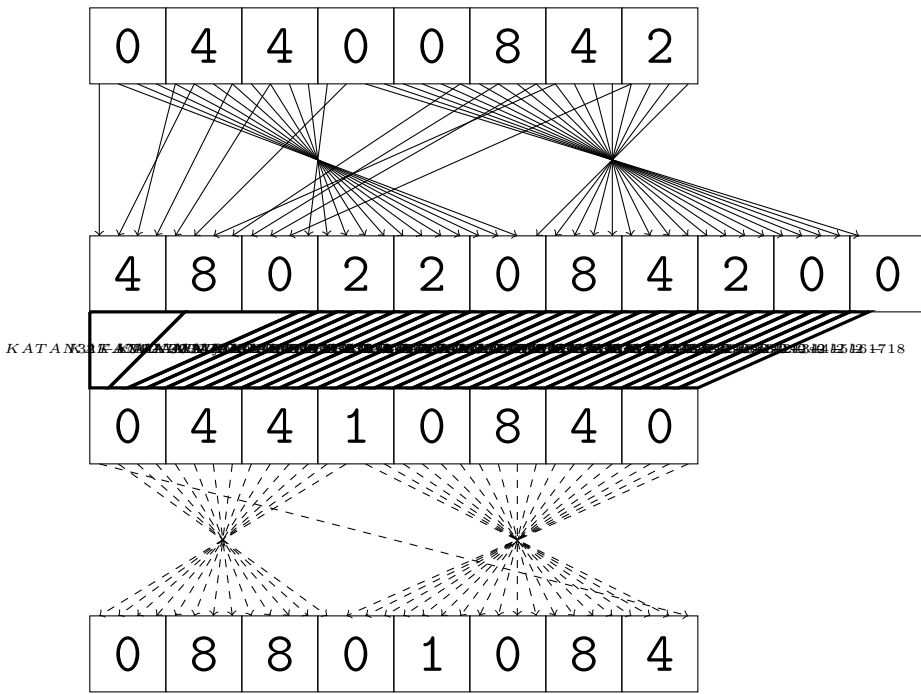
70 KATAN32-r34

Maximal differential probability: $2^{-0.0}$



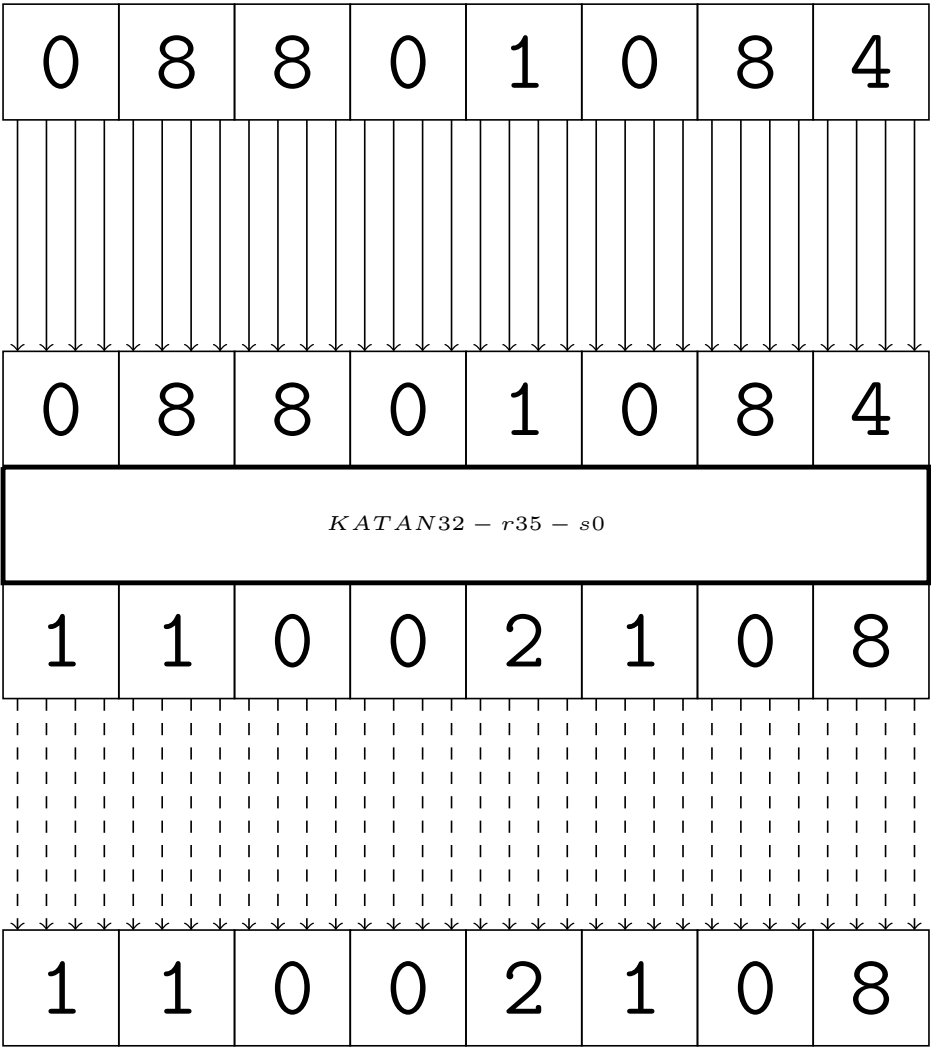
71 KATAN32-r34-s0

Maximal differential probability: $2^{-0.0}$



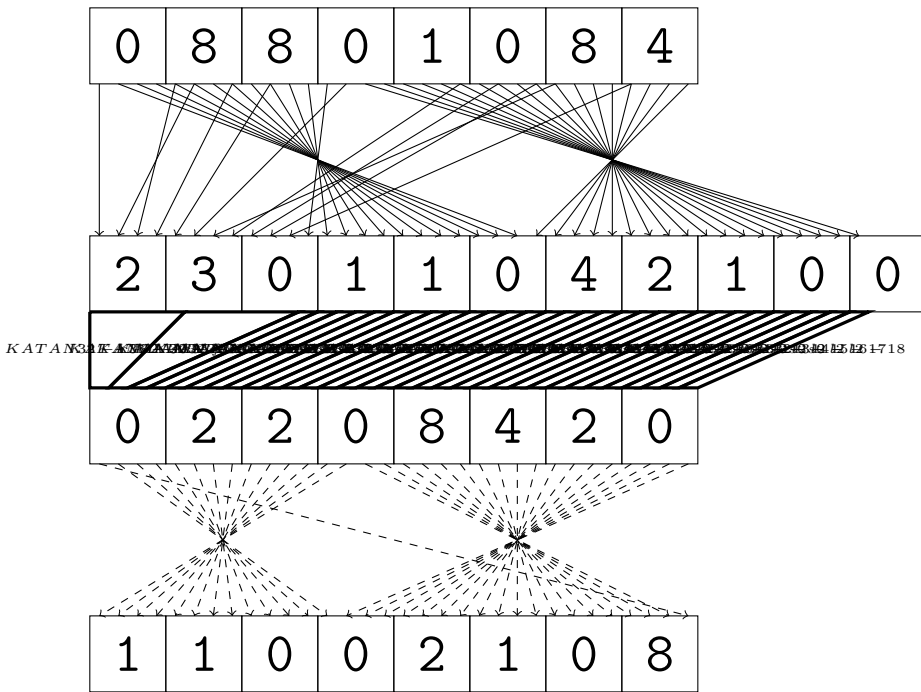
72 KATAN32-r35

Maximal differential probability: $2^{-2.0}$



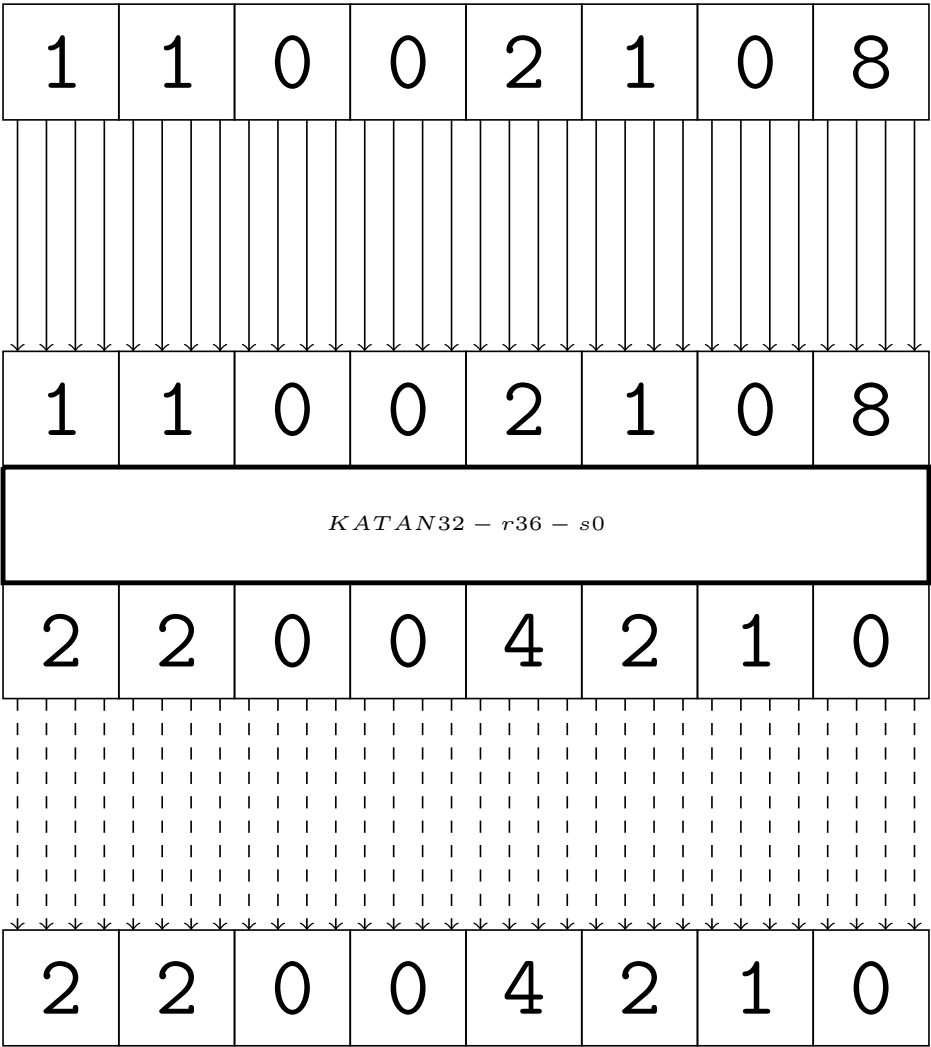
73 KATAN32-r35-s0

Maximal differential probability: $2^{-2.0}$



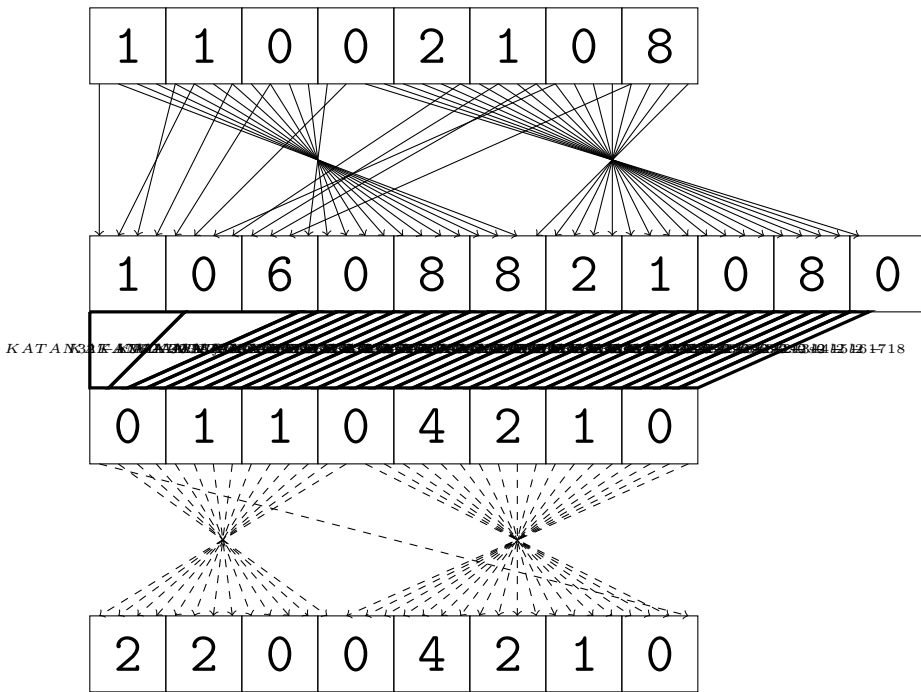
74 KATAN32-r36

Maximal differential probability: $2^{-2.0}$



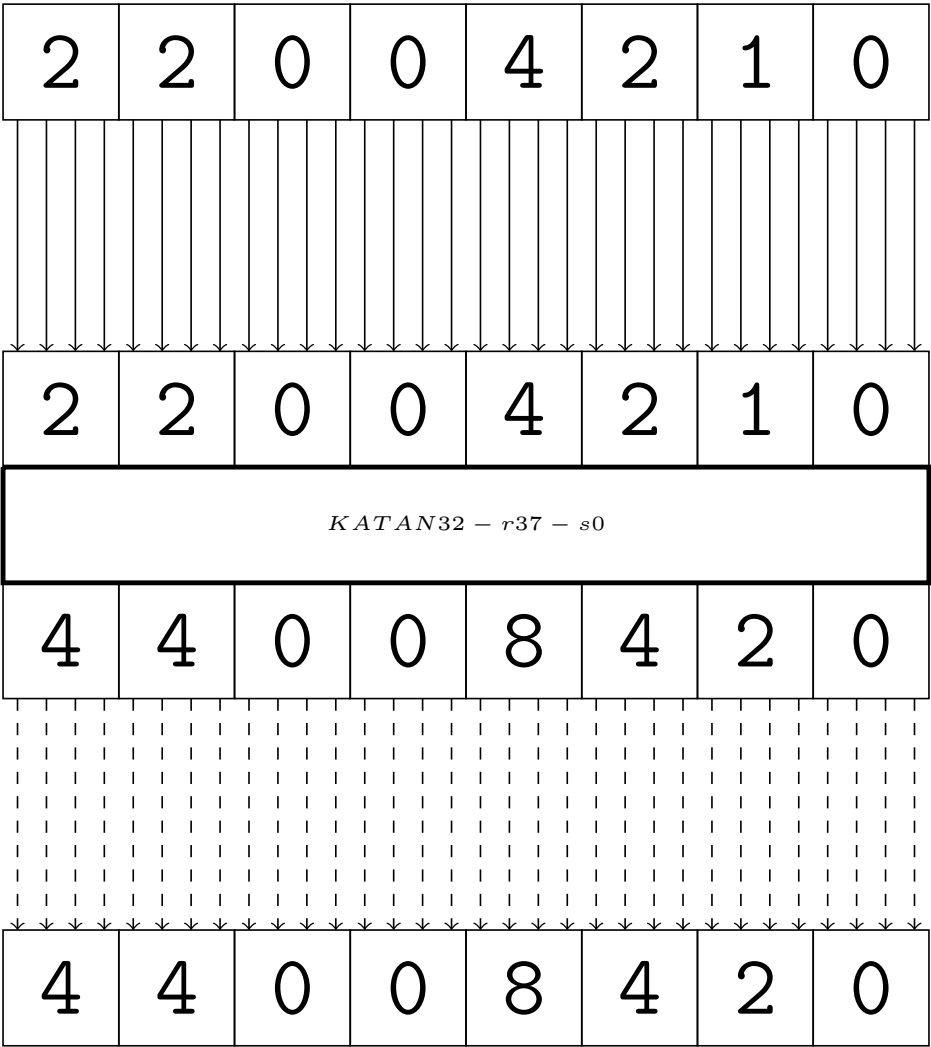
75 KATAN32-r36-s0

Maximal differential probability: $2^{-2.0}$



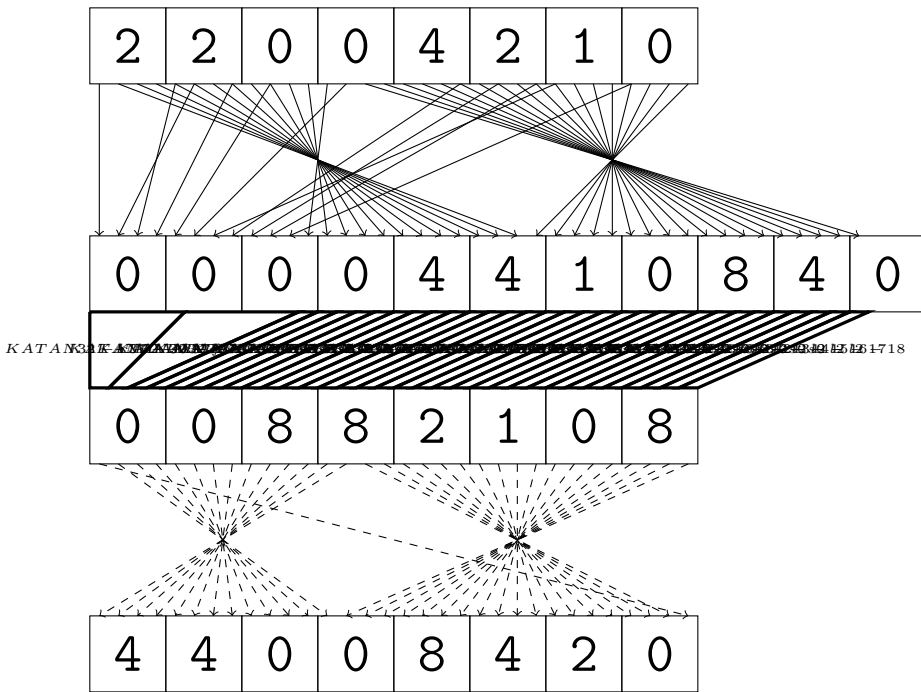
76 KATAN32-r37

Maximal differential probability: $2^{-0.0}$



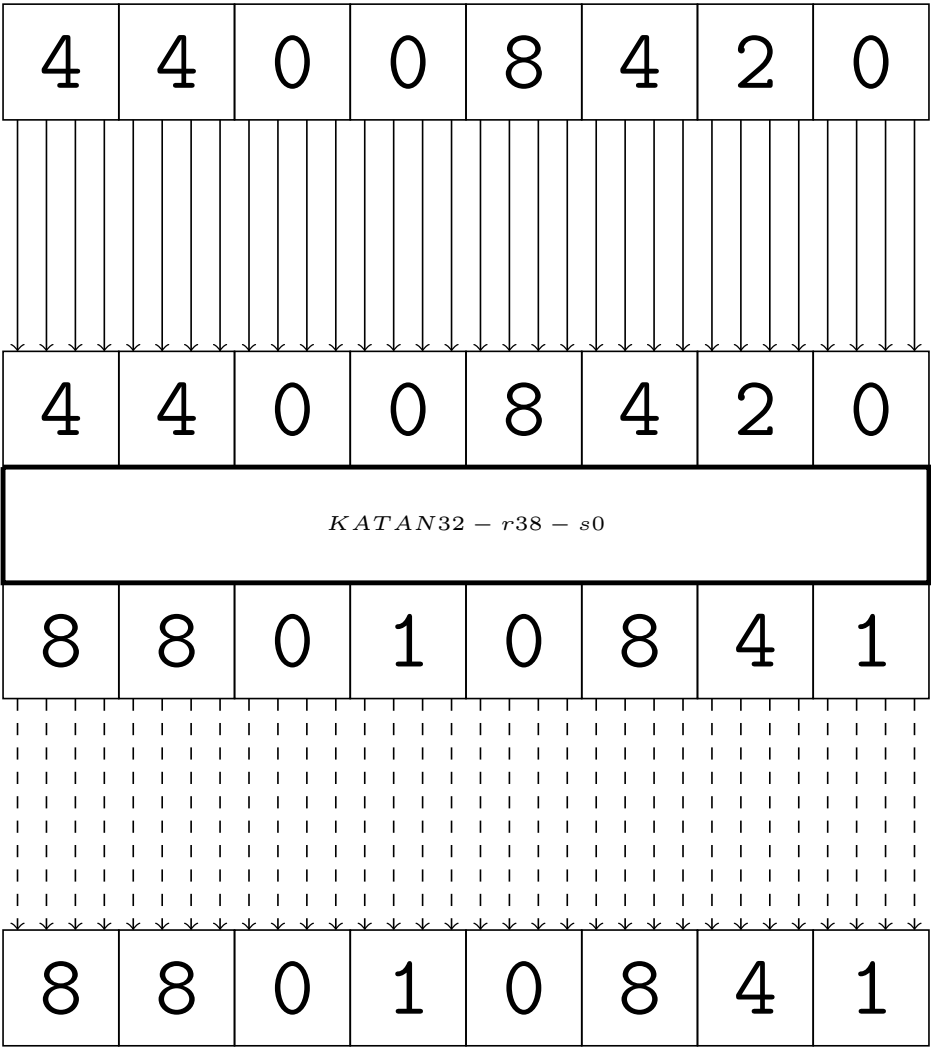
77 KATAN32-r37-s0

Maximal differential probability: $2^{-0.0}$



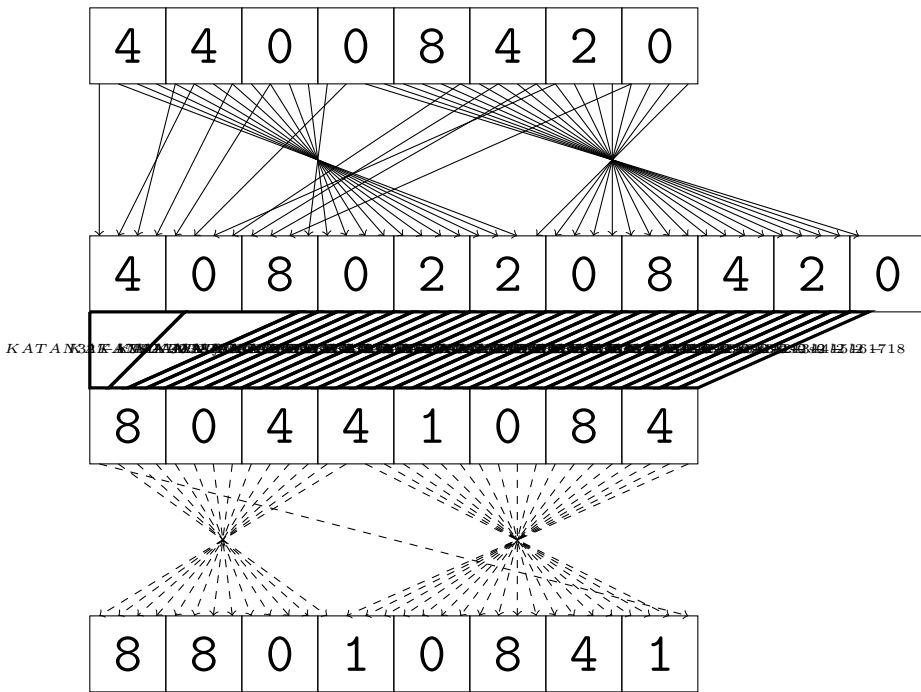
78 KATAN32-r38

Maximal differential probability: $2^{-1.0}$



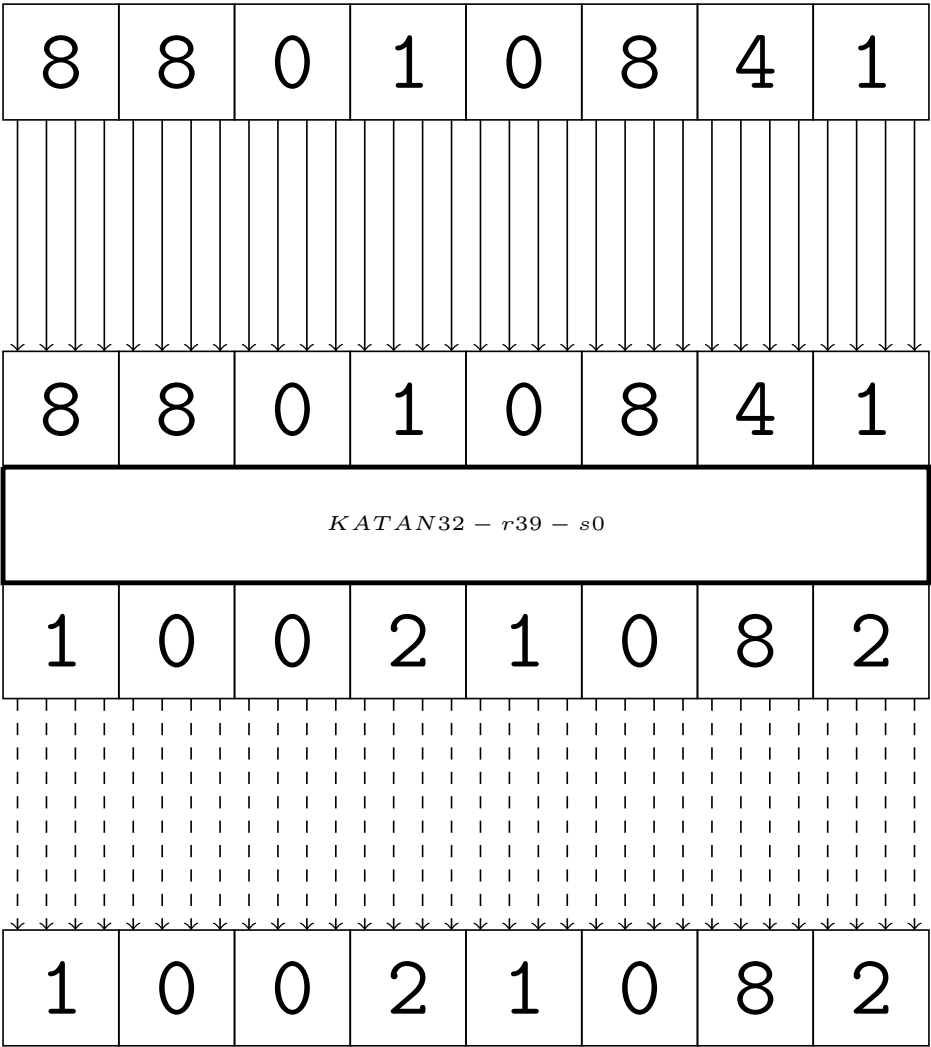
79 KATAN32-r38-s0

Maximal differential probability: $2^{-1.0}$



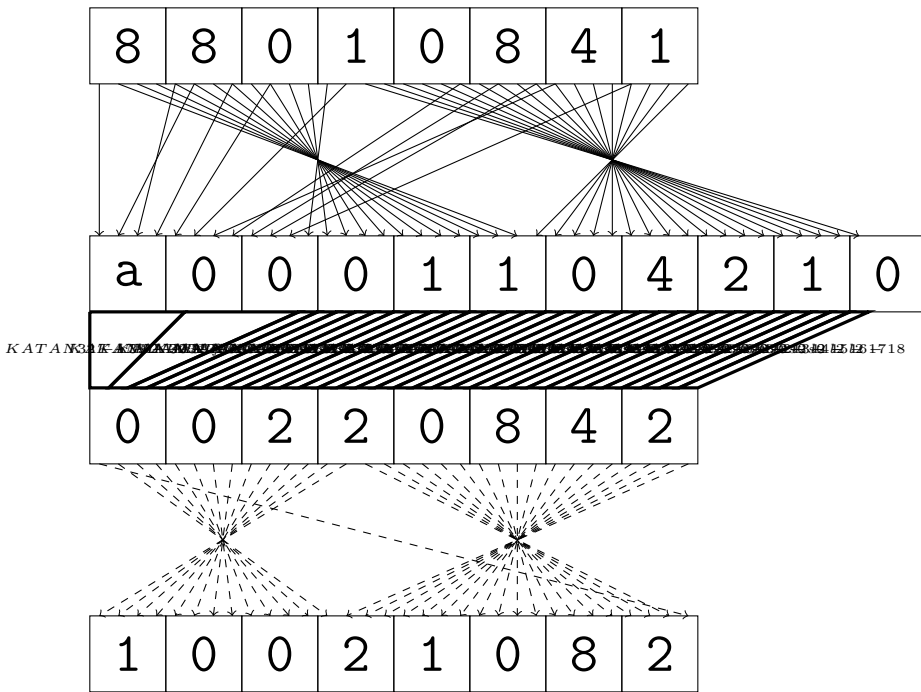
80 **KATAN32-r39**

Maximal differential probability: $2^{-1.0}$



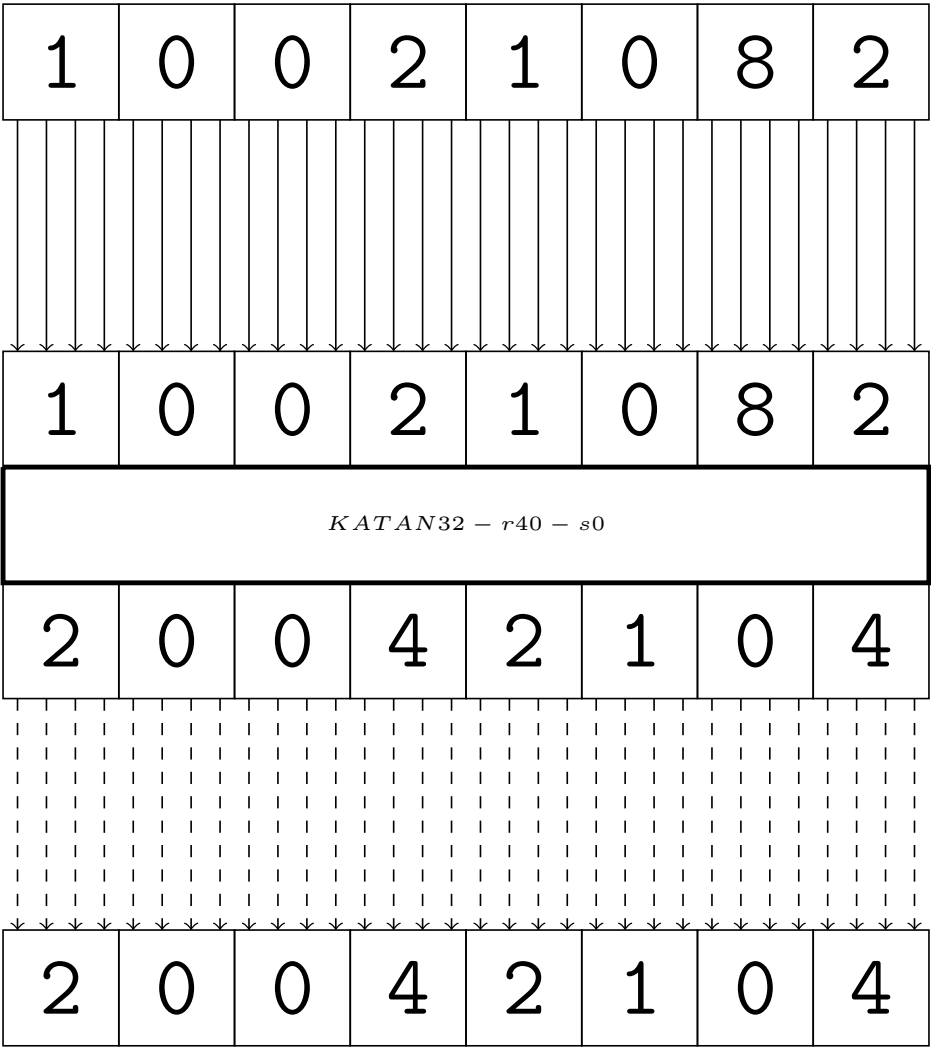
81 KATAN32-r39-s0

Maximal differential probability: $2^{-1.0}$



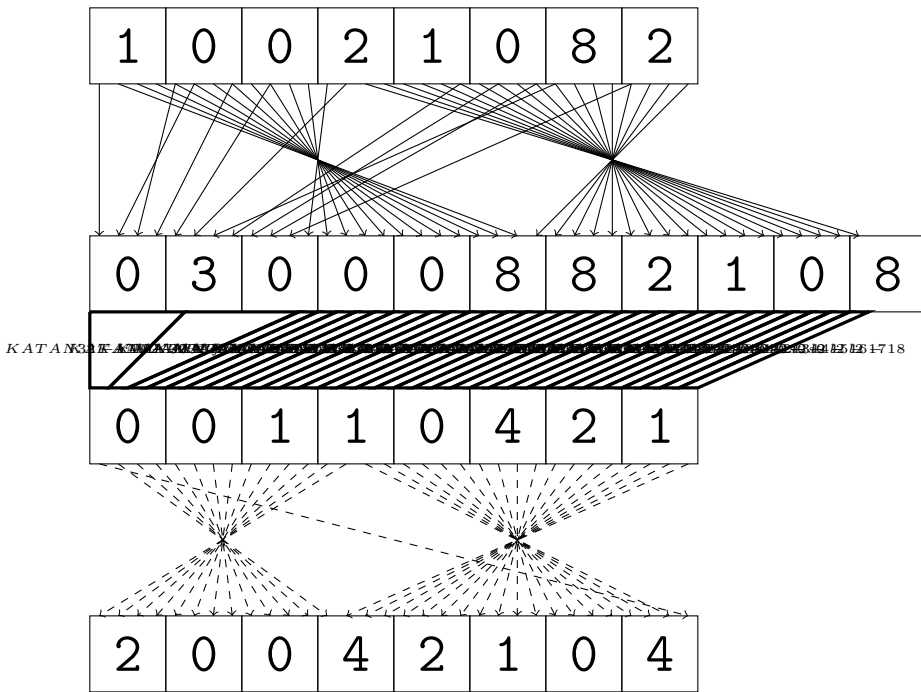
82 KATAN32-r40

Maximal differential probability: $2^{-1.0}$



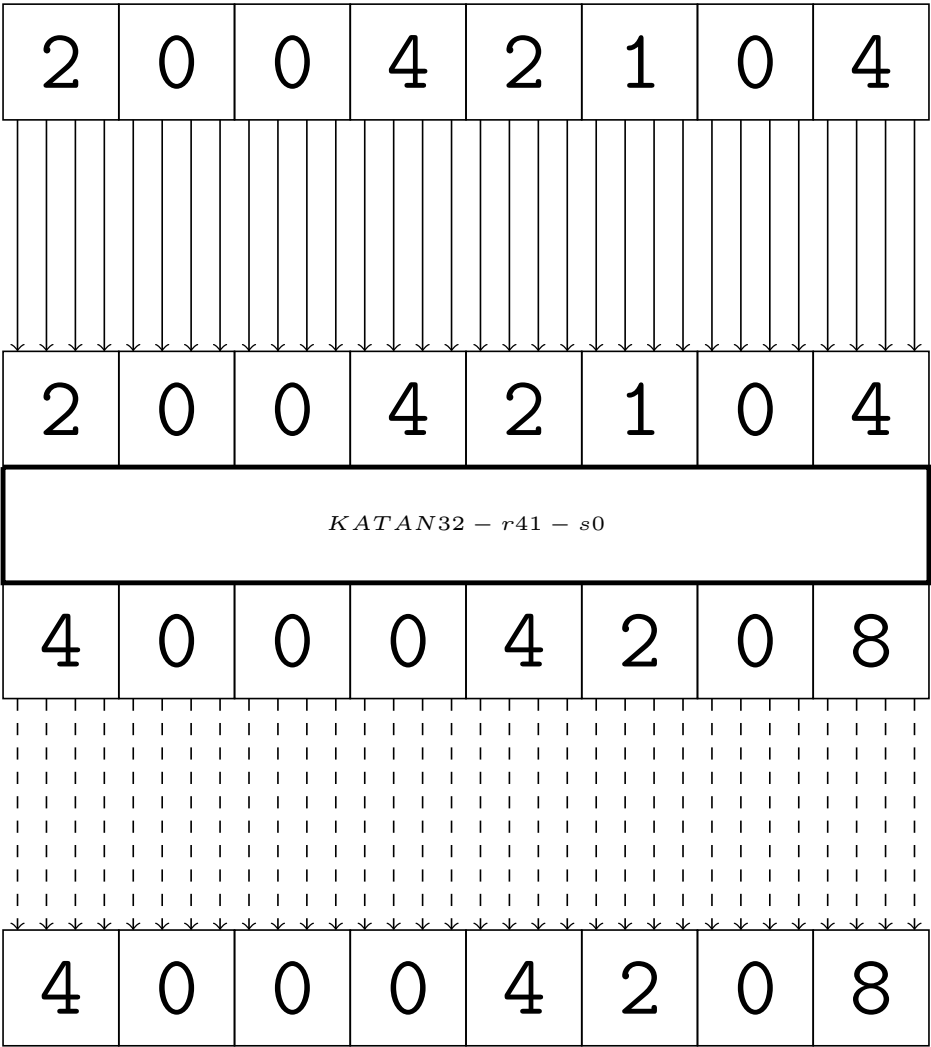
83 KATAN32-r40-s0

Maximal differential probability: $2^{-1.0}$



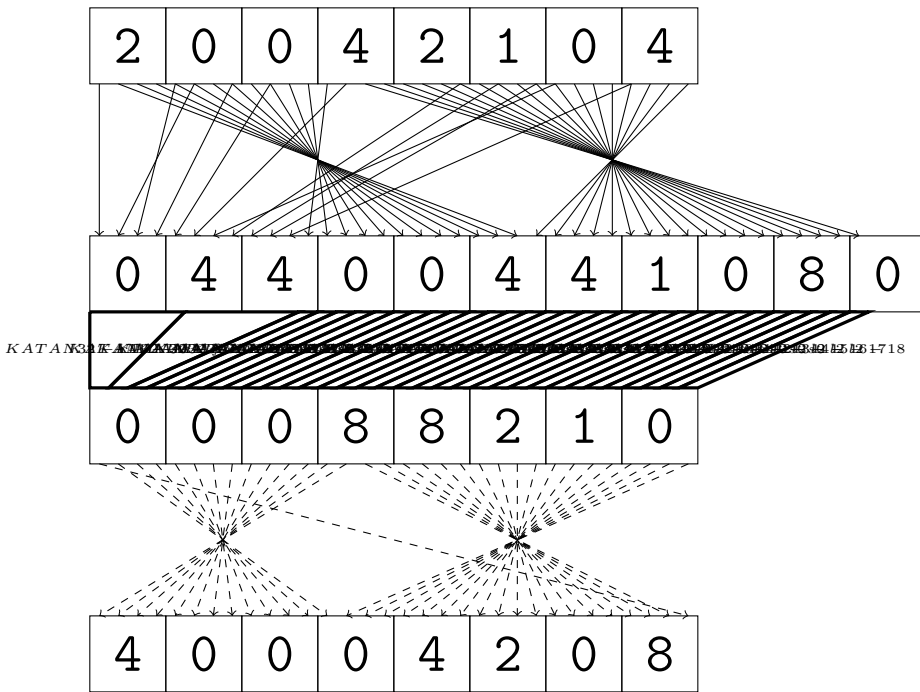
84 KATAN32-r41

Maximal differential probability: $2^{-1.0}$



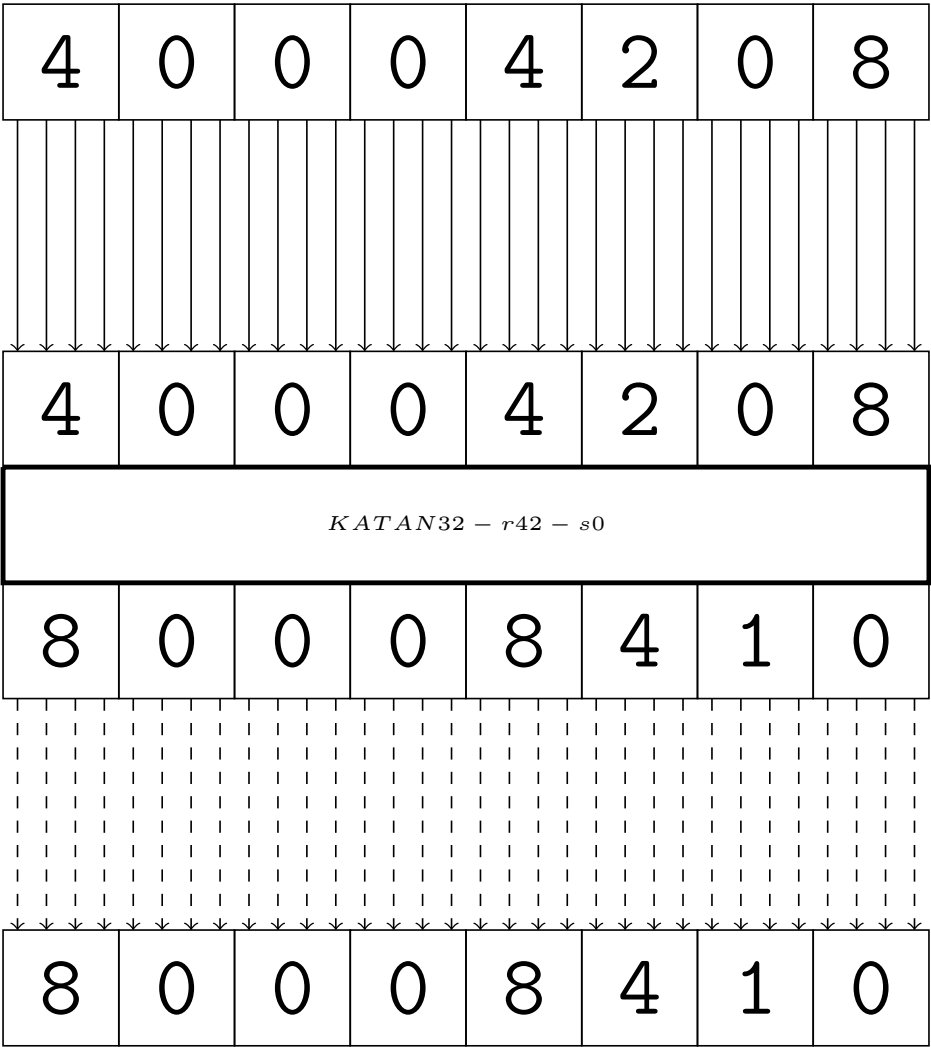
85 KATAN32-r41-s0

Maximal differential probability: $2^{-1.0}$



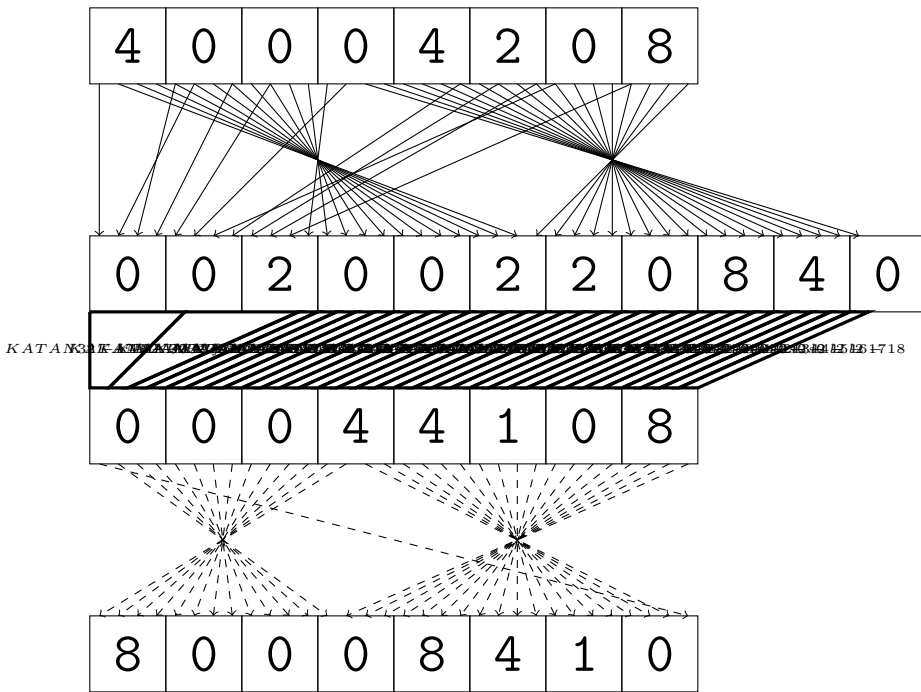
86 **KATAN32-r42**

Maximal differential probability: $2^{-1.0}$



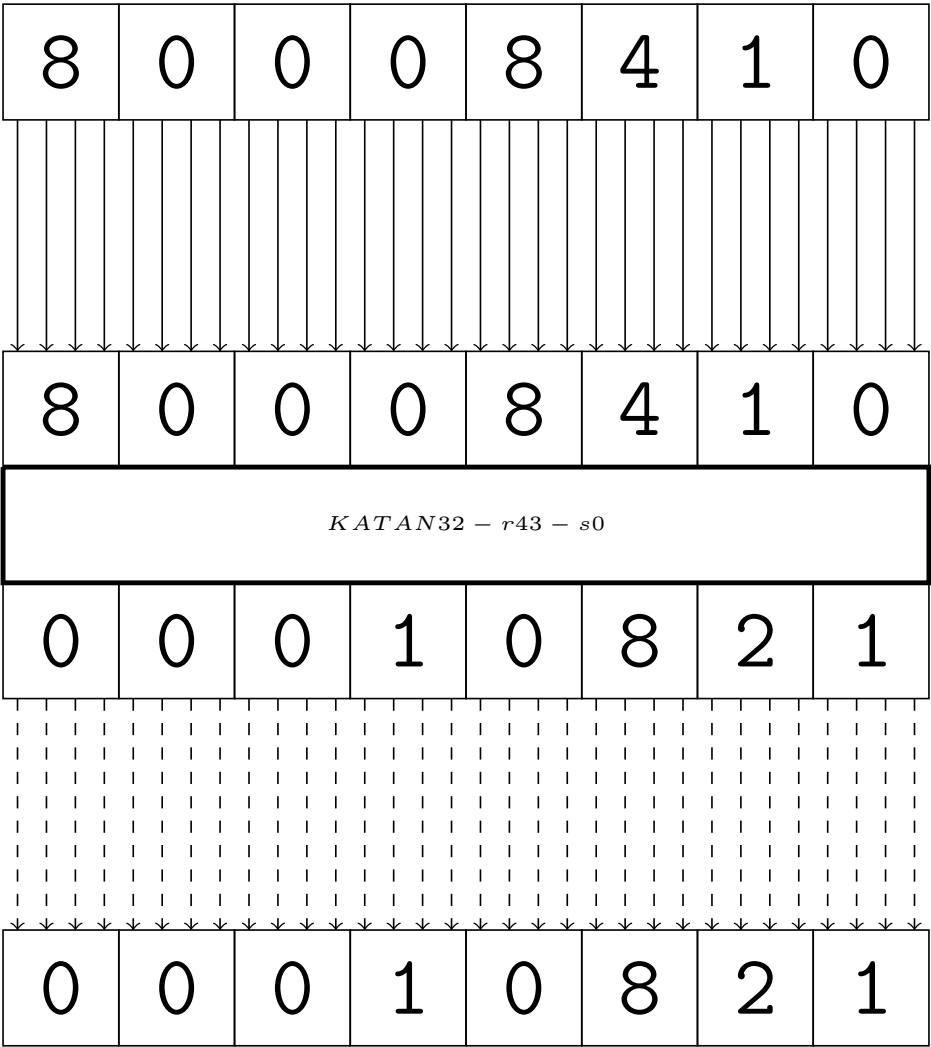
87 KATAN32-r42-s0

Maximal differential probability: $2^{-1.0}$



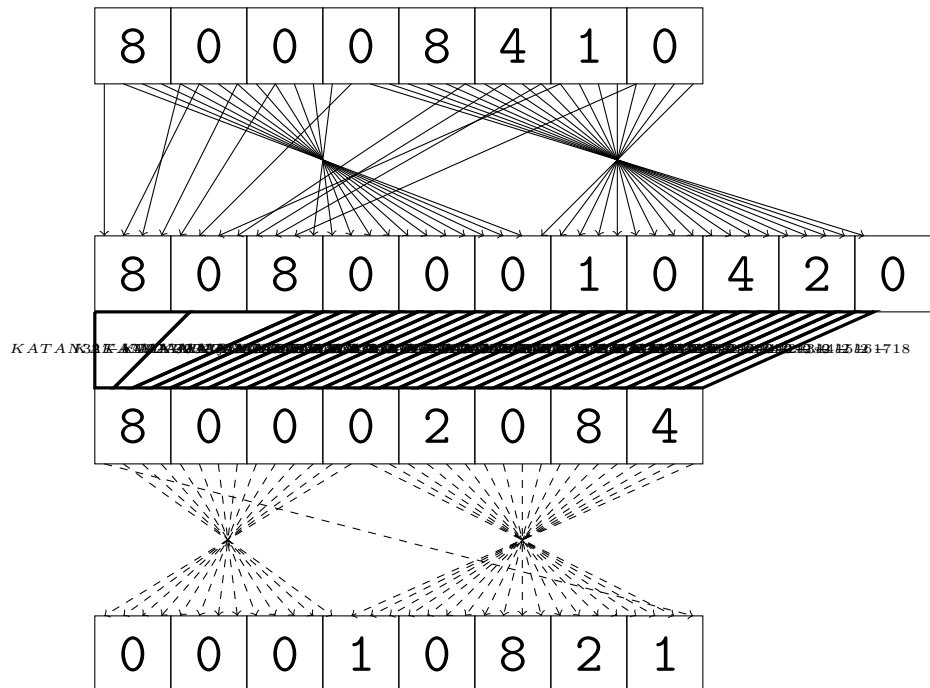
88 **KATAN32-r43**

Maximal differential probability: $2^{-1.0}$



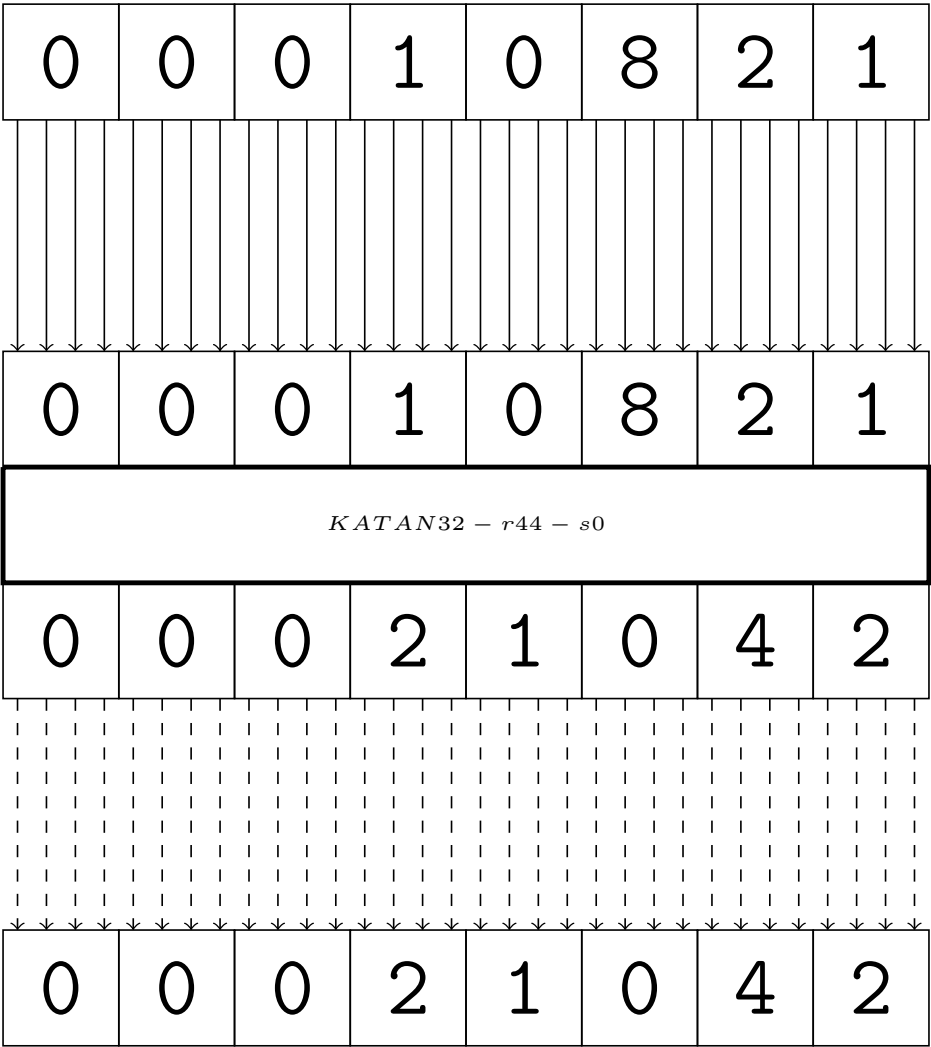
89 KATAN32-r43-s0

Maximal differential probability: $2^{-1.0}$



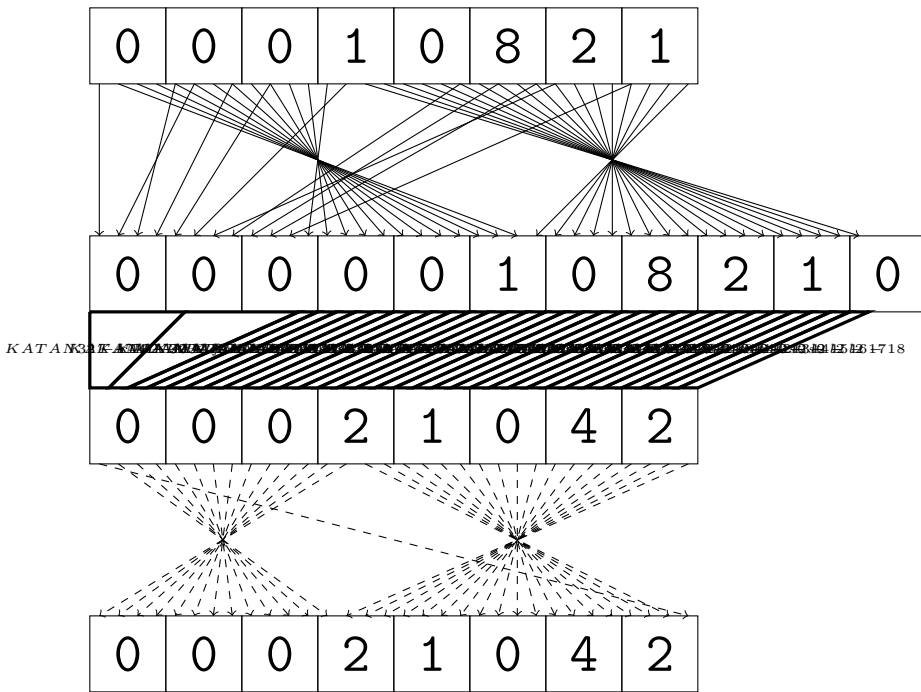
90 KATAN32-r44

Maximal differential probability: $2^{-0.0}$



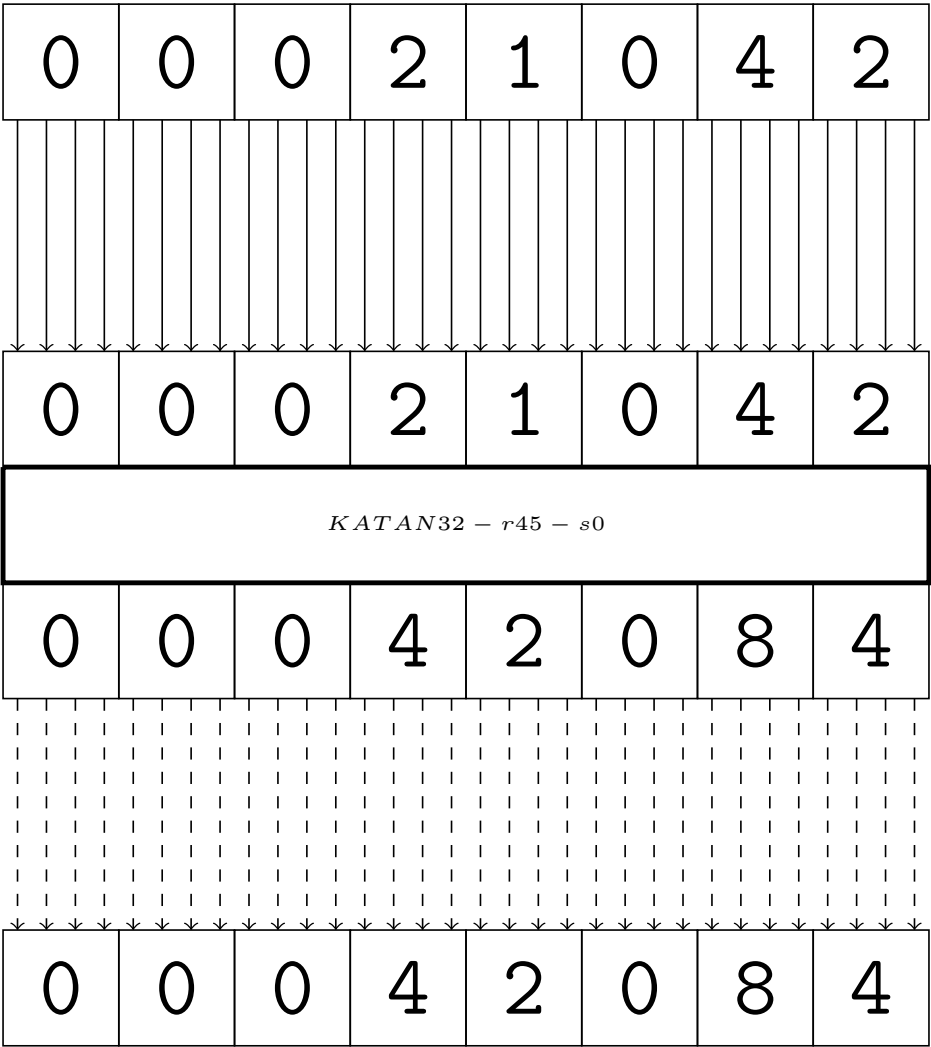
91 KATAN32-r44-s0

Maximal differential probability: $2^{-0.0}$



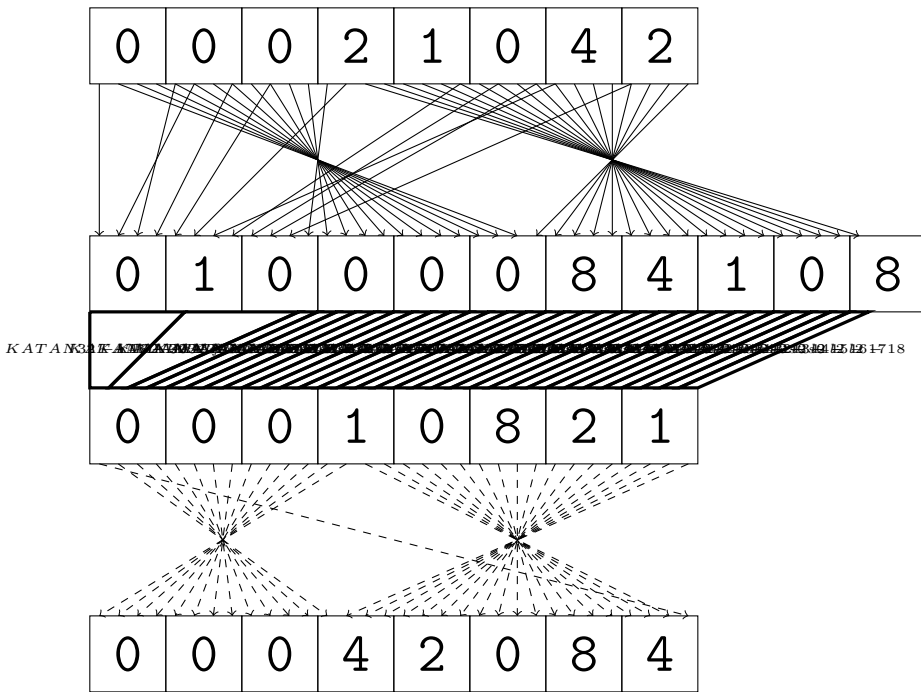
92 KATAN32-r45

Maximal differential probability: $2^{-1.0}$



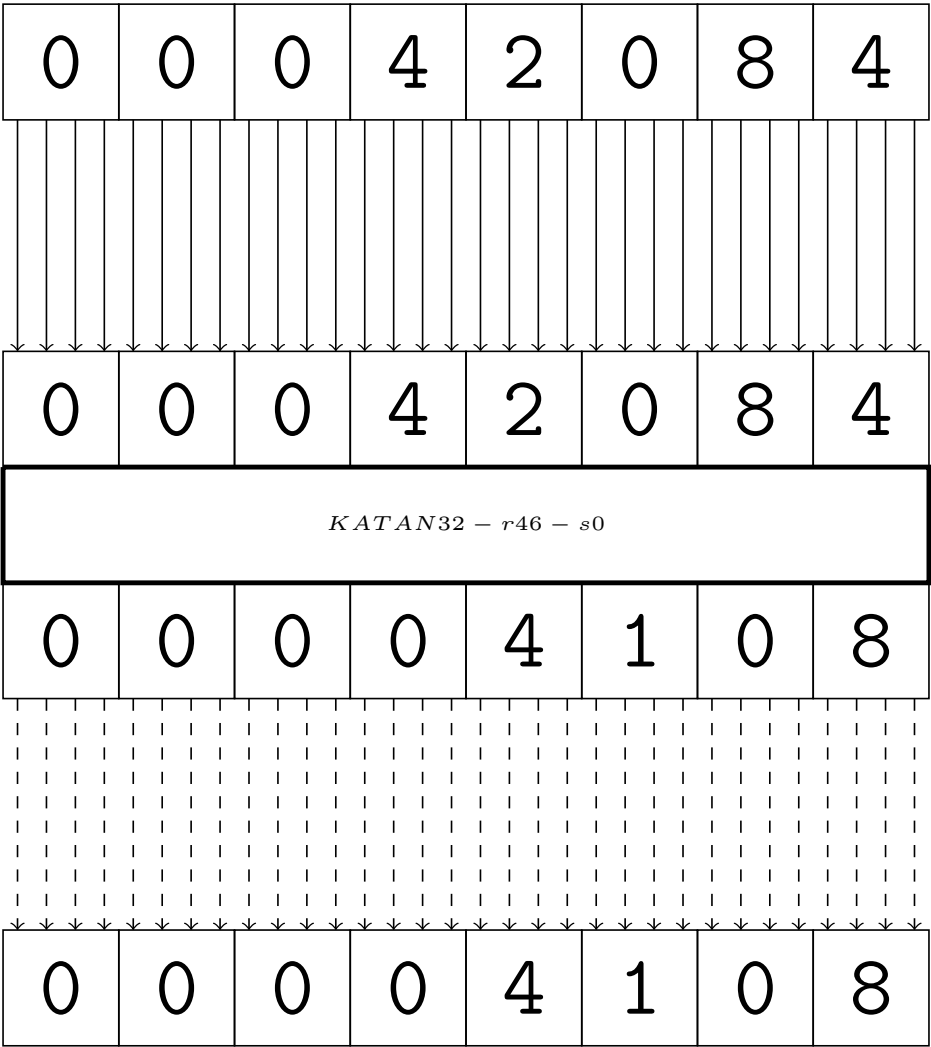
93 KATAN32-r45-s0

Maximal differential probability: $2^{-1.0}$



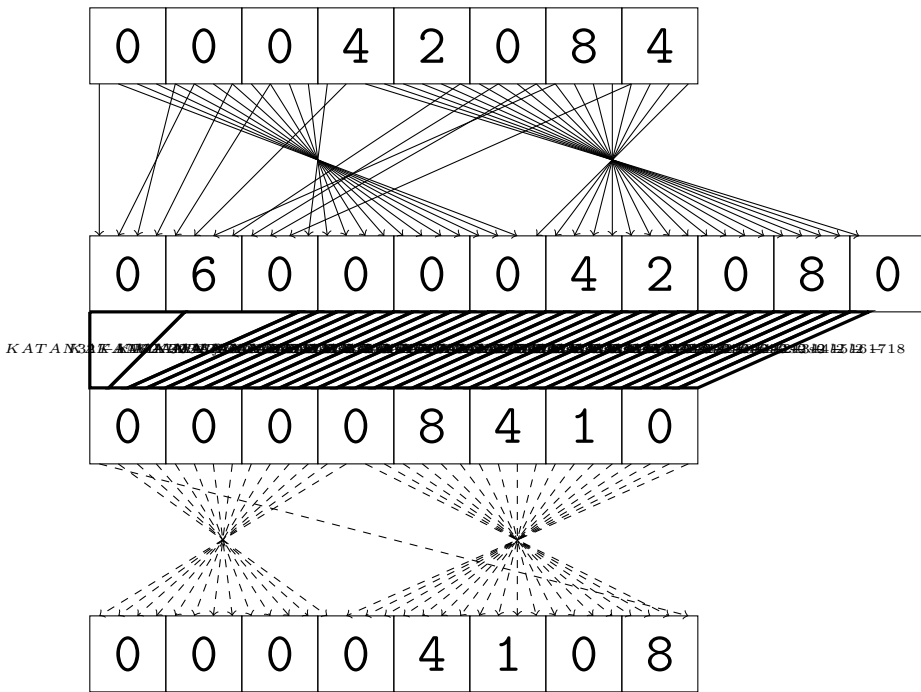
94 KATAN32-r46

Maximal differential probability: $2^{-0.0}$



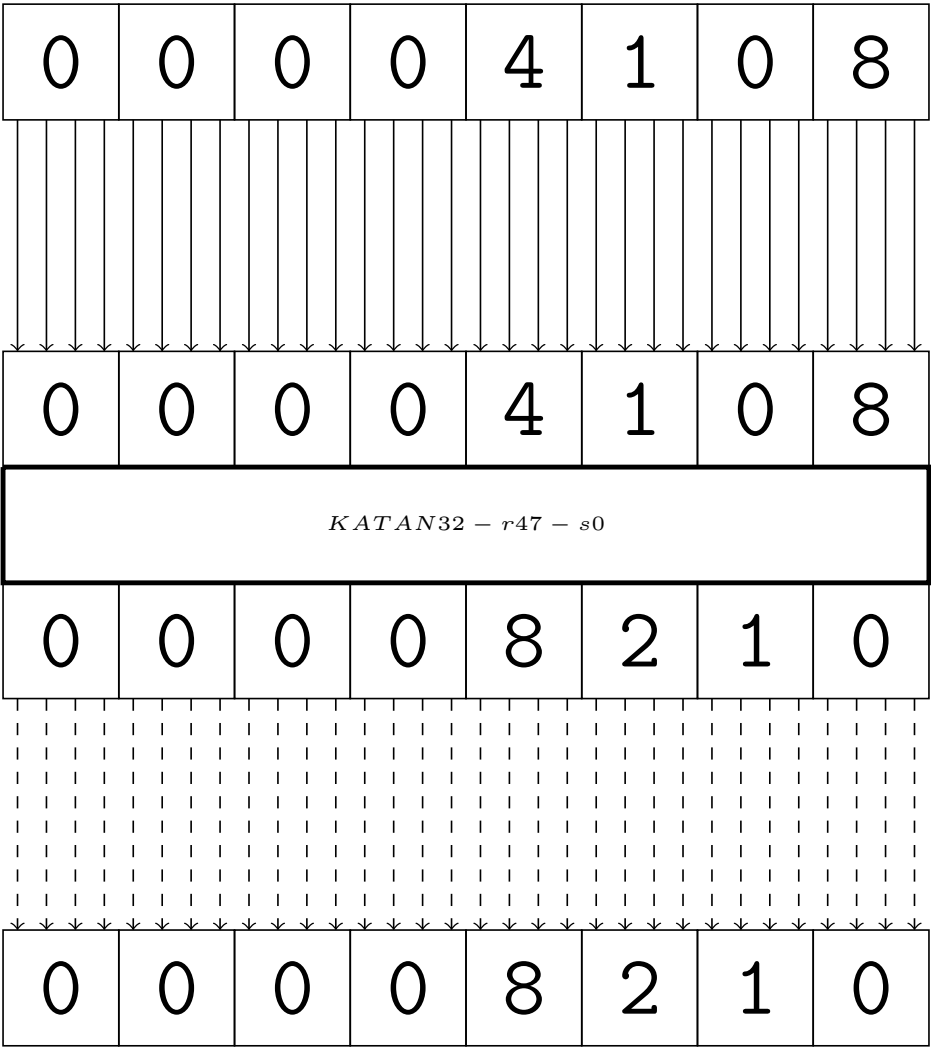
95 KATAN32-r46-s0

Maximal differential probability: $2^{-0.0}$



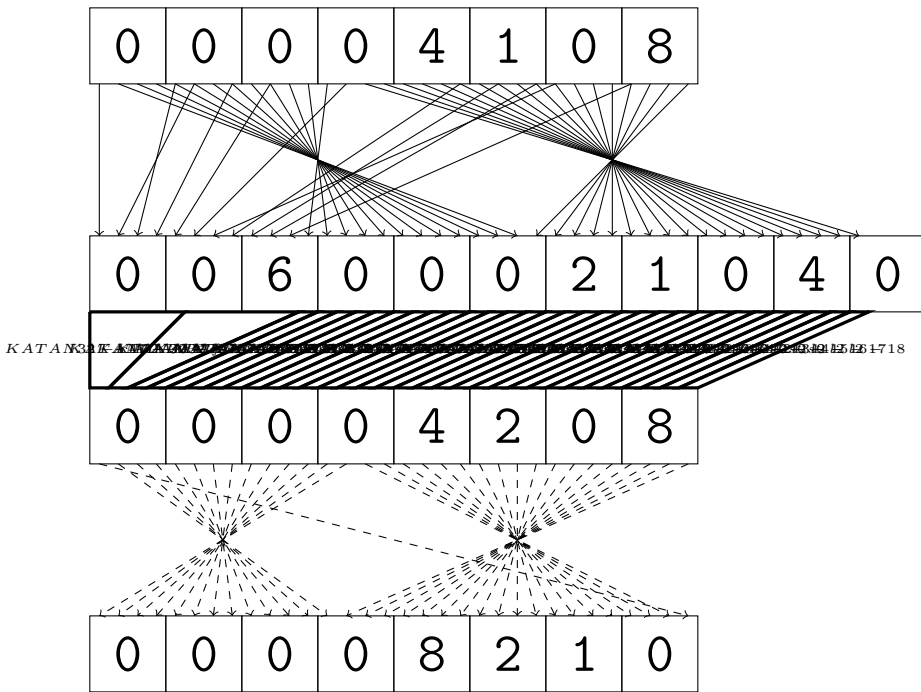
96 KATAN32-r47

Maximal differential probability: $2^{-1.0}$



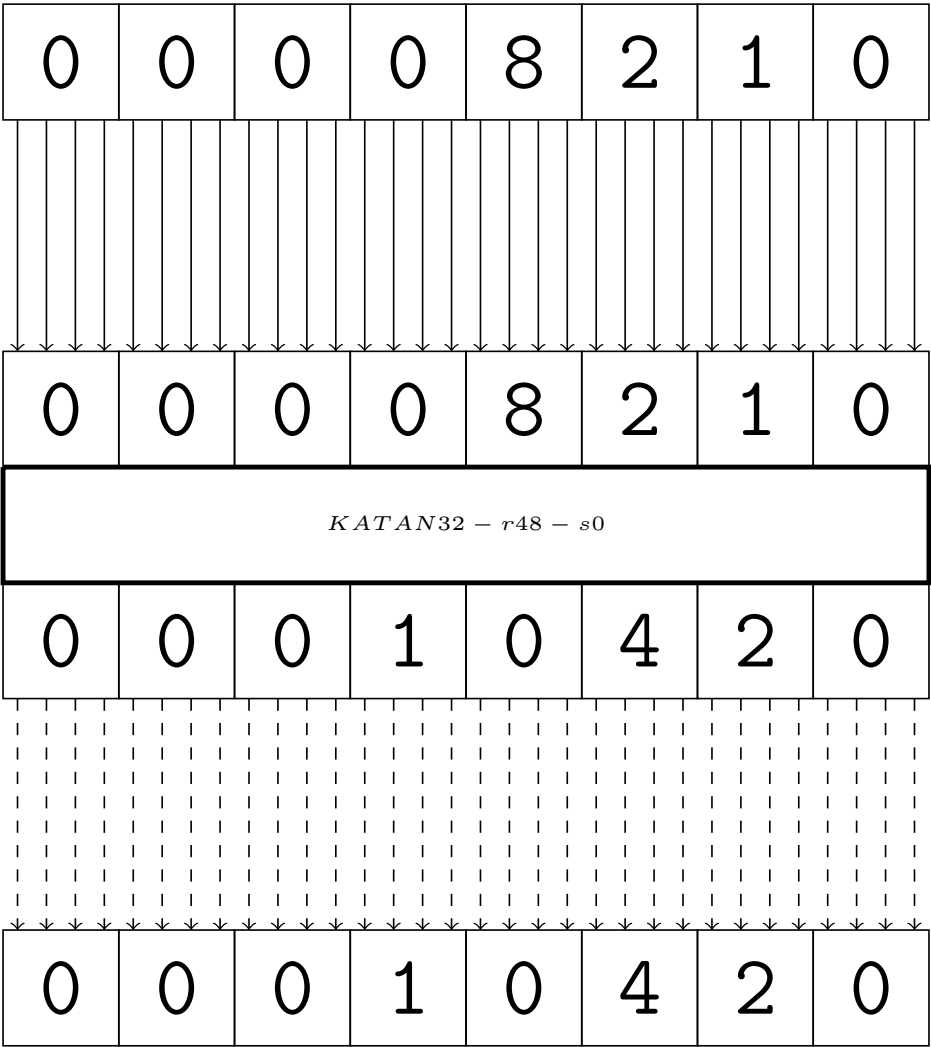
97 KATAN32-r47-s0

Maximal differential probability: $2^{-1.0}$



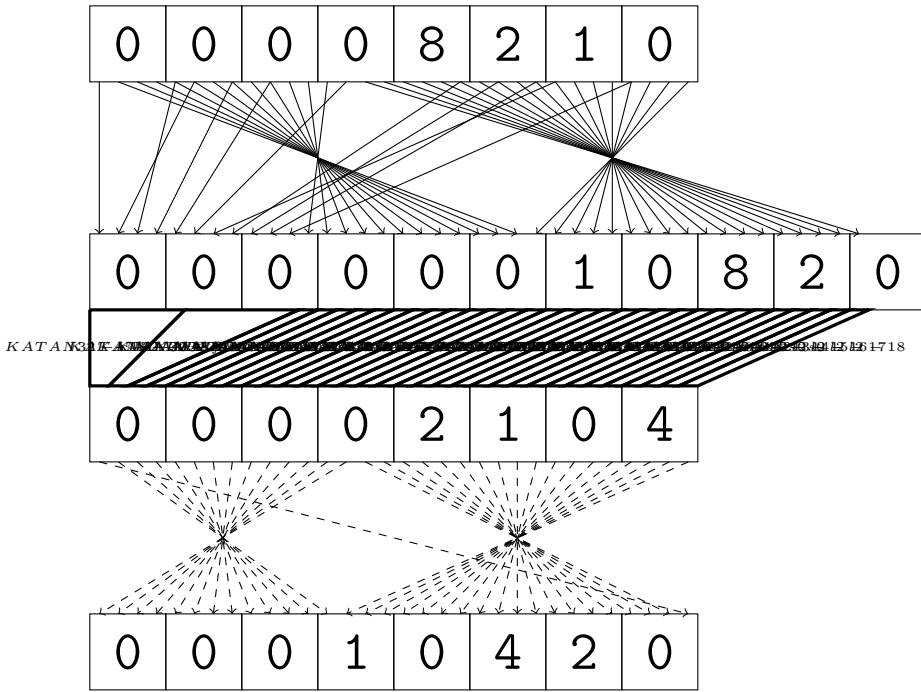
98 KATAN32-r48

Maximal differential probability: $2^{-0.0}$



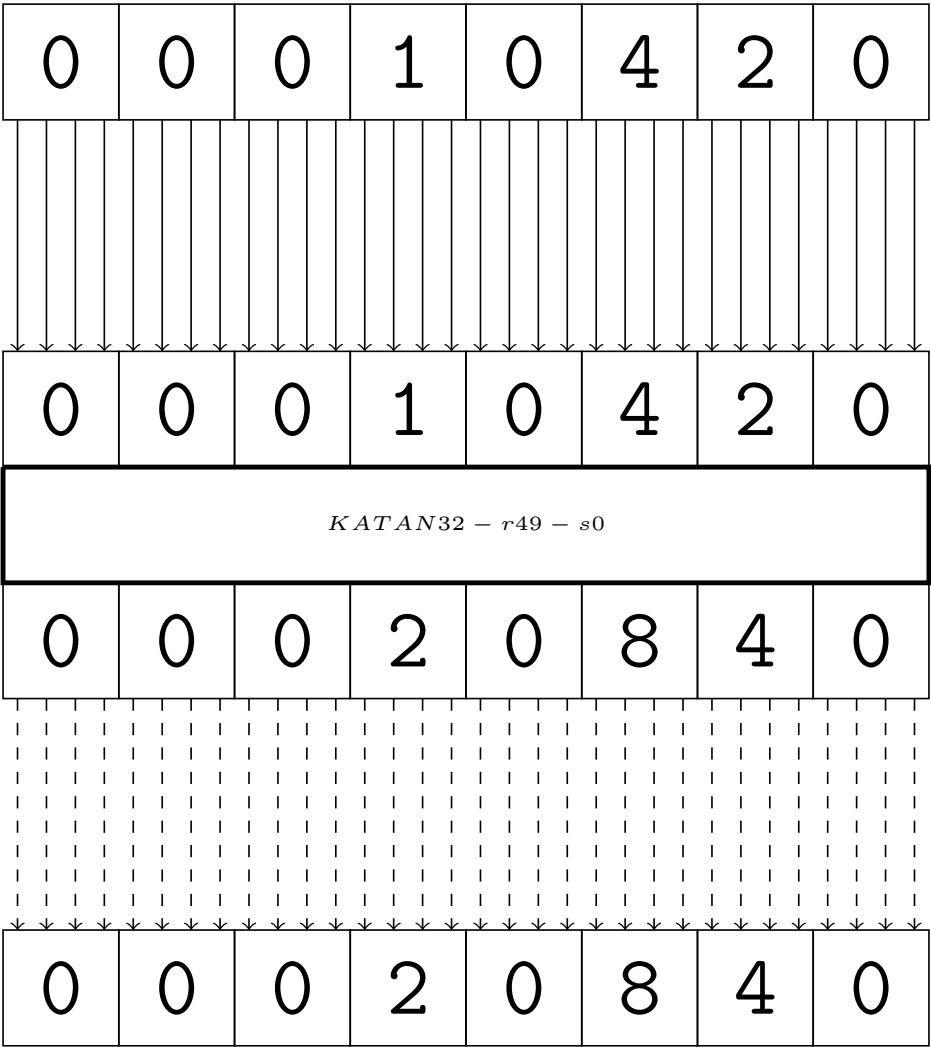
99 KATAN32-r48-s0

Maximal differential probability: $2^{-0.0}$



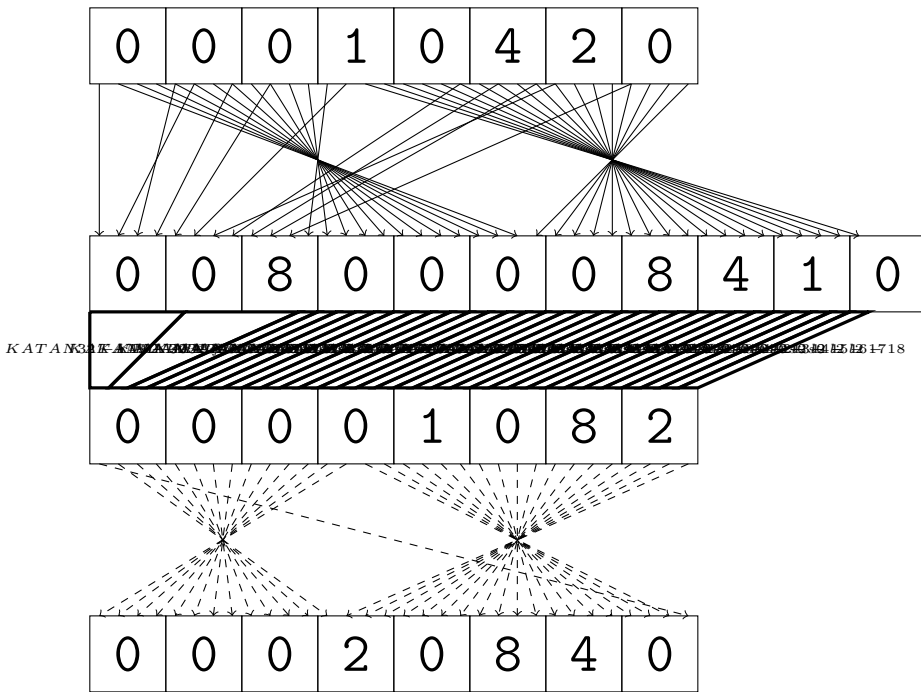
100 KATAN32-r49

Maximal differential probability: $2^{-1.0}$



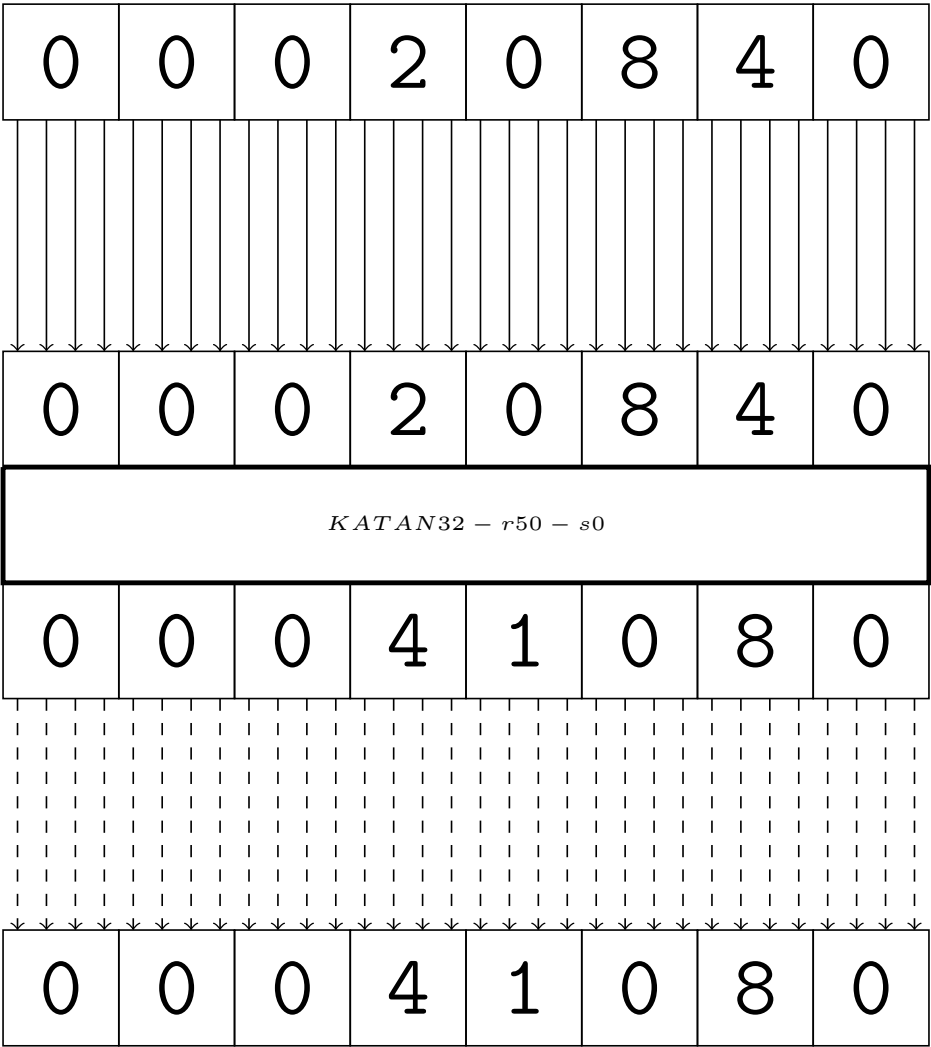
101 KATAN32-r49-s0

Maximal differential probability: $2^{-1.0}$



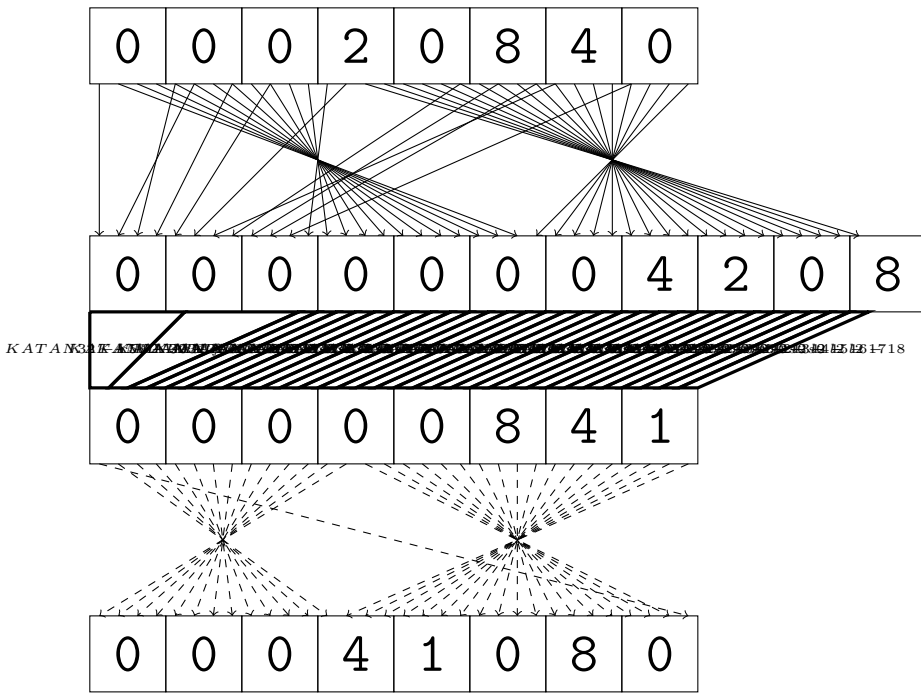
102 **KATAN32-r50**

Maximal differential probability: $2^{-0.0}$



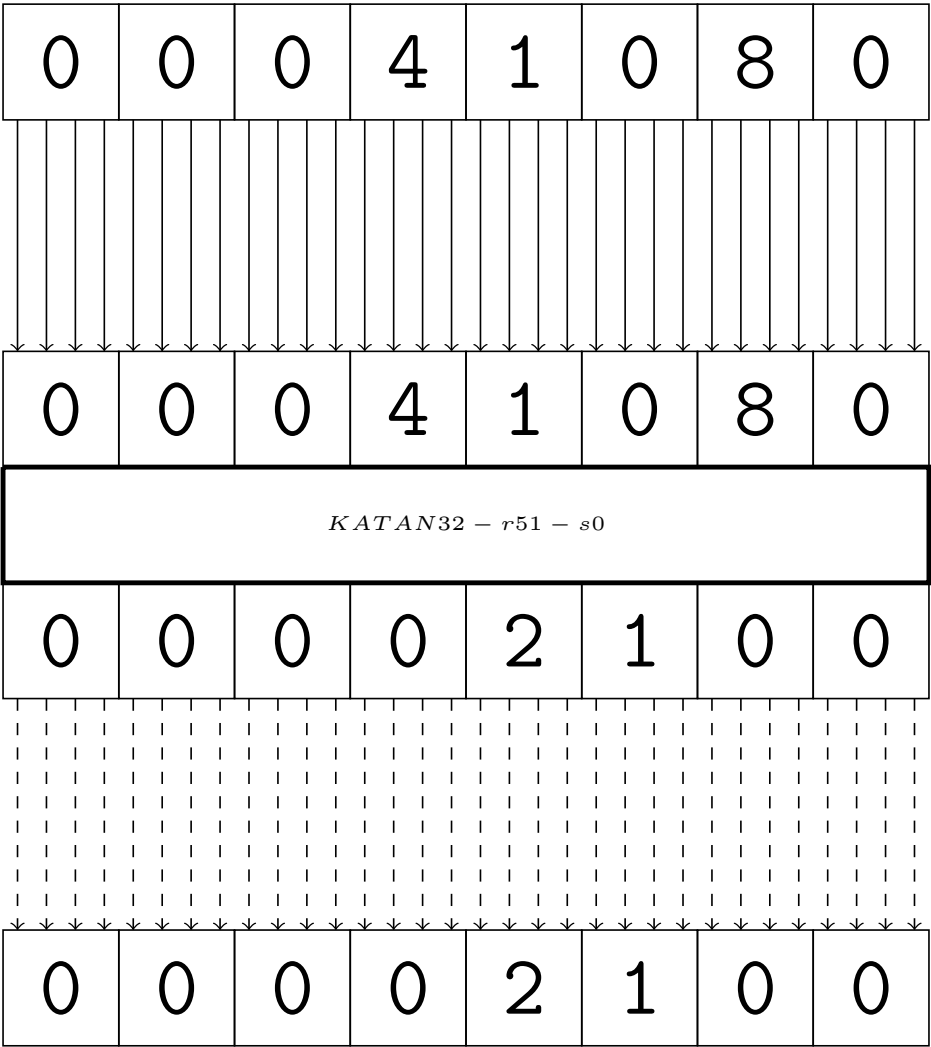
103 KATAN32-r50-s0

Maximal differential probability: $2^{-0.0}$



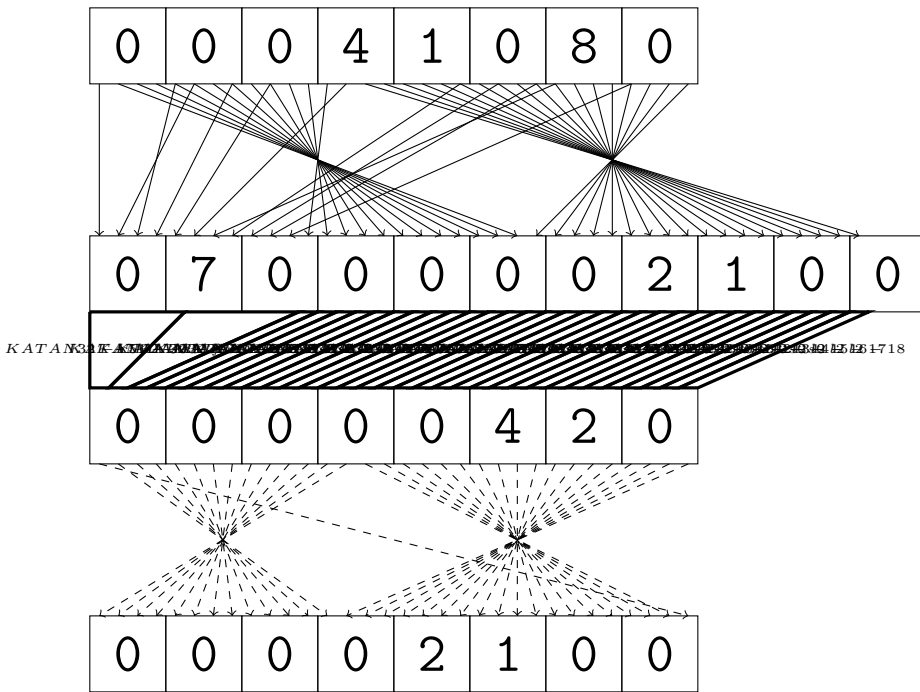
104 **KATAN32-r51**

Maximal differential probability: $2^{-1.0}$



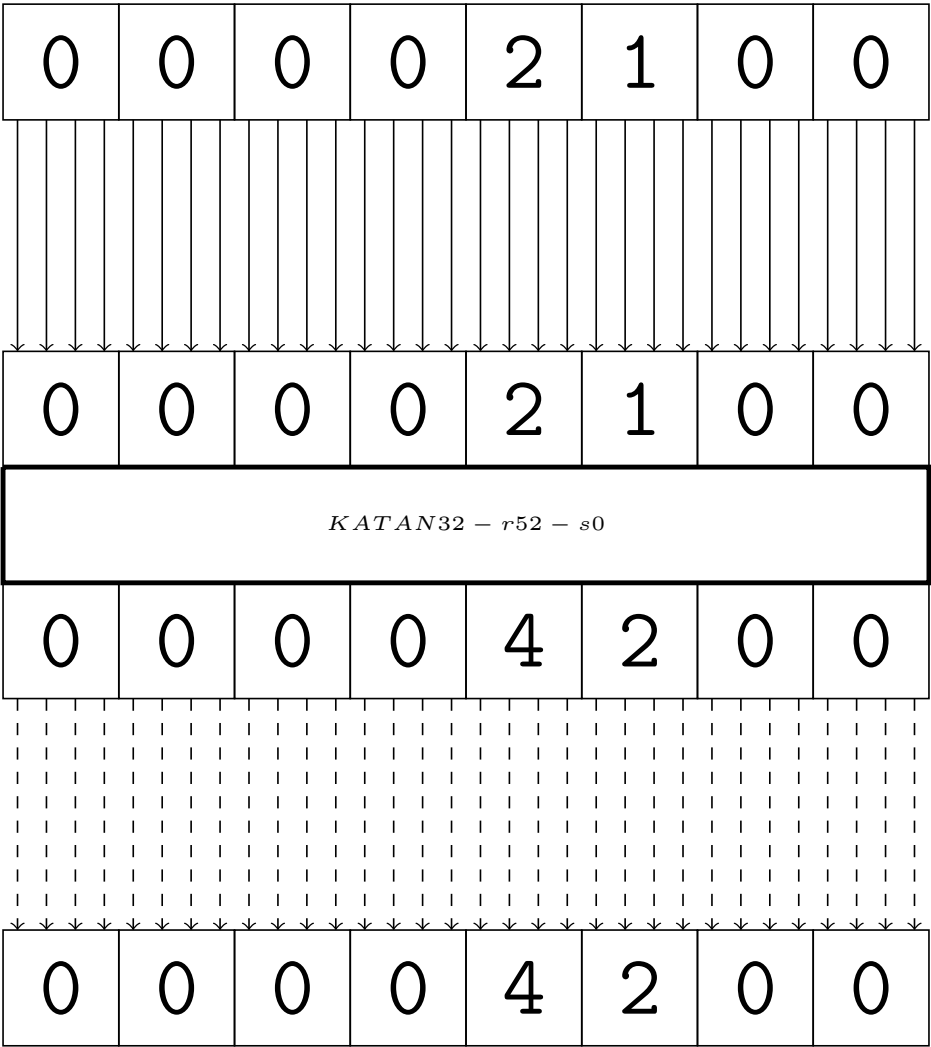
105 KATAN32-r51-s0

Maximal differential probability: $2^{-1.0}$



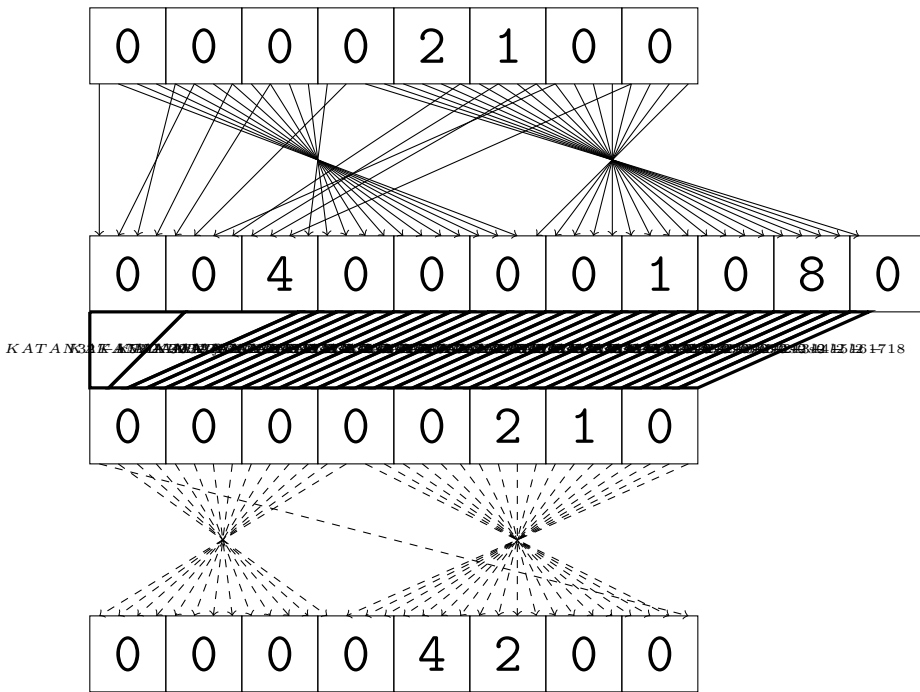
106 **KATAN32-r52**

Maximal differential probability: $2^{-1.0}$



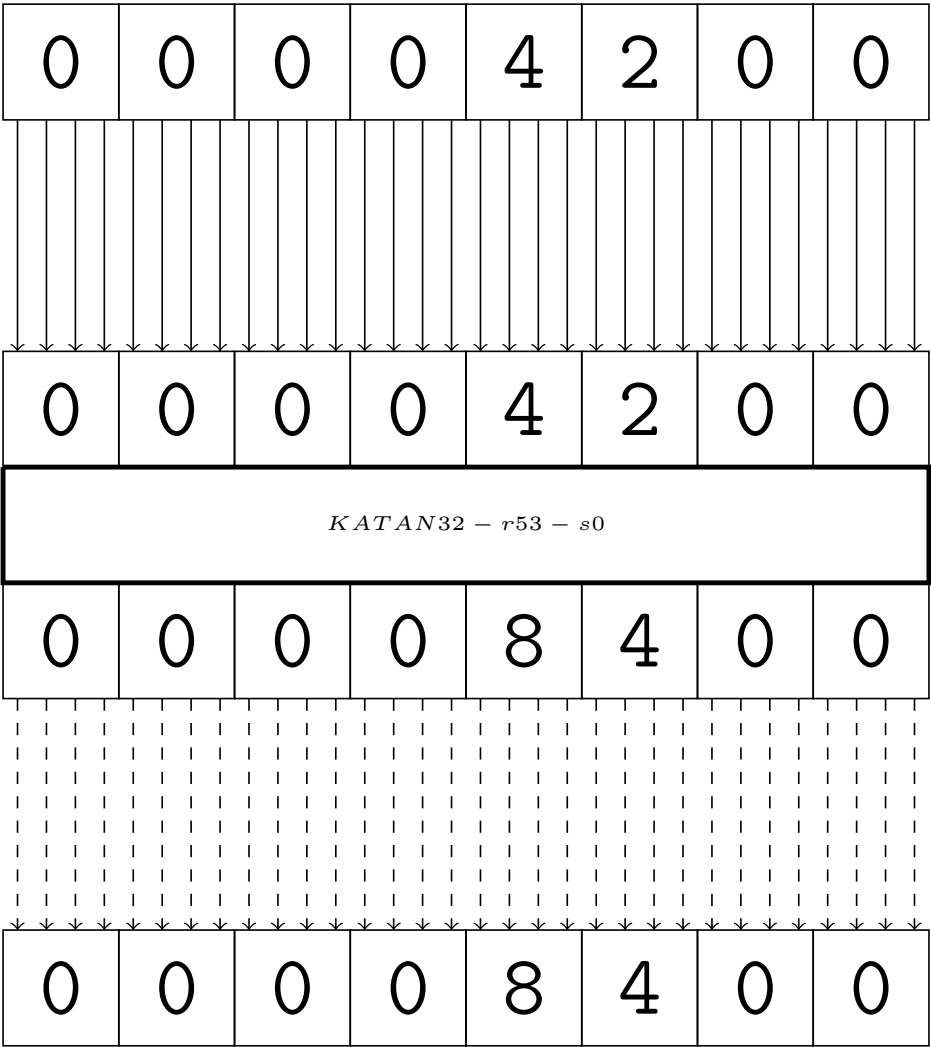
107 KATAN32-r52-s0

Maximal differential probability: $2^{-1.0}$



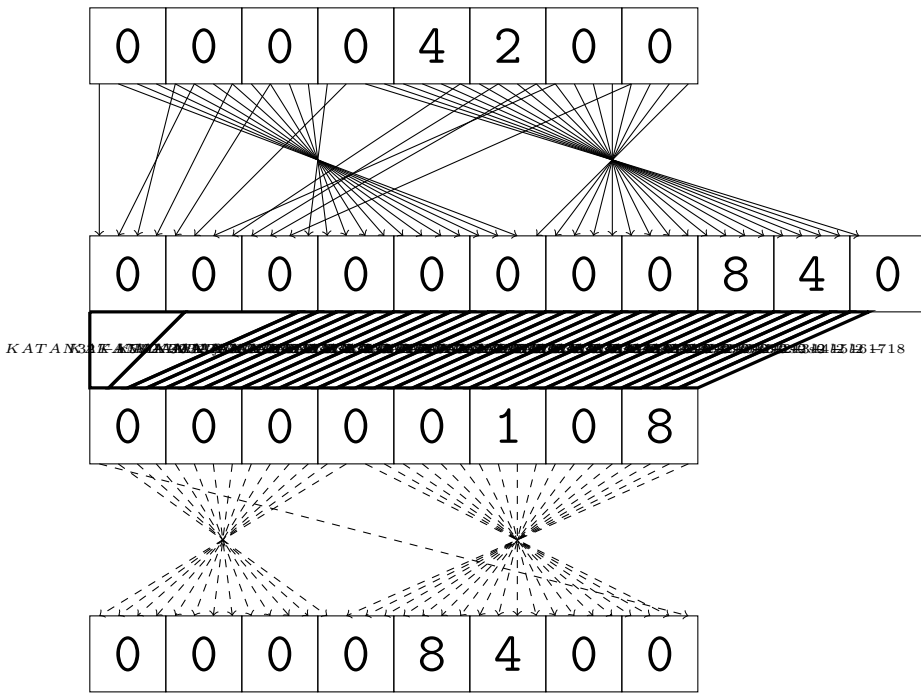
108 **KATAN32-r53**

Maximal differential probability: $2^{-0.0}$



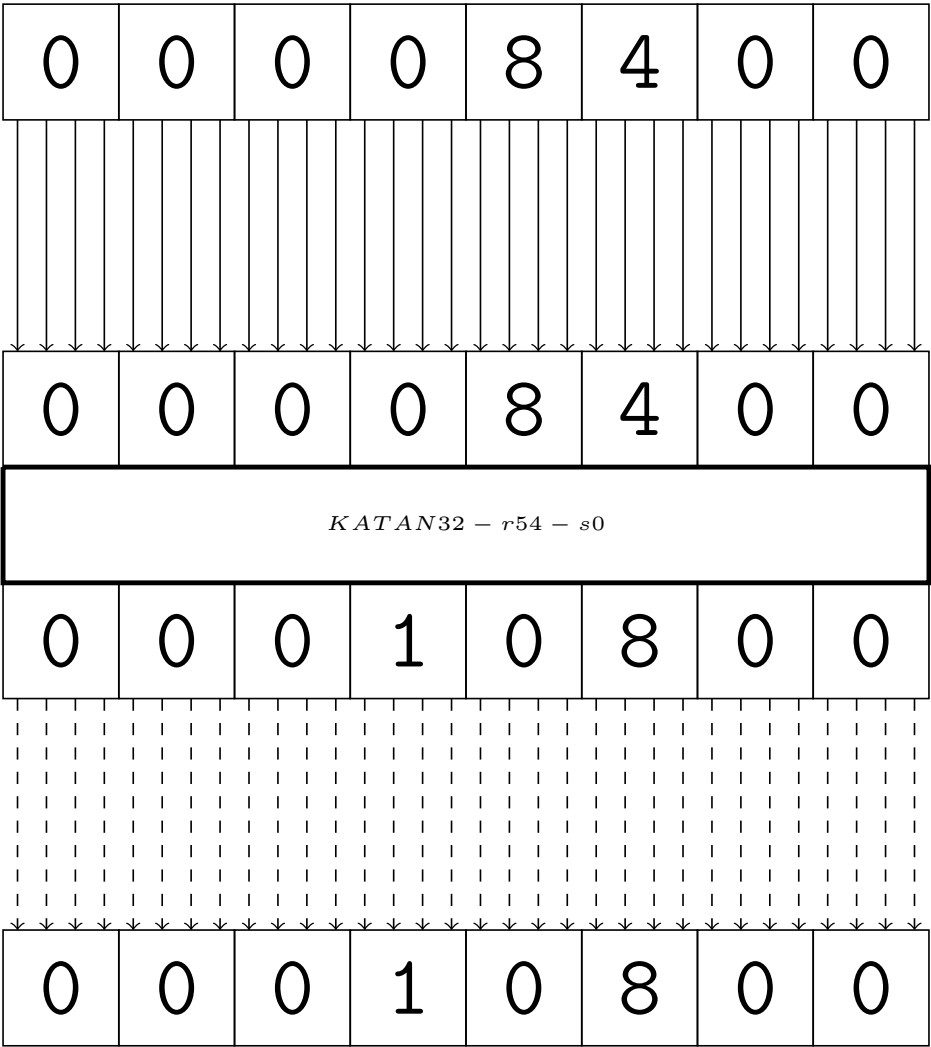
109 KATAN32-r53-s0

Maximal differential probability: $2^{-0.0}$



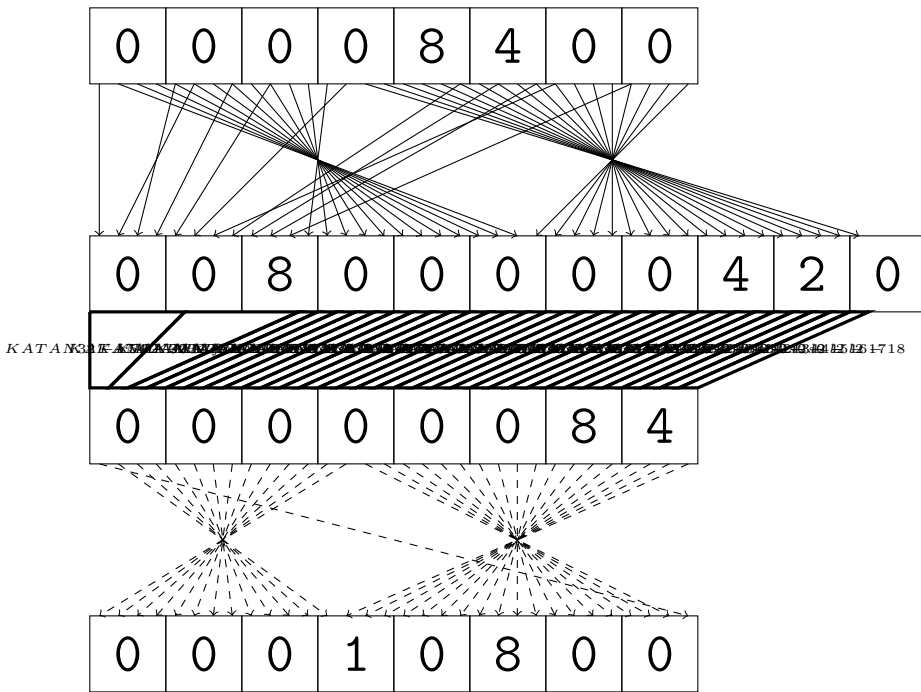
110 KATAN32-r54

Maximal differential probability: $2^{-1.0}$



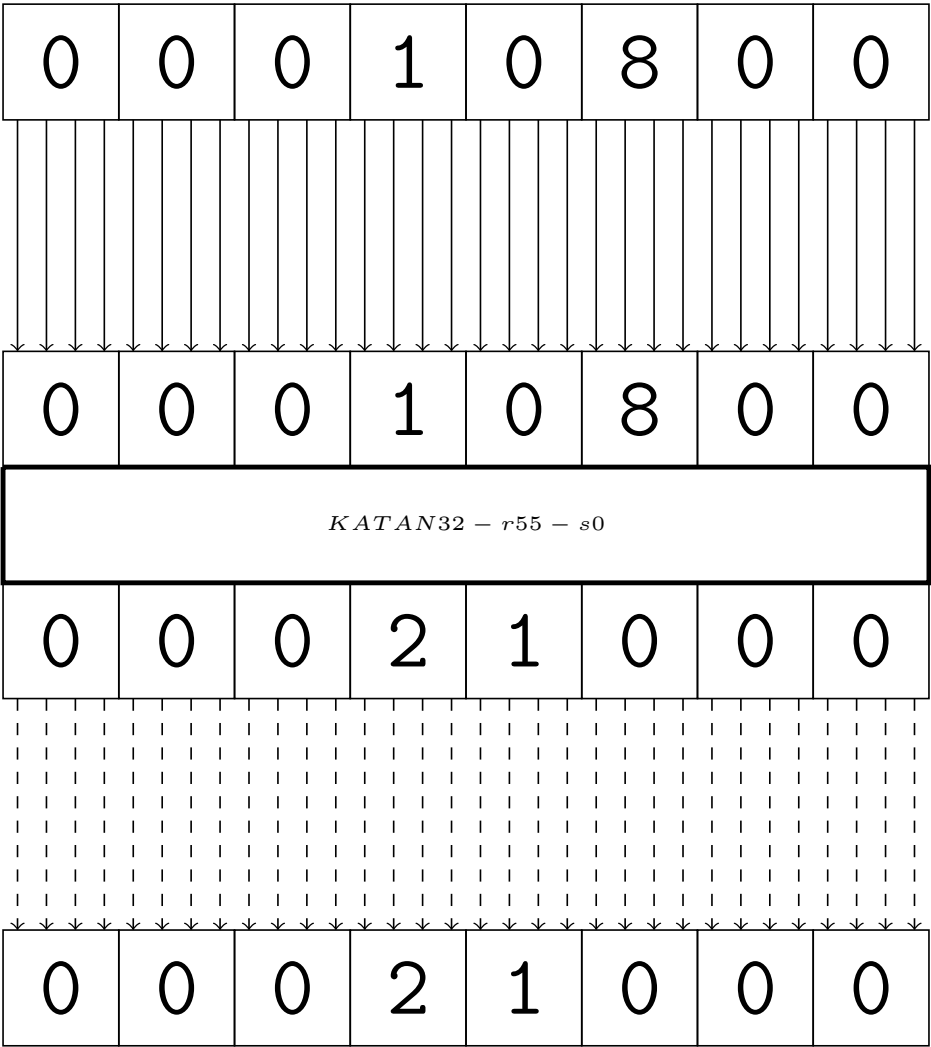
111 KATAN32-r54-s0

Maximal differential probability: $2^{-1.0}$



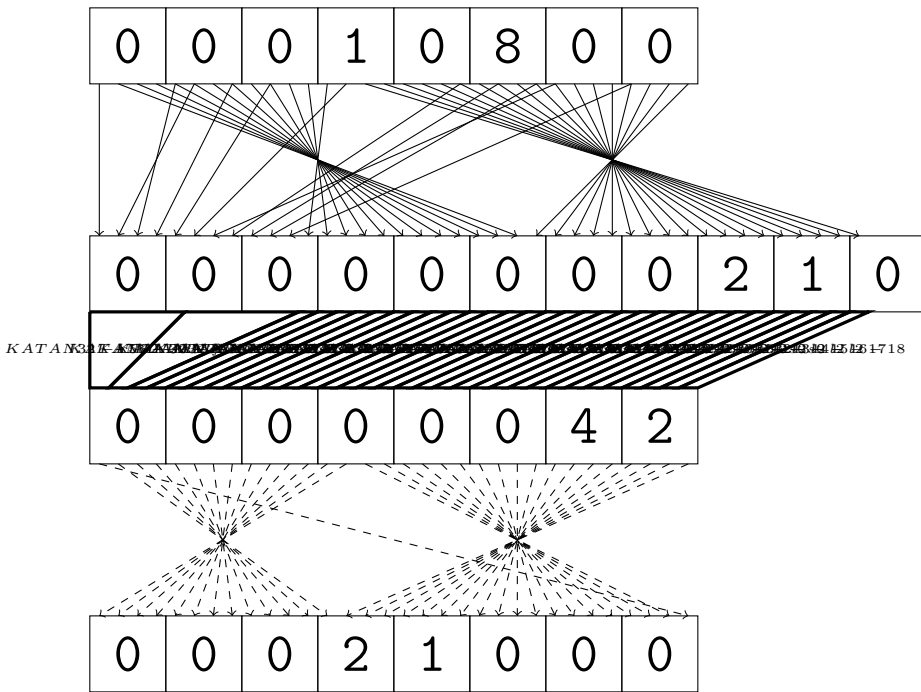
112 KATAN32-r55

Maximal differential probability: $2^{-0.0}$



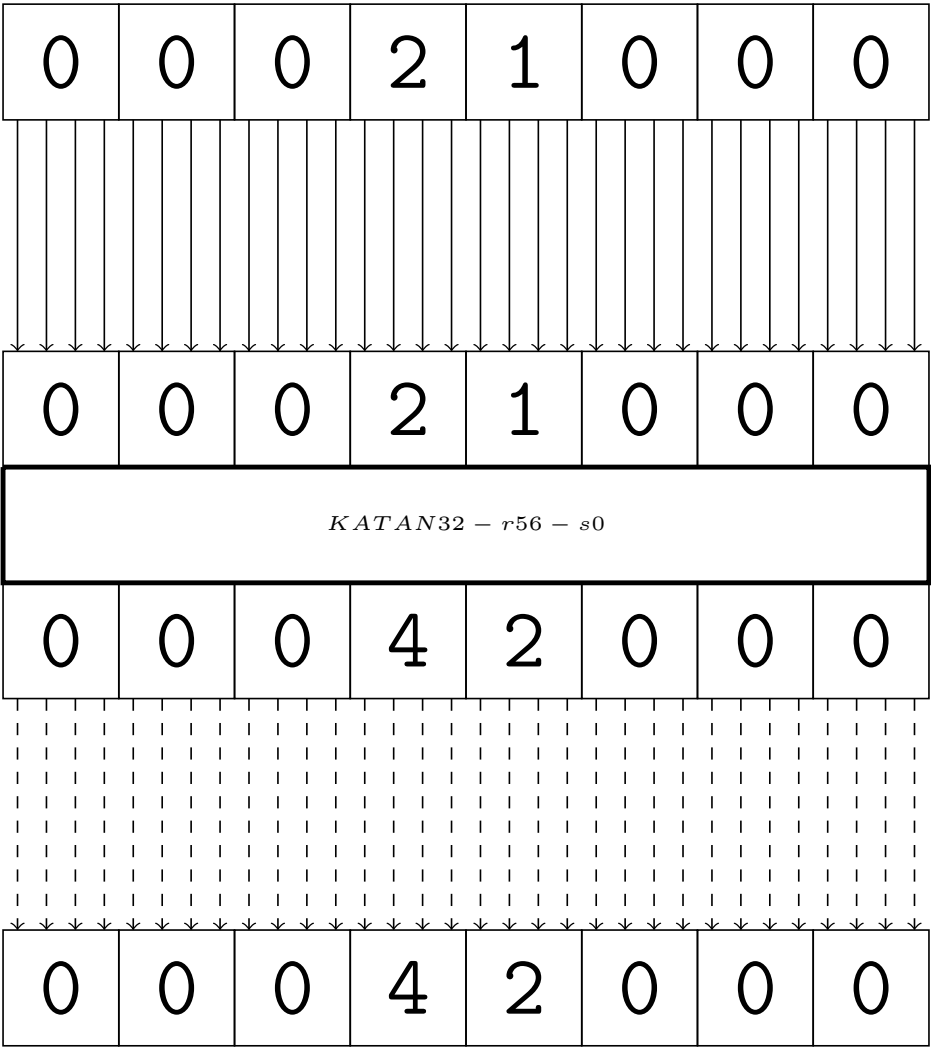
113 KATAN32-r55-s0

Maximal differential probability: $2^{-0.0}$



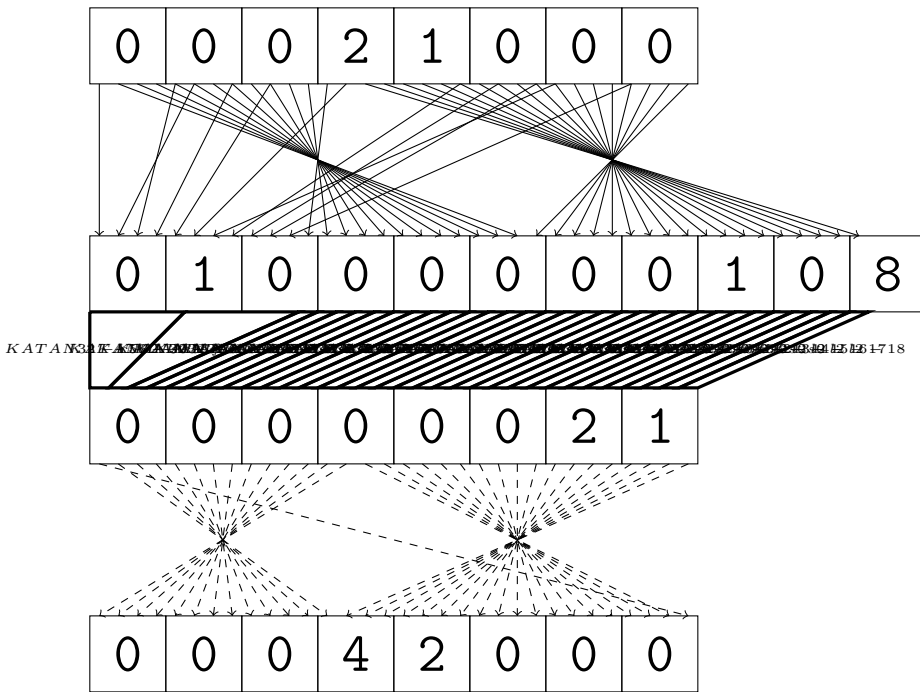
114 **KATAN32-r56**

Maximal differential probability: $2^{-1.0}$



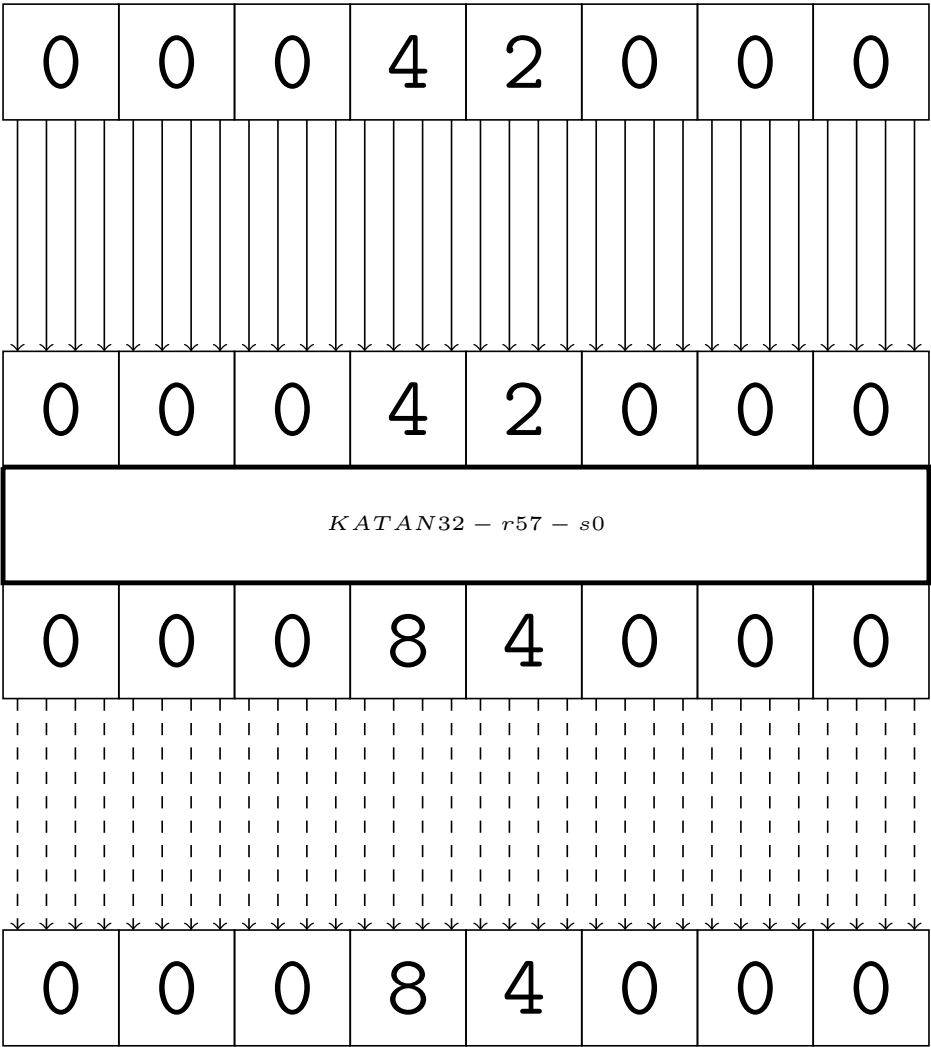
115 KATAN32-r56-s0

Maximal differential probability: $2^{-1.0}$



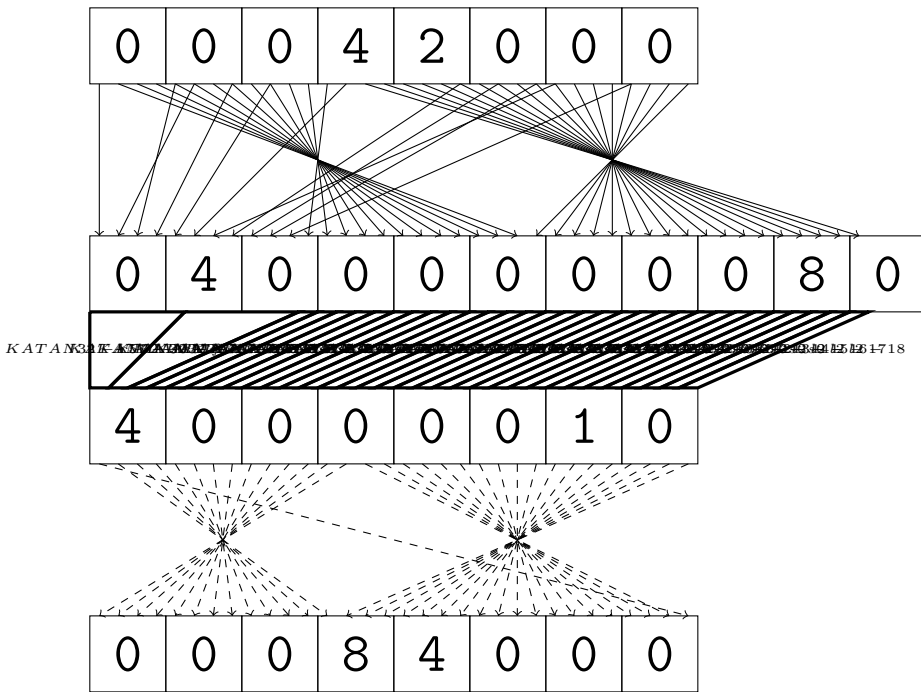
116 KATAN32-r57

Maximal differential probability: $2^{-0.0}$



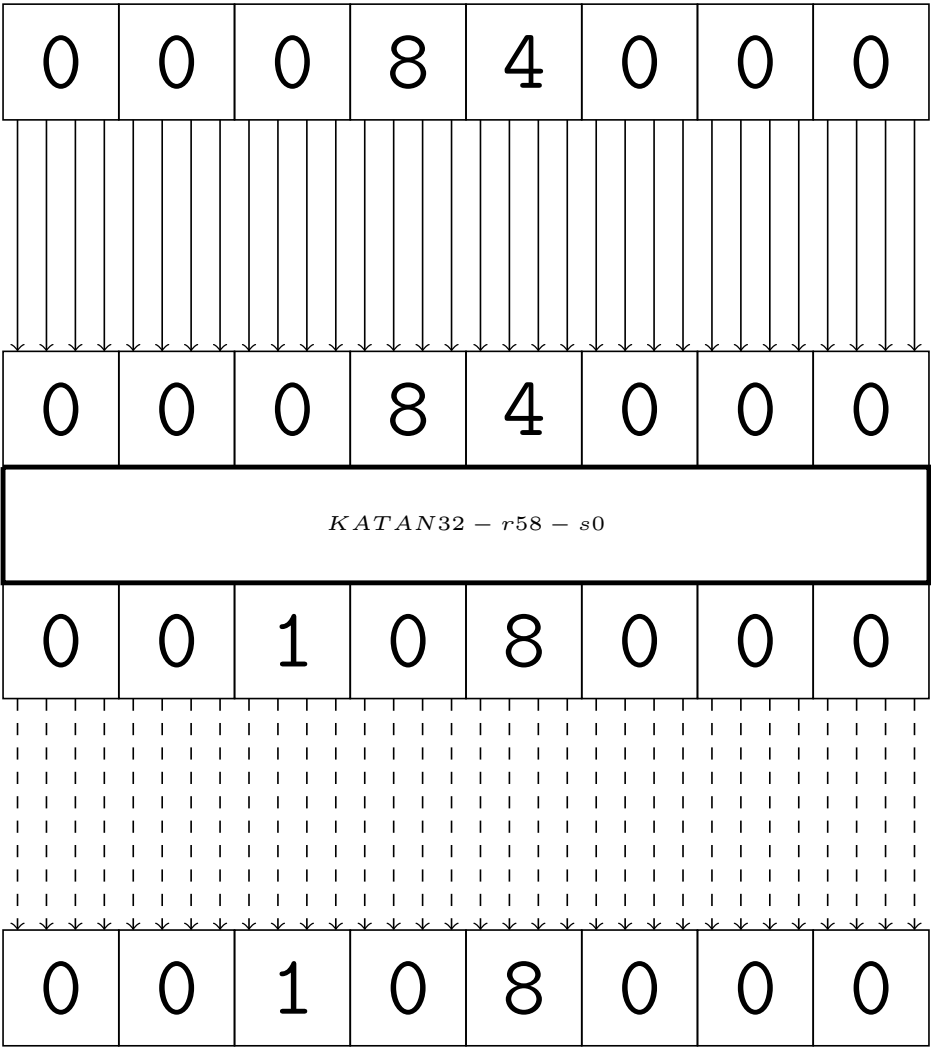
117 KATAN32-r57-s0

Maximal differential probability: $2^{-0.0}$



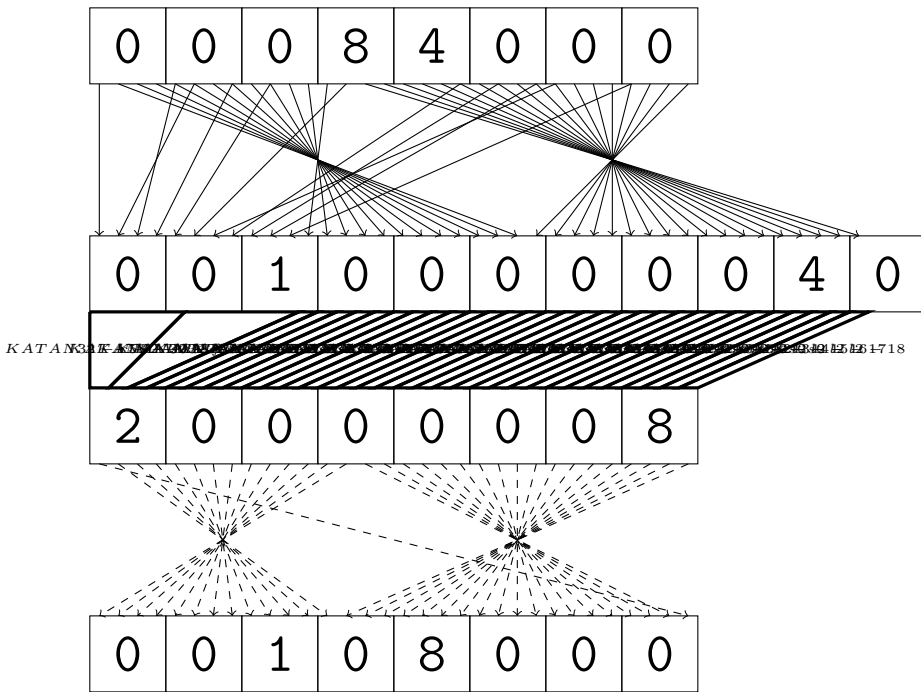
118 KATAN32-r58

Maximal differential probability: $2^{-0.0}$



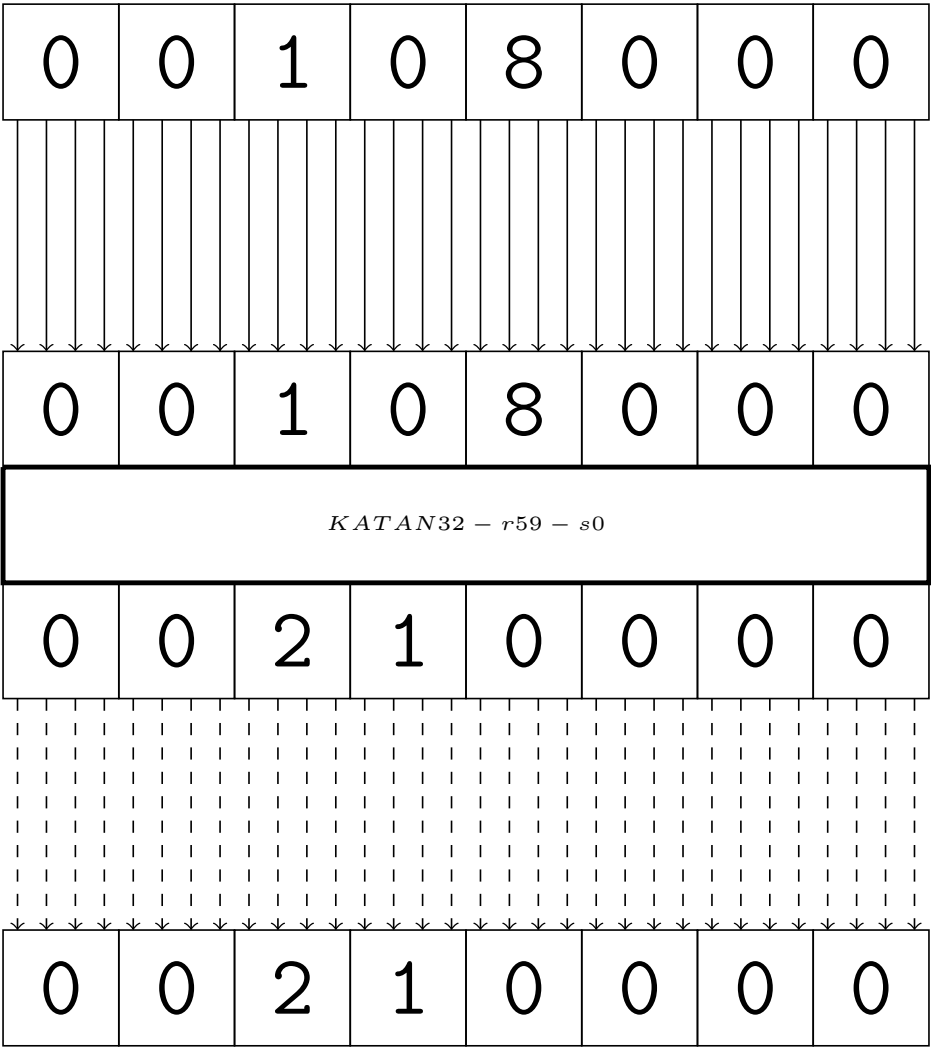
119 KATAN32-r58-s0

Maximal differential probability: $2^{-0.0}$



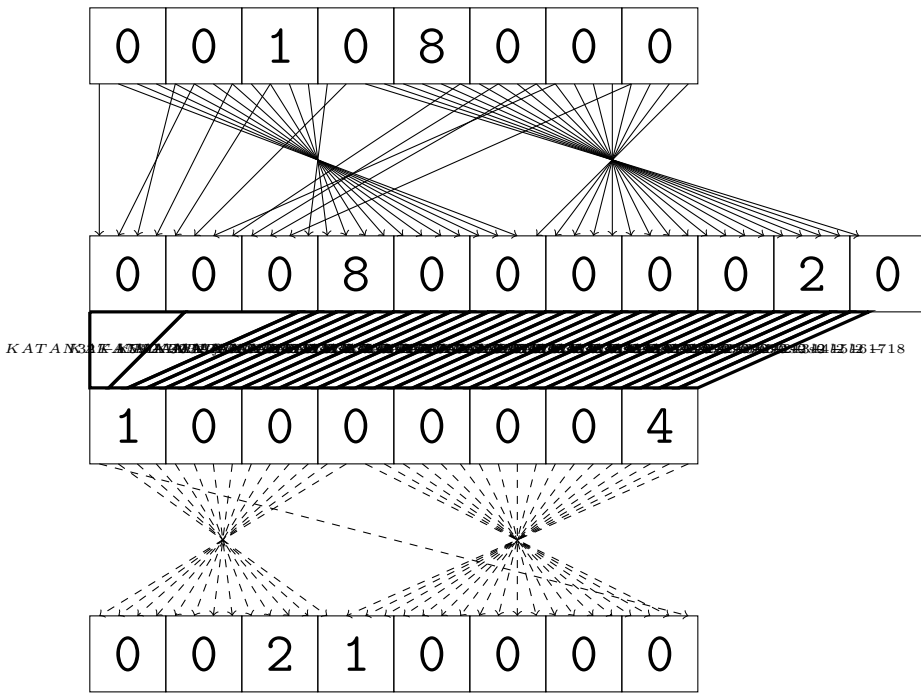
120 **KATAN32-r59**

Maximal differential probability: $2^{-0.0}$



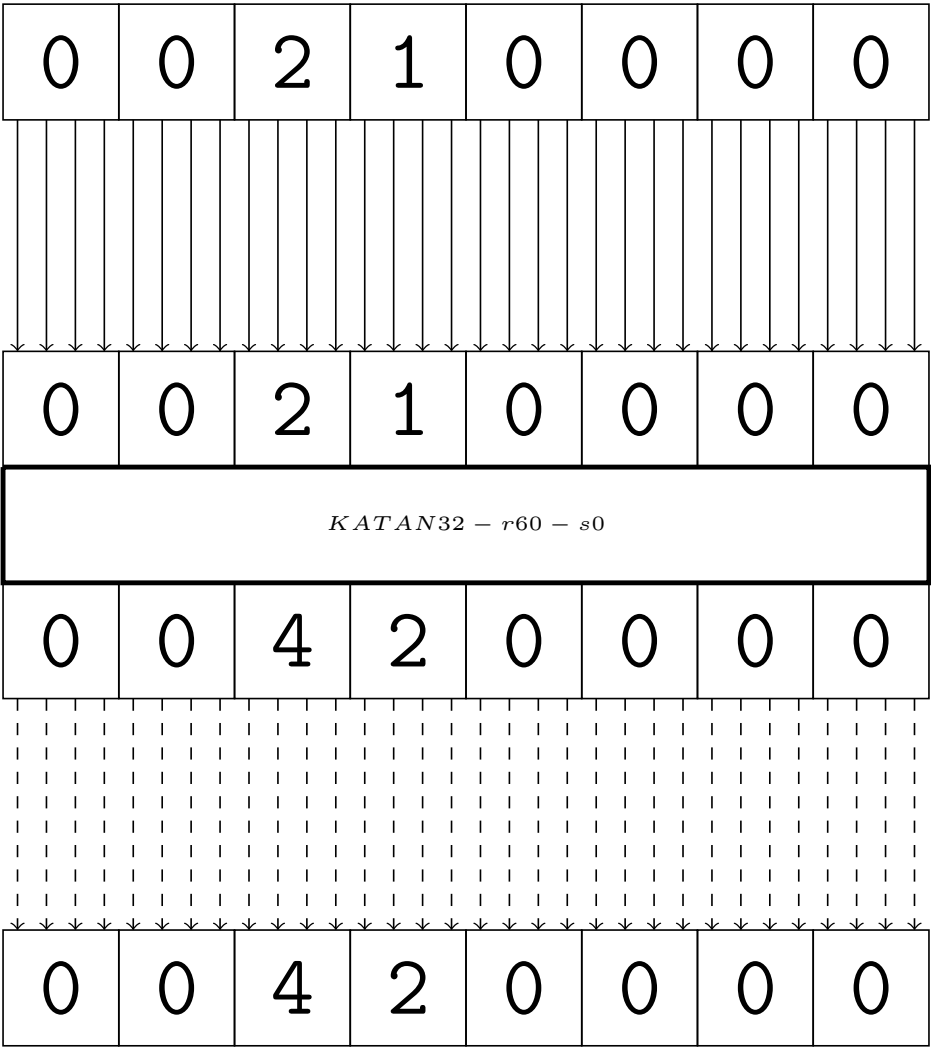
121 KATAN32-r59-s0

Maximal differential probability: $2^{-0.0}$



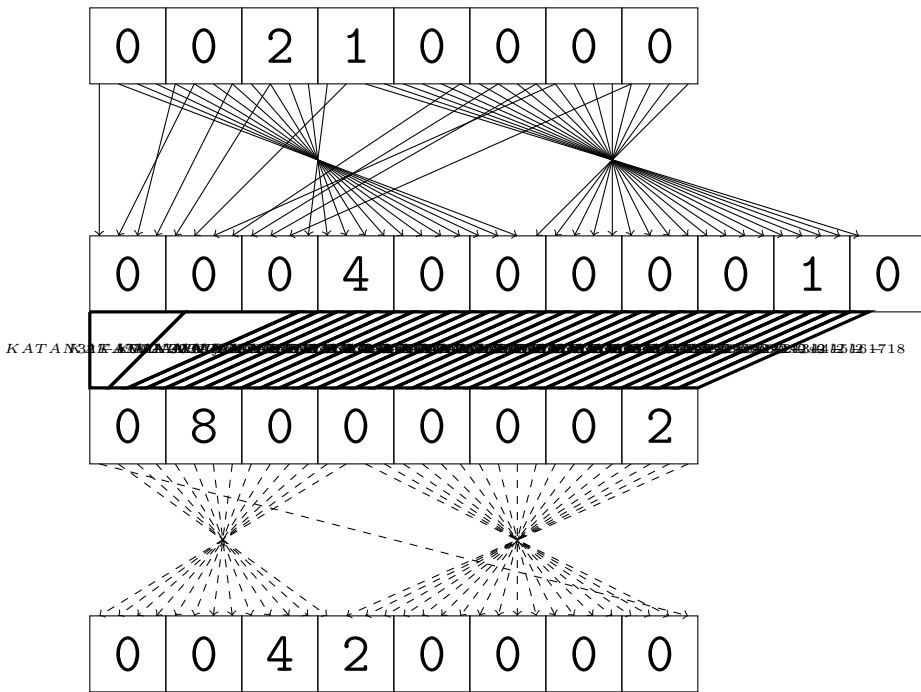
122 KATAN32-r60

Maximal differential probability: $2^{-0.0}$



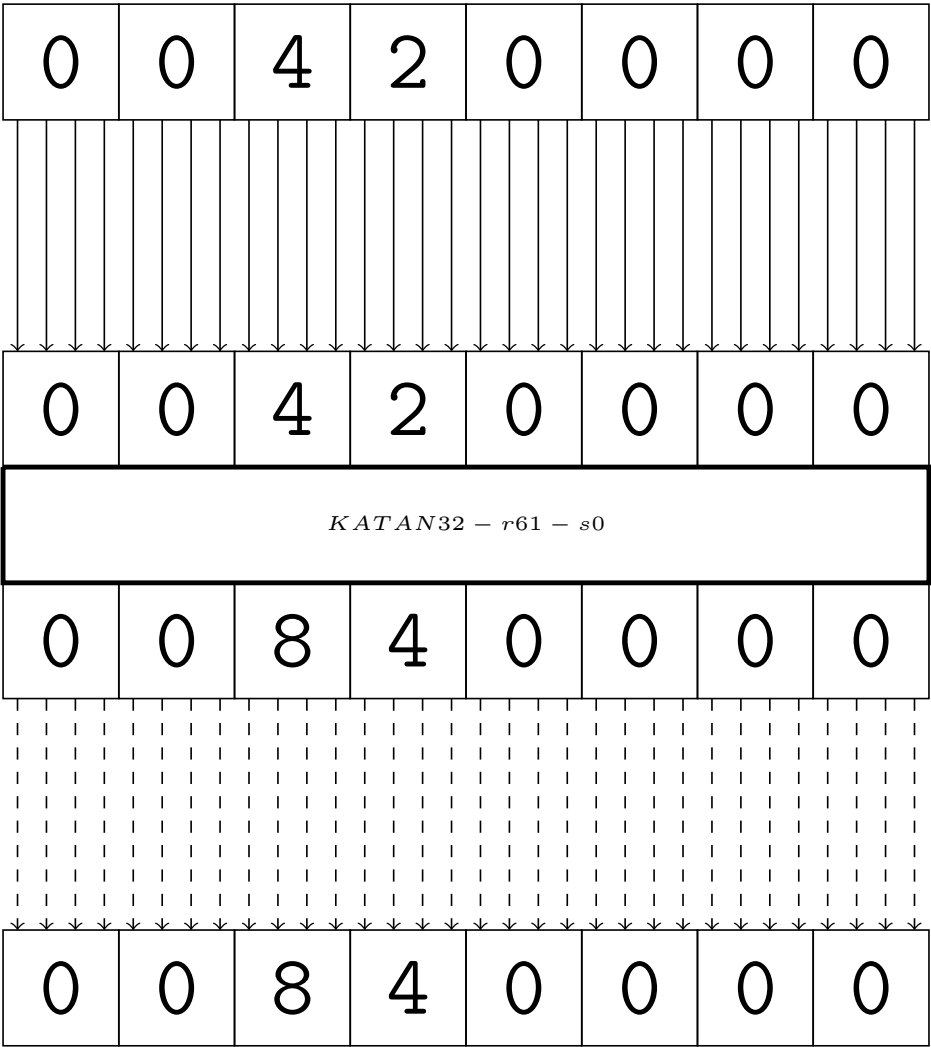
123 KATAN32-r60-s0

Maximal differential probability: $2^{-0.0}$



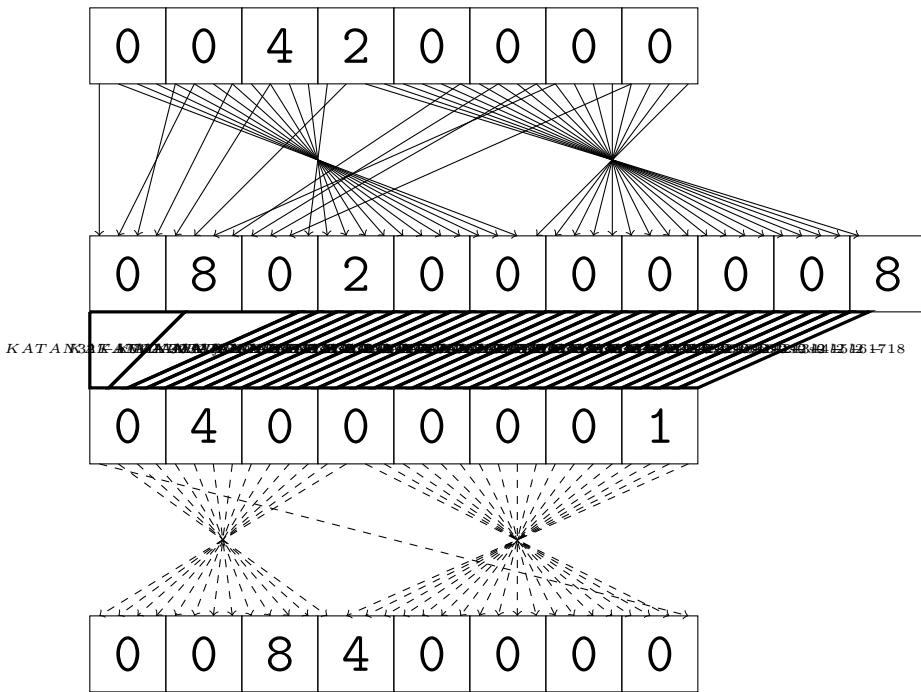
124 **KATAN32-r61**

Maximal differential probability: $2^{-0.0}$



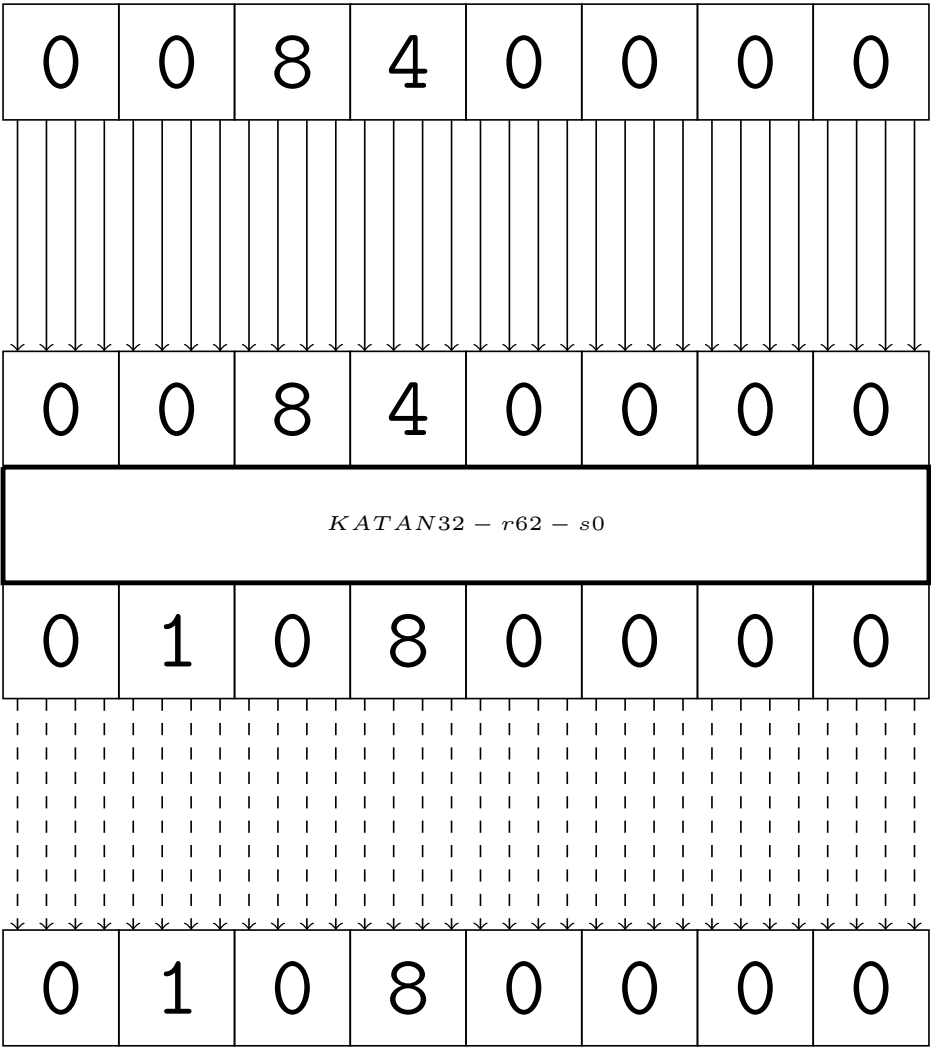
125 KATAN32-r61-s0

Maximal differential probability: $2^{-0.0}$



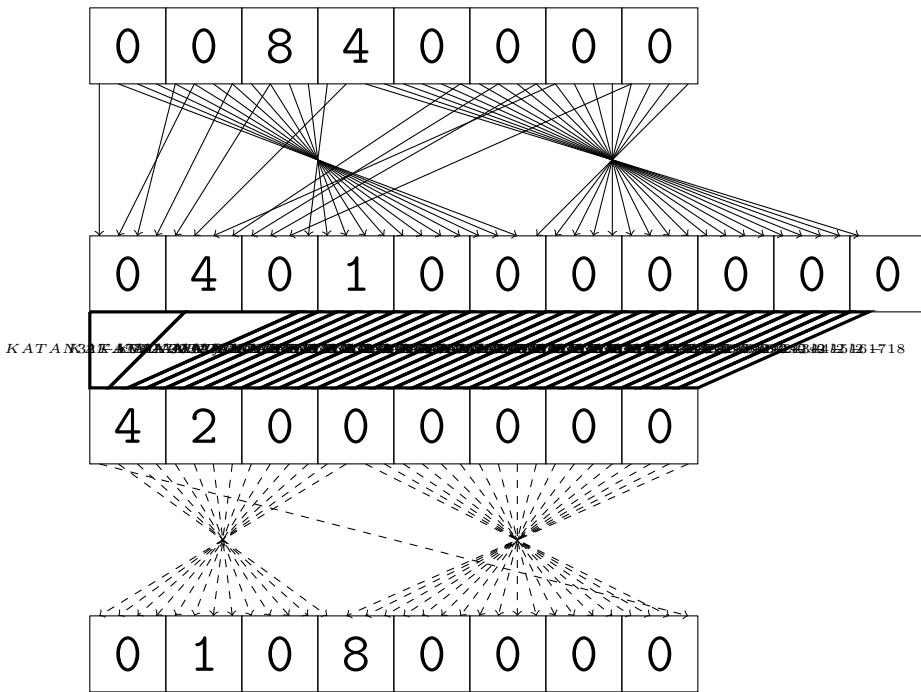
126 KATAN32-r62

Maximal differential probability: $2^{-0.0}$



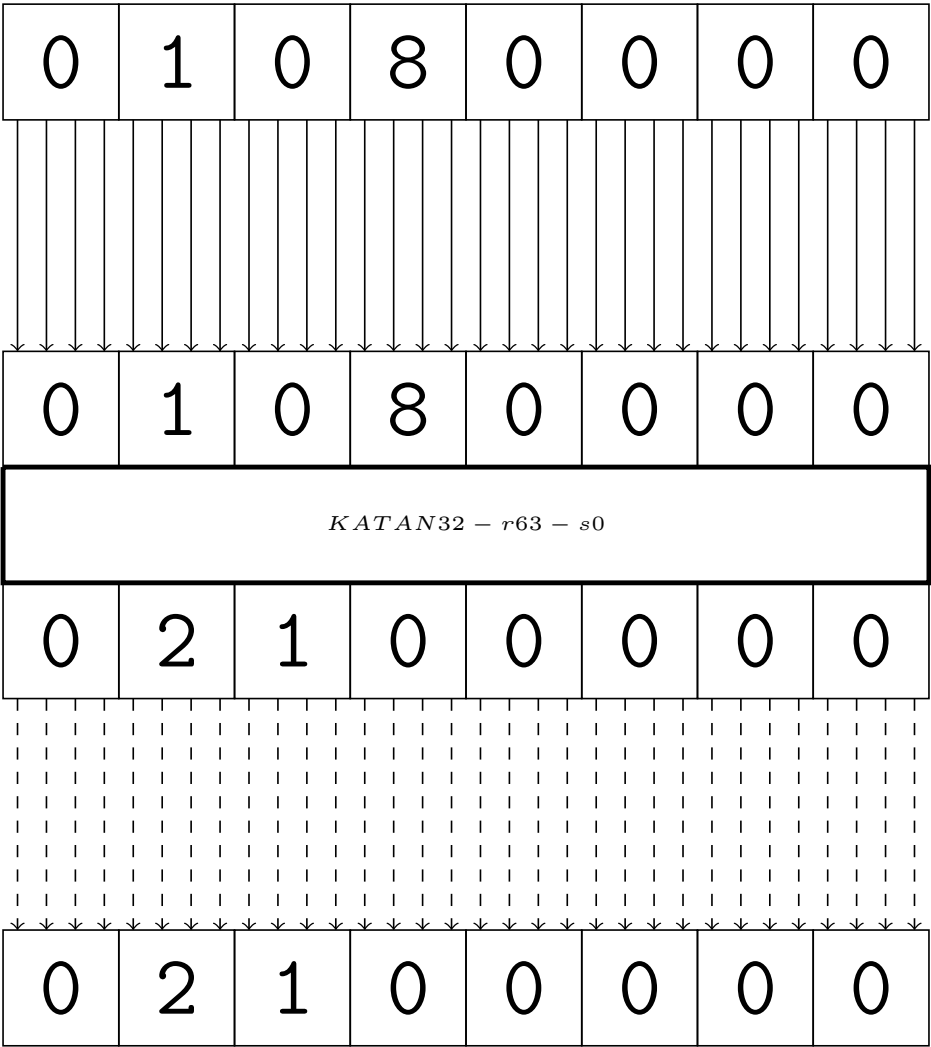
127 KATAN32-r62-s0

Maximal differential probability: $2^{-0.0}$



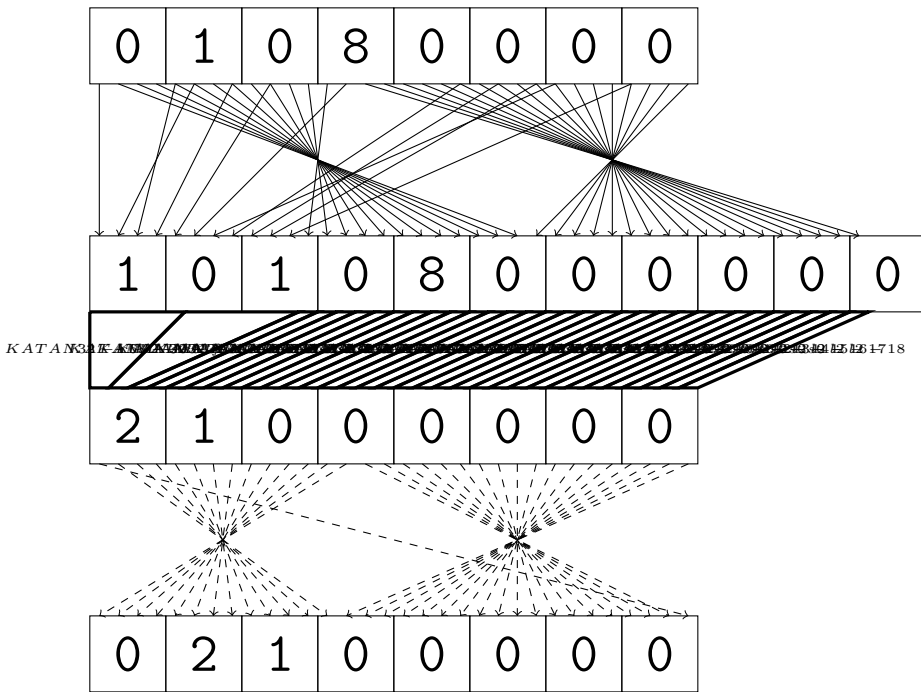
128 **KATAN32-r63**

Maximal differential probability: $2^{-1.0}$



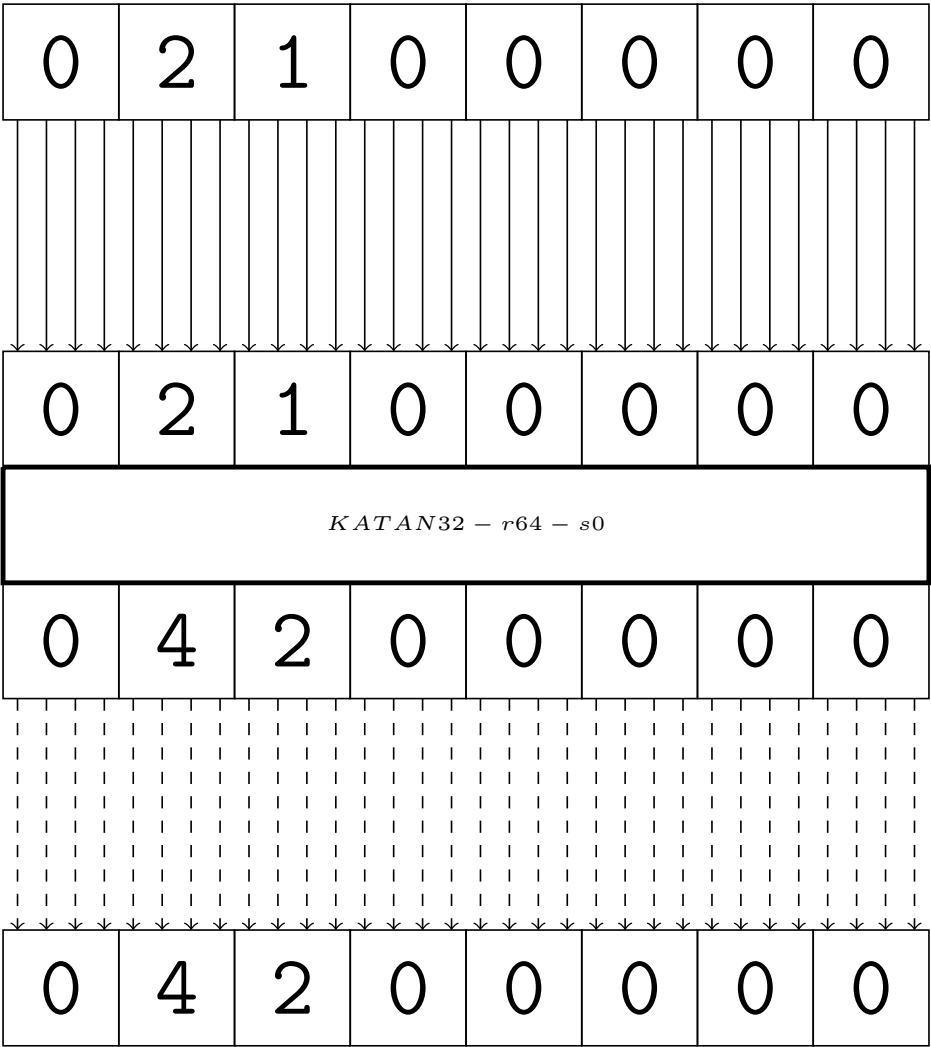
129 KATAN32-r63-s0

Maximal differential probability: $2^{-1.0}$



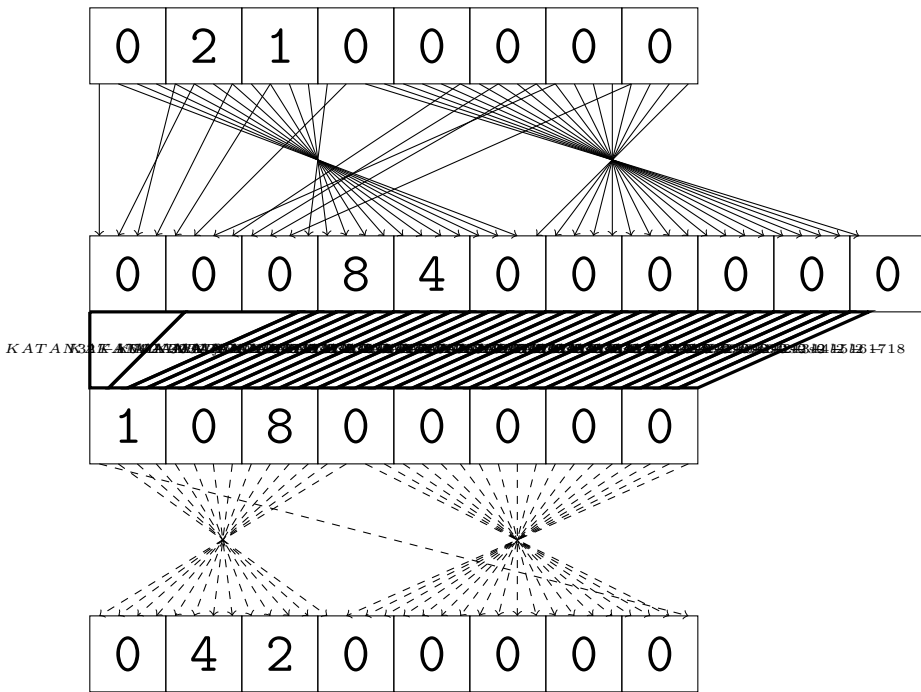
130 KATAN32-r64

Maximal differential probability: $2^{-0.0}$



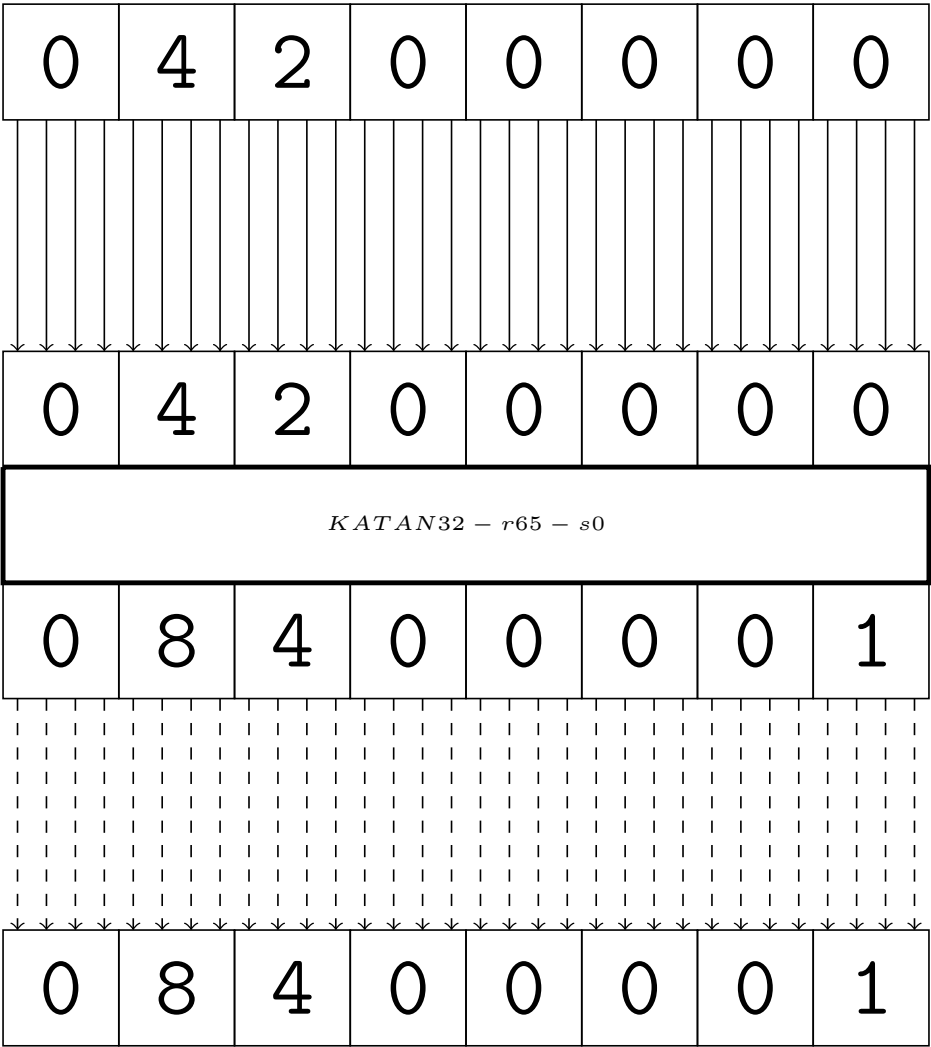
131 KATAN32-r64-s0

Maximal differential probability: $2^{-0.0}$



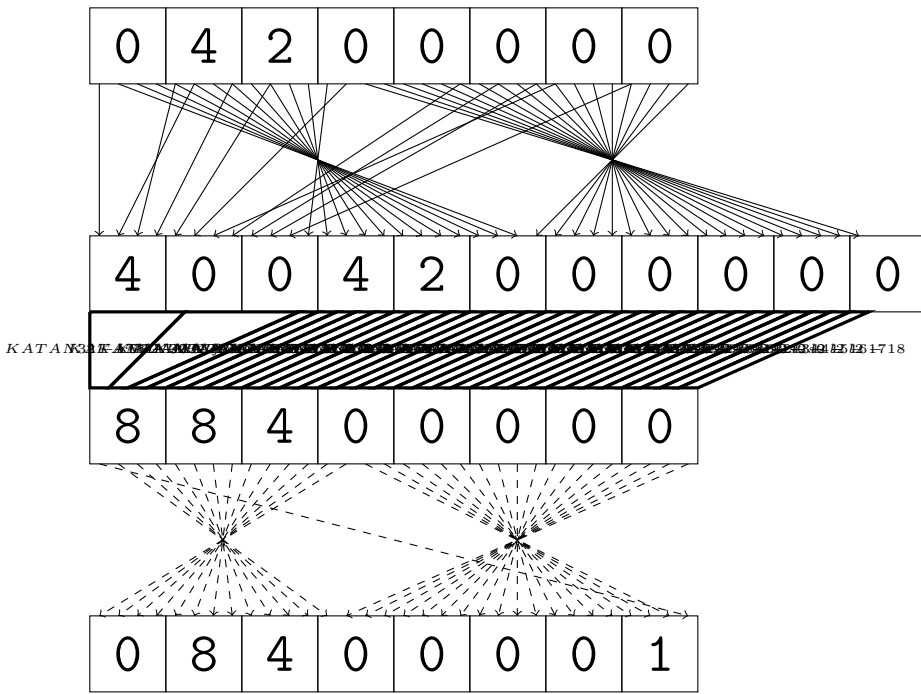
132 KATAN32-r65

Maximal differential probability: $2^{-0.0}$



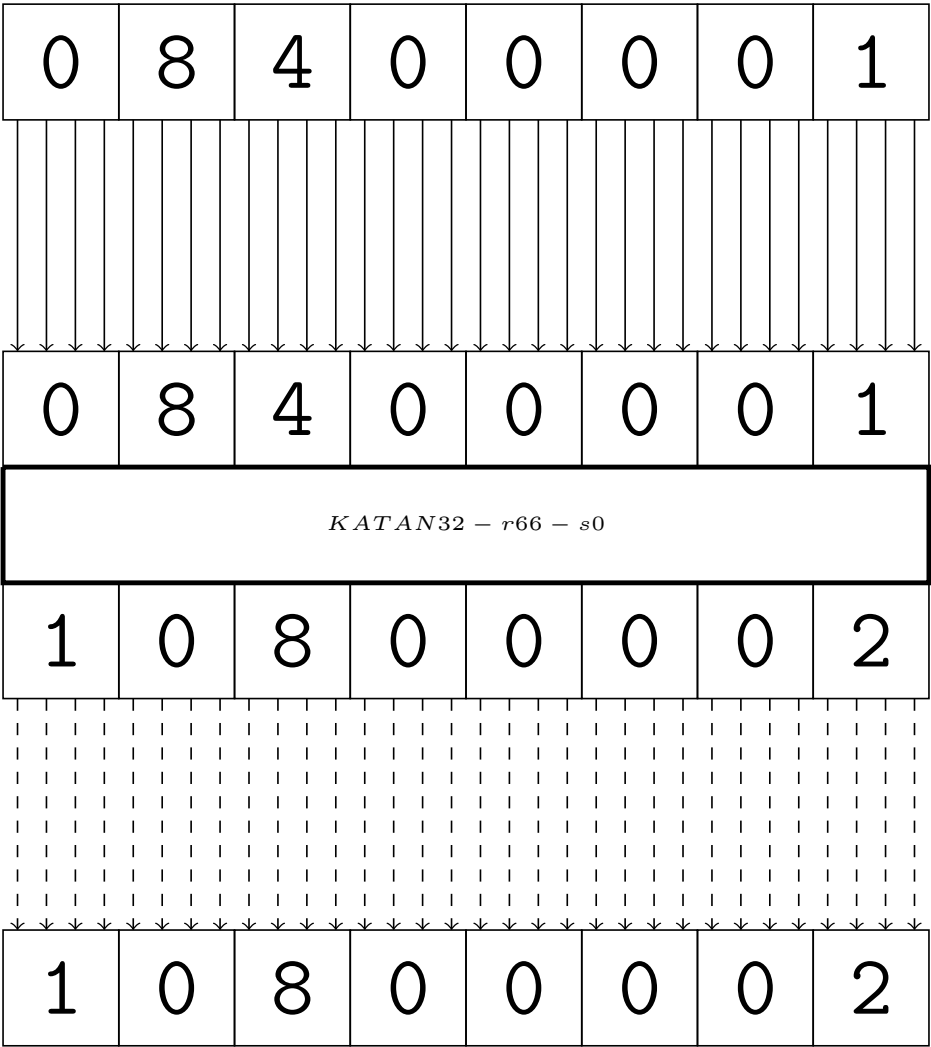
133 KATAN32-r65-s0

Maximal differential probability: $2^{-0.0}$



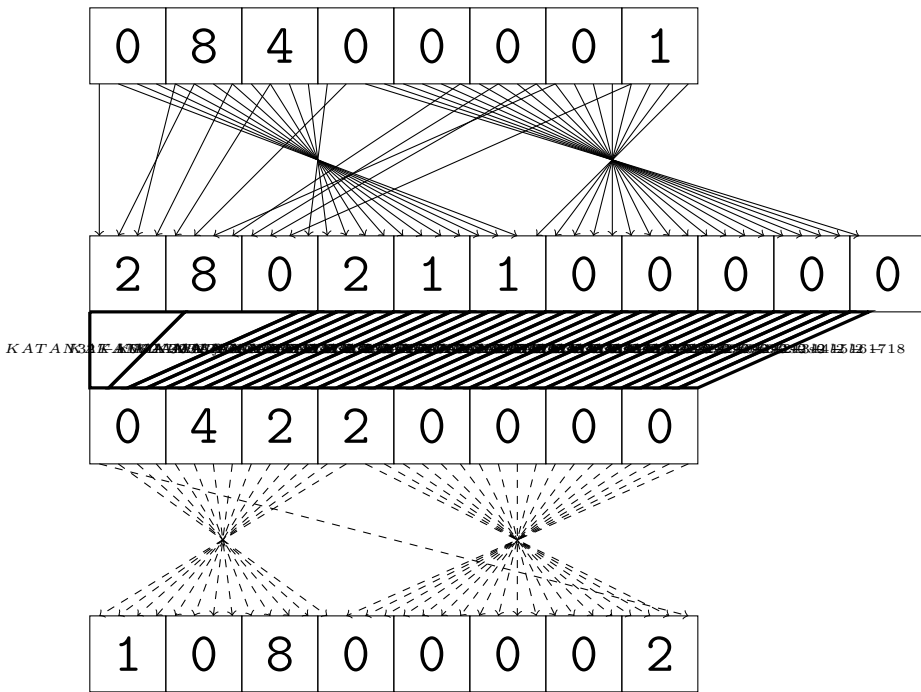
134 **KATAN32-r66**

Maximal differential probability: $2^{-1.0}$



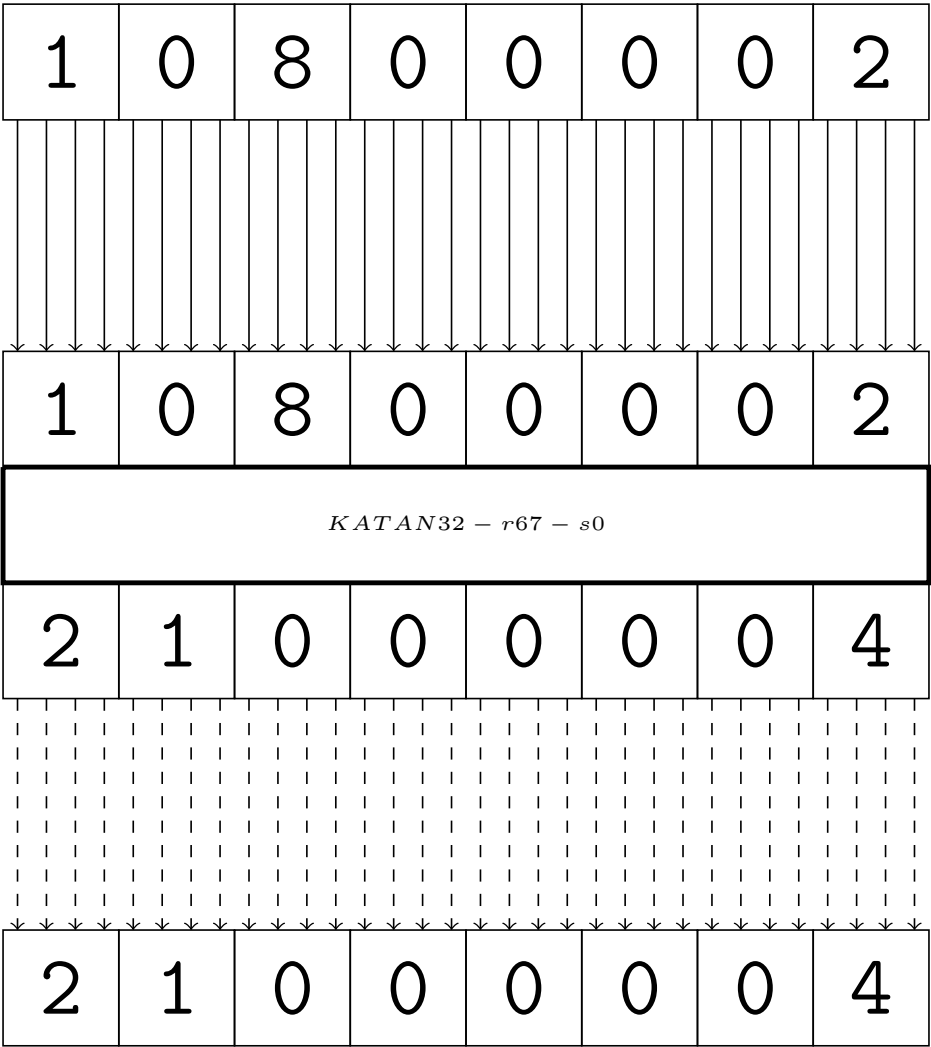
135 KATAN32-r66-s0

Maximal differential probability: $2^{-1.0}$



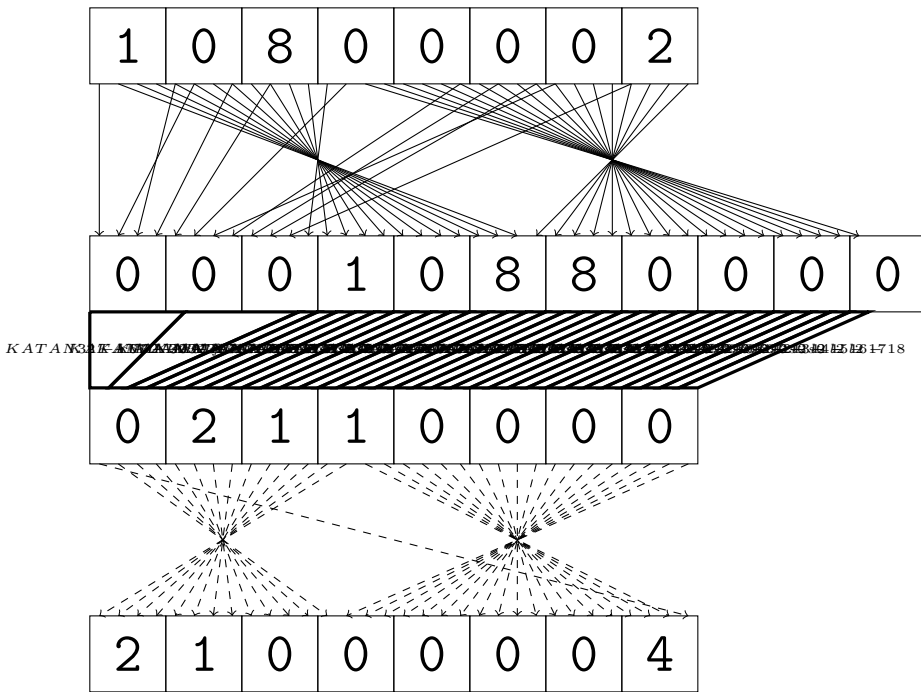
136 **KATAN32-r67**

Maximal differential probability: $2^{-0.0}$



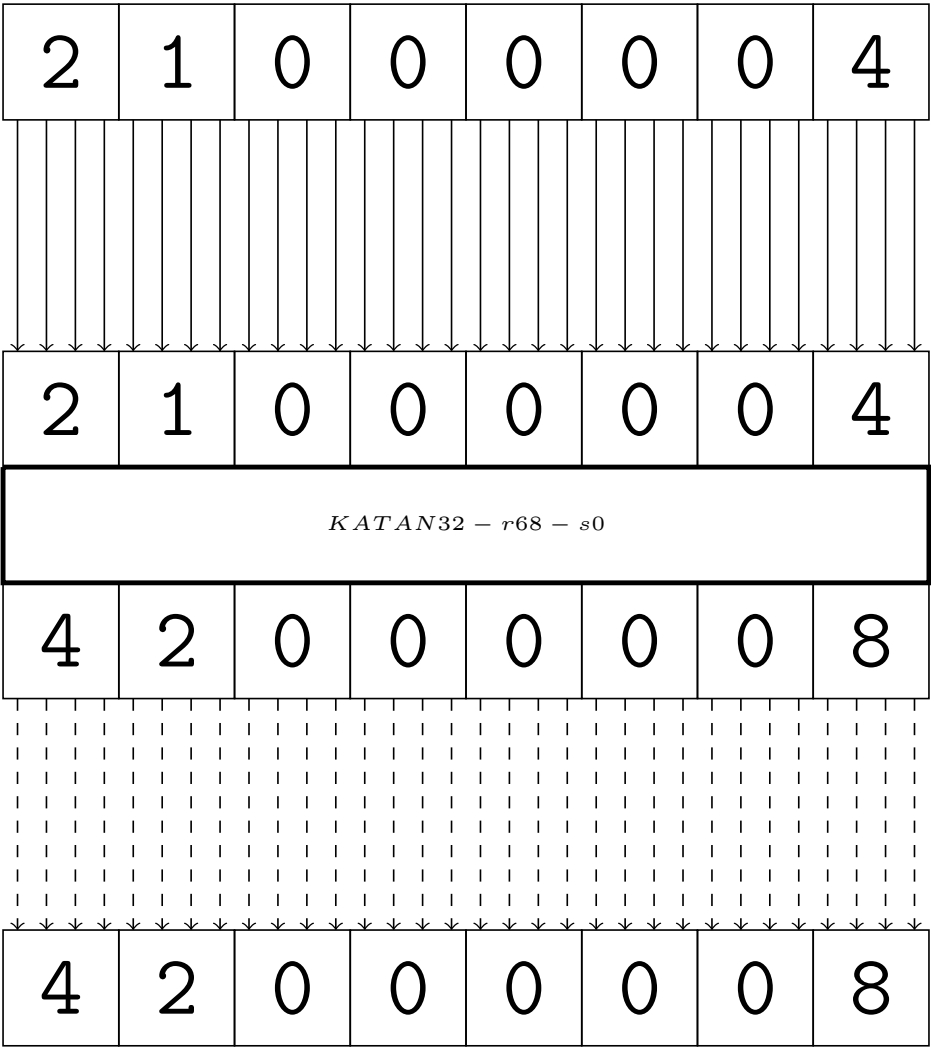
137 KATAN32-r67-s0

Maximal differential probability: $2^{-0.0}$



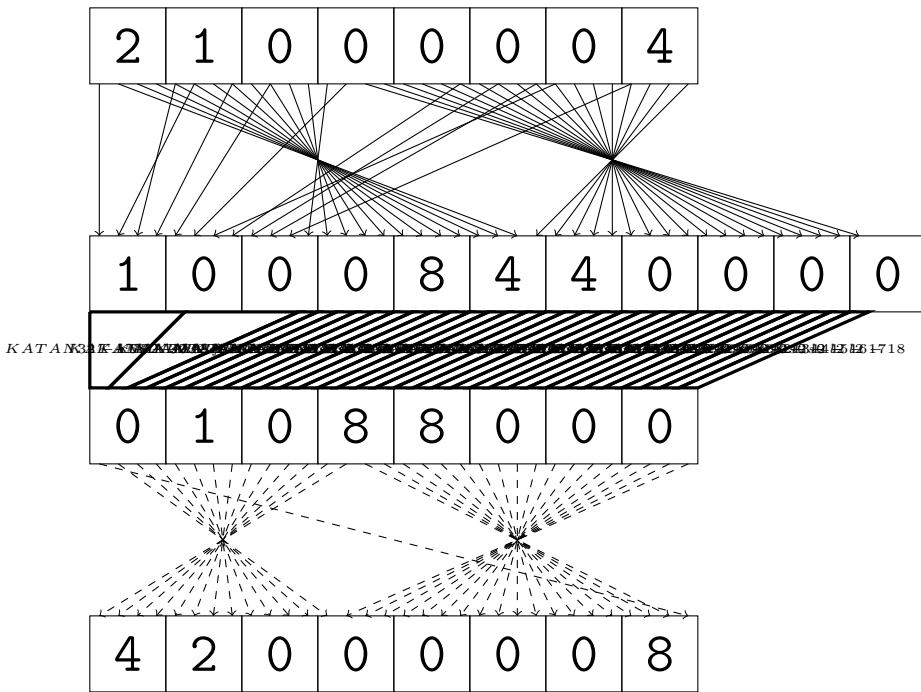
138 **KATAN32-r68**

Maximal differential probability: $2^{-1.0}$



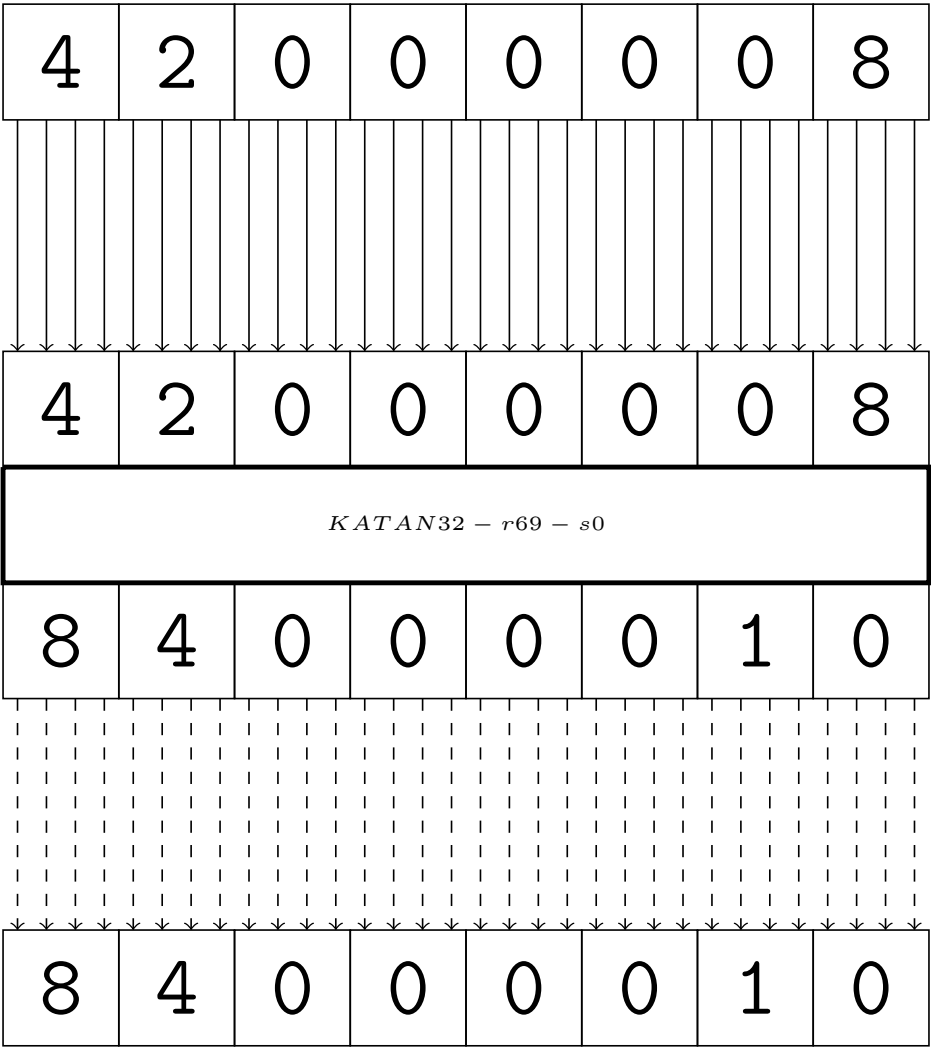
139 KATAN32-r68-s0

Maximal differential probability: $2^{-1.0}$



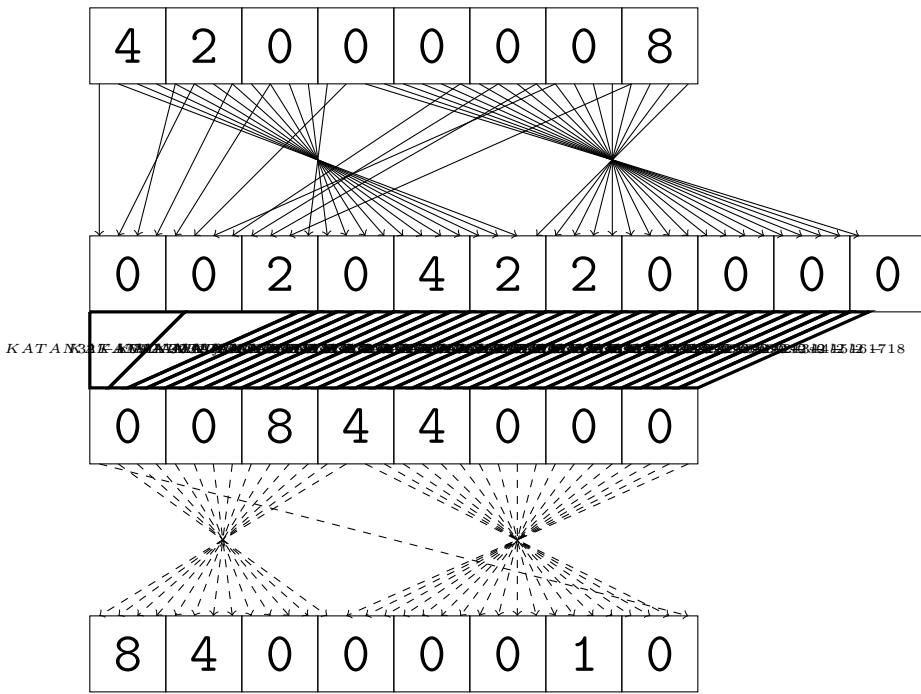
140 **KATAN32-r69**

Maximal differential probability: $2^{-1.0}$



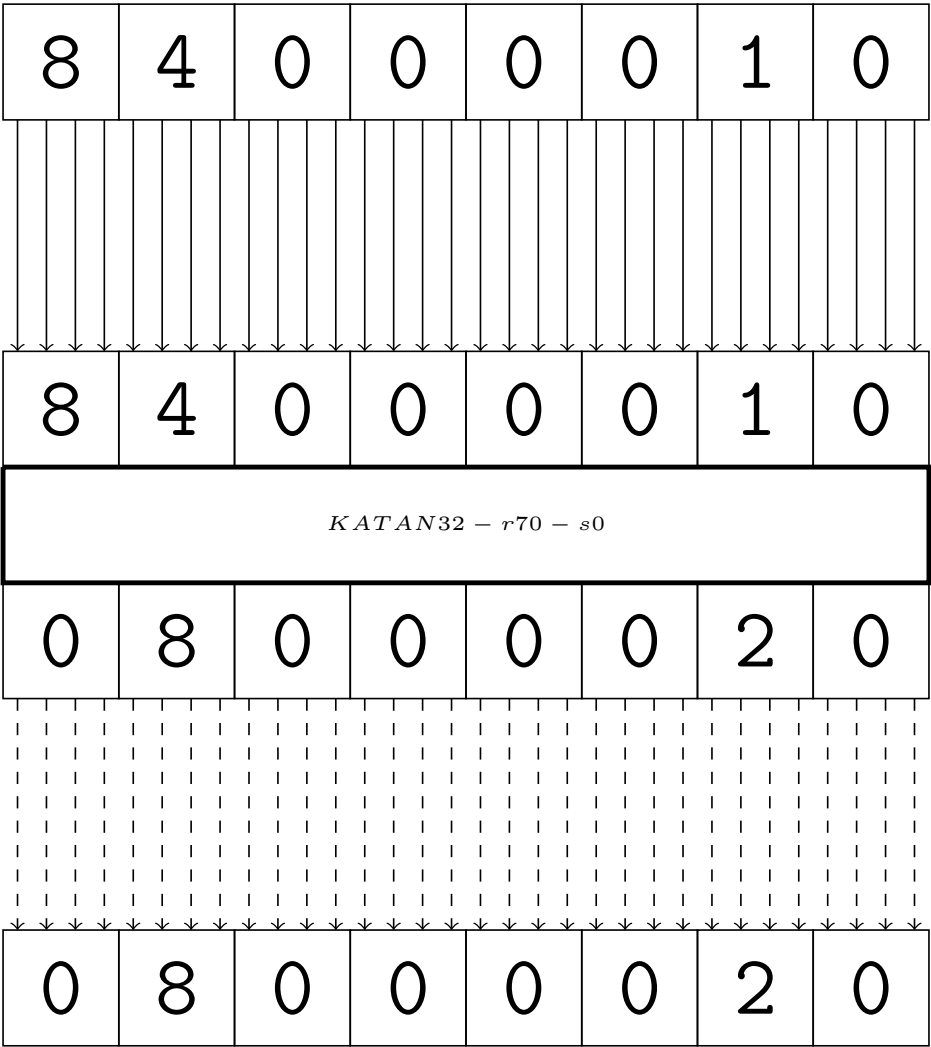
141 KATAN32-r69-s0

Maximal differential probability: $2^{-1.0}$



142 **KATAN32-r70**

Maximal differential probability: $2^{-0.0}$



143 KATAN32-r70-s0

Maximal differential probability: $2^{-0.0}$

